MILITARY SPECIFICATION

MULTIPLEXER, ASYNCHRONOUS DIGITAL COMBINER TD-976( )/G, RESTORER, PULSE FORM TD-982( )/G AND ANCILLARY ITEMS

This specification is approved for use by EL, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification describes the requirements for an Asynchronous Digital Combiner Multiplexer TD-976( )/G, Pulse Form Restorer TD-982( )/G, and ancillary items (see 6.1). The TD-976 and TD-982 are rugged, lightweight, compact, and compatible with associated Shelter Mounted Army Tactical Communications Systems (ATACS) equipment.

1.2 Ancillary items. (See 6.1). The ancillary items used with the TD-976 and TD-982 are as follows:

a. Extender panel for the TD-976.

b. Special adapter cable to interconnect the AN/PTM-7 Test Set and the TD-982.

c. Spare panel assemblies storage case for the TD-976.

FSC 5805

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:58Z
Check the source to verify that this is the current version before use.

2. APPLICABLE DOCUMENTS

2.1 *Issues of documents*. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

NN-P-71	Pallets, Material Handling, Wood, Double Faced, Stringer Construction
QQ-S-781	Steel Strapping, Flat
PPP-B-585	Box, Wood, Wirebound
PPP-B-601	Box, Wood, Cleated Plywood
PPP-B-621	Box, Wood, Nailed and Lock-corner
PPP-B-636	Box, Fiberboard
PPP-B-640	Box, Fiberboard, Corrugated, Triple-Wall
PPP-F-320	Fiberboard, Corrugated and Solid Sheet Stock (Container Grade)
PPP-P-291	Paperboard, Wrapping, Cushioning
PPP-S-760	Strapping, Nonmetallic, (and Connectors)
PPP-T-97	Tape, Pressure-Sensitive Adhesive Filament Reinforced

Military

MIL-P-116	Preservation, Methods of
MIL-L-6363	Lamp, Incandescent, Aviation Service, General Requirements for
MIL-L-7961	Light, Indicator, Press to Test
MIL-P-11268	Parts, Materials, and Processes Used in Electronic Equipment

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:58Z
Check the source to verify that this is the current version before use.

MIL-H-11996	Handset H-60( )/PT, Handset H-156( )/U, and Handset H-165( )/U
MIL-M-13231	Marking of Electronic Items
MIL-F-14072	Finishes for Ground Signal Equipment
MIL-E-15090	Enamel, Equipment, Light-Gray (Formula No. 111)
MIL-P-19644	Plastic Molding Material (Polystyrene Foam, Expanded Bead)
MIL-C-24308	Connector, Electric, Rectangular, Miniature Polarized Shell, Rack and Panel, General Specification for
MIL-C-26482	Connector, Electric, Circular, Miniature, Quick Disconnect, Environment Resisting
MIL-P-26514	Polyurethane Foam, Rigid or Plastic, for Packaging
MIL-M-38510	Microcircuit, General Specification for
MIL-C-55442	Cable Assemblies and Cord Assemblies, Packaging of
MIL-T-55603	Test Set, Telephone AN/PTM-7( )
MIL-C-55661	Connector, Plug, Electrical UG-1870( )/U, Connector, Receptacle, Electrical UG-1837( )/U

Electronics Command

*EL-CP0076-0001A Appendix I	Multiplexer TD-976( )/G and Restorer Pulse Form TD-982( )/G

*Classified Document.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:58Z
Check the source to verify that this is the current version before use.

STANDARDS

Federal

FED-STD-356	Commercial Packaging of Supplies and Equipment
FED-STD-595	Color

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-147	Palletized and Containerized Unit Loads 40 inch x 48 inch Pallets, Skids, or Pallet-Type Base
MIL-STD-252	Wired Equipment, Classification of Visual and Mechanical Defects
MIL-STD-415	Test Provisions for Electronic Systems and Associated Equipment, Design Criteria for
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-461	Electromagnetic Interference Characteristics, Requirements for Equipment
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-471	Maintainability Verification/ Demonstration/Evaluation

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:58Z
Check the source to verify that this is the current version before use.

MIL-STD-781	Reliability Tests; Exponential Distribution
MIL-STD-810	Environmental Test Methods
MIL-STD-1332	Tactical Prime, Precise, and Utility Terminologies for Classification of DOD Mobile Electric Power Engine Generator Set Family
MIL-STD-1472	Human Engineering Design Criteria for Military Systems, Equipments and Facilities
MS-28775	Packing, Preformed, Hydraulic, Plus 275 Deg F, ("O" Ring)

DRAWINGS

Electronics Command

SC-B-693994	Bracket (Card Storage Box)
SM-B-529174	Connector, Receptacle, Electrical
SM-B-530020	Holder, Arrester
SM-B-530036	Cap, Arrester

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:58Z
Check the source to verify that this is the current version before use.

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply.

ANSI Z35.1-1972 Accident Prevention Signs, Specification for

(Copies of ANSI may be obtained from the American National Standards Institute, Inc., 430 Broadway, New York, N.Y. 10018).

*OR 12, 991-1 (U) Nuclear Hardening, EMP Final Report for Multiplexer TD-976( )/G, and Pulse Form Restorer TD-982( )/G: Revised May 1975

(Availability of this document shall be as specified in the contract.)

*Classified Document

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:58Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

3.1 Construction. The equipment shall be constructed to conform with the requirements of this specification and the contract.

3.2 First article. When specified in the contract or purchase order, the contractor shall furnish first article units in accordance with 4.3

3.3 Parts, materials, and processes; general. In addition to the requirements of this specification, the full requirements of MIL-P-11268, including the selection requirements therein, shall apply.

3.4 TD-976 performance requirements.

3.4.1 TD-976 general characteristics. The TD-976 shall be capable of multiplexing and demultiplexing on a time division basis up to eight separate digital bit streams (PCM groups) into a 4.9152 Mb/s bit stream (supergroup). The digital group inputs rate 288/576 kb/s. The equipment shall accept 6/12 channel rate full width PCM signals at its group inputs. The supergroup output shall be processed for transmission over a 1/2 mile of cable without restorers and up to 5 miles with restorers at half-mile intervals. Provisions shall be made for an analog cable order wire signal to be superimposed on the supergroup bit stream and a digital voice order wire signal as part of the supergroup. These shall be terminated at a handset connector on the TD-976, and on its RAU when used and a 4-wire digital voice channel shall also be provided, digitally multiplexed into the supergroup output and available only at the RAU. The OW channels shall not interfere in any way with the group multiplexing operation.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:58Z
Check the source to verify that this is the current version before use.

3.4.2 Cable power supply. The cable power supply, which is a part of the TD-976, shall be capable of providing power for restorer operation for a total of 10 miles to permit loop back operation at a distant terminal. The Remote Alarm Display and Access Unit (RAU) shall be stored within the TD-976 and, when connected by an interconnecting cable, shall be suitable for console mounting.

3.4.3 Frame format. Frame formats for the 48 and 96 channel modes are shown in figures 1a and 1b. These formats shall be used unless alternate formats are approved by the Government.

3.4.4 Group input-output connectors. There shall be 8 group input, 8 group output, and 8 timing output connectors provided for interconnection and operation with the 6 and 12 channel PCM and timing signals of the Multiplexer TD-660( )/G and the Multiplexer TD-204/TD-754( )/G. All connectors shall be on the rear panel of the multiplexer.

3.4.5 Input signals (See 4.17.1). The equipment shall meet all requirements of this specification when the PCM input group bit rates are 288 kb or 576 kb $\pm$.0045%. The nominal pulse widths are either 3.47 microseconds or 1.74 microseconds. Digital "1" is between +1.0 and -0.8 volts relative to chassis ground and Digital "0" is between -1.2 and -3.0 volts relative to chassis ground. The rise and fall times of all input signals shall be a maximum of 150 nanoseconds.

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:58Z
Check the source to verify that this is the current version before use.

3.4.5.1 Input impedance (See 4.17.2). The input impedance of the PCM group IN connections shall be 91 ohms plus or minus 10% up to at least 5 MHz.

3.4.6 Output signals (See 4.17.3). Group output and timing signals shall be as specified in 3.4.6.1 and 3.4.6.2 when terminated in a 91 ohm resistive load.

3.4.6.1 PCM output signals. The PCM group output signals appearing at the 91 ohm termination shall be full width binary pulses whose bit rate is the same as the input bit rate. The pulse width shall be the reciprocal of the bit rate. Digital "1" shall be -0.2 +0.2 volts and digital "0" shall be -2.2 +0.2 volts relative to chassis ground. Peak ripple voltage plus noise shall be less than 0.4 volts peak-to-peak. Rise & Fall times shall be 120 nanoseconds, maximum.

3.4.6.2 Output timing signals. The timing signal appearing at the 91 ohm termination shall be a positive going periodic pulse train having a nominal repetition rate of 576 kb. Pulse width shall be 80 to 120 nanoseconds. Signal baseline shall be -2.2 +0.2 volts relative to chassis ground. The pulse peak shall be 2.0 +.2 volts above the baseline. Rise and fall times shall be 120 nanoseconds maximum. Peak ripple voltage shall be less than 0.4 V peak-to-peak. The positive going edge of the timing signal shall lead the transition of the PCM signal by 10 to 70 nanoseconds measured at the 50% amplitude points.

Source: https://assist.dla.mil -- Downloaded: 202640-26T03:58Z
Check the source to verify that this is the current version before use.

3.4.6.3 Output dummy signal. The dummy pattern, appearing at the 91 ohm termination in lieu of the PCM signal, shall have the same voltage levels at 576 kb/s as in 3.4.6.1 with a unique pattern (see 3.4.15.5).

3.4.6.4 Output jitter (See 4.17.4). The TD-976 shall be capable of interfacing with the 40 mile cable system employing TD-206's using either a TD-754 or TD-204 at the ends. When the cable system equipment operates error free over a frequency range of 576 kb ±40 bits per second, the addition of the TD-976 to this system shall not decrease this range to less than ±17 bits per second.

3.4.7 Group bit count integrity (BCI) (See 4.17.5). The equipment shall exhibit absolute group bit count integrity for 24 hours in an environment of randomly distributed errors up to 1 in 1000 in the supergroup bit stream.

3.4.7.1 Time to group bit count integrity. Group bit count integrity shall be attained within one-half second after applying a PCM input signal with the TD-976 in either the 48 or 96 channel mode. Switching from 48 to 96 channels shall not increase this time beyond the one half second, maximum.

3.4.8 System operation (See 4.17.7 and 4.17.8). The TD-976 group output signal characteristics shall permit proper operation over a 40 mile PCM cable system. It shall also permit proper operation through 20 TD-976's, i.e., 10 tandem hops. Supergroup input/output characteristics shall permit proper operation over a 0 to 5 mile cable system employing TD-982's with another TD-976 or the AN/GRC-144 radio. Proper operation shall include meeting the error rate requirements of 3.4.8.1 for an eight (8) hour period with no loss of BCI during this period.

Source: https://assist.dla.mil -- Downloaded: 202640-26T03:58Z
Check the source to verify that this is the current version before use.

3.4.8.1 Group error rate (See 4.17.33). Under all operating conditions the error rate between the PCM group input and output terminals on a zero to five mile path through a pair of TD-976's, or in tandem operation using up to 20 TD-976's, shall not exceed 1 per minute. Applying the analog order wire signal, 1600 Hz ring, shall not increase the error rate.

3.4.9 Super group output-input signals (See 4.17.11 and 4.17.33). The TD-976 shall interface with Cable Assembly Special Purpose, Electrical CX-11230. Circuits shall be provided to drive up to 1/2 mile length at a 4.9152 Mb/s rate without restorers. A selectable build out network shall simulate the attenuation of 1/4 mile of cable. Access to this attenuator shall be available from the front panel. The TD-976 shall provide cable power for the TD-982 for the entire 5 mile cable system including loop-back. A front panel switch shall remove or apply this cable power or loop-back the power from a distant TD-976.

3.4.9.1 Supergroup output signals (See 4.17.9 and 4.17.33). The signals appearing at the output connector of the TD-976 shall be a composite of 1/2 baud bi-polar data, analog voice orderwire, and constant current restorer power. When there are greater than four (4) inputs to the TD-976, the information output shall always be at the 96-channel information rate. When there are less than or equal to four (4) inputs, the information output shall be at either the 48 or 96 channel rate. The information rate shall be manually switch selected by the operator. When the TD-976 is transmitting at the 48 channel information rate, each bit shall be transmitted twice, i.e., shall appear in two consecutive slots.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:58Z
Check the source to verify that this is the current version before use.

3.4.9.1.1 Data (See 4.17.9). The data signal at the termination is specified with the output connector to the cable system terminated into 56 ohms and the building out network attenuator set for 0 attenuation. Digital "1" signals shall be alternate positive and negative half baud pulses. Pulse amplitude shall be +0.9 ±0.1 volts for positive pulses and -0.9 ±0.1 volts for negative pulses. Pulse width shall be 100 ±10 nanoseconds measured at the 50% amplitude points. Rise and fall times shall be 25 nanoseconds maximum. Pulse droop shall be 0.2 volts maximum. Backswing shall be 0.2 volts maximum. Ripple plus noise shall be less than 5% of pulse amplitude. The above parameters are defined in Figure 2. Digital "0" signals shall be at the ac zero level and the residual ripple plus noise shall not exceed 0.05V from the baseline.

3.4.9.1.2 Orderwire (See 4.17.10). The cable orderwire signal appearing at the output connector to the cable system shall be as follows when the connector is terminated in 880 ohms. With 0 dBm at 1000 Hz applied to the microphone terminals of the handset connector and the switches set for analog voice orderwire, the cable orderwire signal shall be 2.75 volts rms ±10%. Witb no signals applied, the residual noise level shall be less than 20 millivolts rms.

3.4.9.2 Supergroup input signals. The signals appearing at the input connector of the TD-976 from the cable system shall be a composite of the bi-polar data, analog voice orderwire, and constant current restorer power after these signals have passed through the zero to five mile restorer cable system.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:58Z
Check the source to verify that this is the current version before use.

3.4.9.3 Dynamic range (See 4.17.13). The TD-976 and TD-982 shall provide proper operation when the input signal level from the cable is varied by ±4 dB of the nominal value at room temperature.

3.4.9.4 Lightning protection. All supergroup transmit and receive circuits of the TD-976 and TD-982 shall be protected from high voltage surges caused by lightning or static discharge. Lightning protectors consisting of cap arrester per SM-B-530036, holder arrester per SM-B-530020 and packing, preformed, "0" Ring per MS-28775-012 shall be provided.

3.4.9.5 Supergroup loop back switch (See 4.17.12). A switch located on the front panel of the TD-976 shall be provided to enable the supergroup output stream to be internally patched to the supergroup input circuitry. This switch shall also loop back the incoming supergroup signal to the supergroup output.

3.4.10 Master oscillator (See 4.17.15). The master oscillator provided in the transmit section shall have a frequency accuracy of 9830400 Hz ±.003 percent.

3.4.11 Frame supergroup synchronization time (See 4.17.14). Frame synchronization of the supergroup signal shall not require more than 30 milliseconds following application of the S.G. IN signal to the unit in an error free environment. A frame synchronization time of 50 ms shall not be exceeded more than 10% of the time with a supergroup input error rate of one error per 1000 bits distributed randomly in the supergroup signal with a confidence level of 90%.

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:58Z
Check the source to verify that this is the current version before use.

3.4.12 PCM group switches (See 4.17.16 and 4.17.29). A two position manual switch shall be provided for each of the 8 PCM groups to permit transmission of data with or without a PCM frame pattern. This card edge mounted switch shall be available from the front of the equipment when the trap door is open.

3.4.12.1 PCM group switch "ON". With the group switch in the "ON" position, the equipment shall perform as specified in 3.4.15.

3.4.12.2 PCM group switch "OFF". With the group switch in the "OFF" position, the action of the equipment alarms for these specific functions, including the insertion of dummy, shall be inhibited. An idle pattern which is either the complement of the dummy pattern or is made up of all "1"s shall be multiplexed into the supergroup stream.

3.4.13 Order wires. The TD-976 shall have provision for the following order wires:

a. Analog voice - cable orderwire.

b. Digital voice orderwire.

c. Digital data/TTY orderwire.

3.4.13.1 Analog voice. This orderwire shall be superimposed on the supergroup cable signal to the radio or a remote TD-976. It shall not be switched or otherwise interconnected between adjacent TD-976's. At the radio terminal it may be interconnected with the radio order wire and extended to a distant station. A front panel switch shall be provided for signalling, plus both audible and visual indicators for incoming calls.

Source: https://assist.dla.mil -- Downloaded: 202640-26T03:58Z
Check the source to verify that this is the current version before use.

3.4.13.2 Digital voice. This orderwire shall be included as part of the supergroup bit stream. It will be encoded A/D and decoded D/A solely at the TD-976 and may be switched or otherwise interconnected between multiplexers on a 4 wire analog basis. A total of six (6) analog-digital-analog conversions may be utilized on an operating circuit. A hybrid arrangement shall be provided by the TD-976 to permit full duplex communication between the local operator and distant terminal operators. A front panel switch for signalling plus both audible and visual indicators for incoming calls, is required. Facilities for the 4 wire interconnection shall only be a part of the Remote Alarm Display and Access Equipment of 3.4.14.2.

3.4.13.3 Digital data/TTY. This order wire information shall be included as part of the supergroup bit stream. The input and output signal shall be 75 baud nonsynchronous teletype with provision for future transmission of 1200 b/s data. The orderwire information signals are not synchronized with the clock of the TD-976. This orderwire shall not be switched or interconnected between adjacent TD-976's. The terminating teletype equipment will be an AN/FGC-25X or equivalent. The teletype signal shall be 20 or 60 milliamperes +130 volts neutral. Loop battery shall not be furnished by the Multiplexer TD-976. Signalling and receive-send signals for this order wire shall be provided by the TD-976. Facilities for this interconnection are only available at the RAU.

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:58Z
Check the source to verify that this is the current version before use.

3.4.14 Orderwire appearances (See 4.17.16 and 4.17.17). Orderwire appearances shall be both on the front panel of the TD-976 and on the Remote Alarm Display and Orderwire Access Unit (RAU) which is a part of the TD-976. This unit normally is stored within the TD-976 when it is not in use. Where functions are on both the front panel and the remote access unit and both are used, these shall be in parallel. Incoming calls on DVOW or AVOW shall be displayed on both units and a response from either shall remove the audible and visual signals from both. Access to the TTY terminals and the 4 wire DVOW shall be only on the RAU.

3.4.14.1 Front panel. A handset connector and associated controls shall be located on the front panel of the TD-976. Both the analog voice and digital voice orderwire channels shall use the same handset through appropriate switches. Interconnection between the two shall not be possible.

3.4.14.1.1 Signalling. A signalling detector and indicator, both audible and visual, shall be associated with each orderwire to indicate the presence of an incoming call. Signalling shall be possible over each of the two orderwire channels.

3.4.14.1.1.1 Signalling circuits. The signalling oscillator shall operate at 1600 +30 Hz for signalling purposes for the analog orderwire. The detector shall operate with tone signals originating from a 1600 Hz oscillator at the opposite end of the link and it shall be protected from speech "talk down". When the detector is activated, an audible signal shall sound. The activated conditions shall also light a lamp labeled CABLE O.W. Answer by the operator at the TD-976 front panel or at the remote alarm display and orderwire access equipment (see 3.4.14.2) shall remove the signal.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:58Z
Check the source to verify that this is the current version before use.

3.4.14.1.2 Handset. Handset H-156( )/U in accordance with MIL-H-11996 shall be used to interface with the handset connector. Talking battery voltage shall be supplied by the TD-976.

3.4.14.1.3 Analog voice (See 4.17.18, 4.17.19 and 4.17.20). The analog voice orderwire characteristics shall be as follows when the front panel switches are set to connect the handset connector to the analog voice orderwire:

a. Input impedance: 50 ohms ±10% at 1000 Hz.

b. Output impedance: 150 ohms ±10% at 1000 Hz.

c. Input level: 0 dBm (1000 Hz reference).

d. Sidetone level: -18 dBm ±4 dB.

e. Output level: -18 ±1 dBm (1000 Hz) in a 5 mile system.

f. Frequency response relative to 1000 Hz when two Multiplexers are connected as a system with 0 to 5 miles of cable:

±0.5 to -1 dB from 300 Hz to 1400 Hz

±0.5 to -4 dB from 200 Hz to 1700 Hz

3.4.14.1.4 Digital voice (See 4.17.18, 4.17.21 and 4.17.22). The digital voice orderwire characteristics shall be as follows when the front panel switches are set to connect the handset connector to the digital voice orderwire:

a. Input impedance: 50 ohms ±10% at 1000 Hz.

Source: https://assist.dla.mil -- Downloaded: 202640-26T03:58Z
Check the source to verify that this is the currentversion before use.

b. Output impedance: 150 ohms ±10% at 1000 Hz.

c. Input level: 0 dBm (1000 Hz reference).

d. Sidetone level: -18 dBm ±4 dBm.

e. Output level: -15 dBm ±2 dBm (1000 Hz)

f. Frequency response relative to 1000 Hz when two TD-976's are connected as a system with 0 to 5 miles of cable:

+0.5 to -1.5 dB from 300 to 3500 Hz

+0.5 to -6.0 dB from 200 to 3640 Hz

3.4.14.2 Remote alarm display and access unit. An equipment for remote display of the alarms and access to the orderwire circuits shall be provided. This unit shall duplicate the functions described by 3.4.14.1, 3.4.14.1.1, 3.4.14.1.1.1, 3.4.14.1.2, 3.4.14.1.3, 3.4.14.1.4, and 3.4.15. Also, this unit shall provide access to the TTY terminals and the 4 wire DVOW. Interconnection between the TD-976 and its RAU shall be in accordance with 3.4.14.3 and 3.4.14.3.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:58Z
Check the source to verify that this is the current version before use.

3.4.14.2.1 Digital voice (See 4.17.21, 4.17.23, 4.17.24 and 4.17.25). The access unit shall provide an additional digital voice appearance. The second appearance shall be on a four (4) wire, 600 ohm basis. This appearance shall be physically and electrically isolated from the associated operator's handset appearances. The effect on the extension circuit of connecting and/or disconnecting the operator's position shall be minimized. The characteristics of the digital voice orderwire extension shall be as follows:

a. Input impedance: 600 ohms ±10% balanced.

b. Output impedance: 600 ohms ±10% balanced.

c. Input level: -4 dBm (1000 Hz reference).

d. Output level: -4 ±1 dBm.

e. Frequency response relative to 1.0 kHz when two TD-976's are connected as a system with 0 to 5 miles of cable.

+0.5 to -1.5 dB from 300 to 3500 Hz

+0.5 to -6.0 dB from 200 to 3640 Hz

f. Input-output characteristics: Shall be linear within ±1 dB for signals from -4 to -30 dBm.

g. Signal to noise ratio: Shall exceed 55 dB for F1A weighting for one A/D-D/A conversion. For six (6) A/D-D/A conversions signal to noise ratio shall be greater than 42 dB.

h. Distortion: The signal to quantizing distortion ratio shall not be less than 30 dB below the fundamental for one (1) A/D-D/A conversion. For six (6) A/D-D/A conversions the signal to quantizing distortion shall not be less than 24 dB below the fundamental

i. Longitudinal balance: Shall not be less than 40 dB.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:58Z
Check the source to verify that this is the current version before use.

3.4.14.2.2 Digital data/TTY (See 4.17.26, 4.17.27 and 4.17.28). The access unit shall also provide a digital data appearance on a four (4) wire full duplex (FDX) basis.

a. The TD-976 output shall operate as part of a 20 or 60 milliampere +130 volt DC neutral loop, and in addition shall provide the capability for future accommodation of a low level polar, ±1 milliampere ±6 volt loop. The low level loop may be accommodated by separate appearances or internal jumpering or switching by the operator. One signal level will be utilized at a time.

(1) Max speed: Neutral - 75 bits per second.

Polar - 1200 bits per second.

(2) Code: Neutral - Baudot (stop-start).

Polar - ASC II.

(3) Impedance: Polar - 100 ohms max.

(4) Bias distortion: Neutral - ±10% max.

Polar - ±10% max.

(5) Longitudinal balance: 40 dB minimum.

(6) Loop power: External. Provision is made at the Multiplexer TD-976 for connection of polar loop supply on the transmit side.

(7) Terminal equipment: Neutral - AN/FGC-25X.

Polar - Unknown

(8) The receive error rate shall not exceed 1 pulse in 10^8 bits.

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:58Z
Check the source to verify that this is the current version before use.

b. The TD-976 input circuits shall operate as part of a 20 or 60 milliampere +130 volt DC neutral loop, and in addition, shall accommodate a low level, ±1 milliampere ±6 volt loop. The low level loop shall be accommodated by separate appearances. One signal level will be utilized at a time.

(1) Max speed: Neutral - 75 bits per second.

Polar - 1200 bits per second.

(2) Code: Neutral - Baudot (start-stop).

Polar - ASC II.

(3) Impedance: Neutral - 150 ohms resistive.

Polar - 6000 ohms min (non inductive)

max capacitance 2500 picofarads.

(4) Bias distortion: Neutral - ±35% max.

Polar - ±49% max.

(5) Longitudinal balance: 40 dB min.

(6) Loop power: External.

(7) Terminal equipment: Neutral - AN/FGC-25X.

Polar - Unknown

Source: https://assist.dla.mil -- Downloaded: 20261O-26T03:58Z
Check the source to verify that this is the current version before use.

3.4.14.2.3 Signal. All orderwire circuits shall have an associated remote signal lead appearance. This lead will normally be connected to ground. Detection of "ring" on a particular orderwire shall cause a voltage to be applied to this lead. This voltage shall be removed with the cessation of "ring". Answer by the operator at the TD-976 or remote alarm display and orderwire access equipment shall remove any locked up signals.

3.4.14.2.4 Control. All orderwire circuits shall have an associated remote control lead appearance. This lead shall normally be open. Application of ground to the control lead at the remote location shall cause the multiplexer to transmit "ring" on the associated orderwire. Removal of ground shall cause ring transmission to cease. Sufficient release delay shall be provided in the multiplexer to insure that the duration of ring signal at the distant station will be sufficient to activate detection circuits. The input impedance of the multiplexer circuit shall be a minimum of 1000 ohms with a current carrying capacity of 30 mA minimum.

3.4.14.2.5 Squelch circuits. A squelch shall be incorporated in order to quiet the D.V.O.W. circuits and TTY interface circuits while the frame of the TD-976 is not synchronized. The TTY signal shall produce a Mark hold condition.

3.4.14.3 Cable connectors for RAU AND TD-976 (See 4.4). Connector M24308/24-35, in accordance with MIL-C-24308 shall be provided on the rear of the remote orderwire access unit and connector MS3114E-22-55S with protective cover, MS3181-22N, in accordance with MIL-C-26482 on the rear of the TD-976.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:58Z
Check the source to verify that this is the current version before use.

3.4.14.4 RAU display power. The orderwire and alarm display lamps on the RAU shall be powered by an external 24V DC (nominal) power source.

3.4.14.4.1 Display power connector (See 4.4). Connector M24308/24-31, in accordance with MIL-C-24308 shall be mounted on the rear of the RAU.

3.4.14.5 Display lamp assembly (See 4.4). Alarm displays and call indicating signals shall utilize lamp assemblies in accordance with MIL-L-7961 and midget flange base bulbs per MS-18209 of MIL-L-6363 or equivalent. A single test switch to activate all the lamp assemblies may be used. Operation of the test switch shall not cause errors in the signal.

3.4.14.6 Display size. The front panel outer dimensions and fastening provisions shall be in accordance with the engineering sketch as furnished in the bid request and contract.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:58Z
Check the source to verify that this is the current version before use.

3.4.15 Alarms (See 4.17.29 and 4.17.33). The following conditions shall be detected and displayed on the front panel:

a. Loss of framing identification in the super group signal.

b. Loss of cable transmit signal.

c. Absence of PCM activity on group input signals.

d. Absence of PCM activity on group output signals.

e. Presence of a dummy signal.

All alarm detection circuits shall be operational within 5 seconds after power application to the TD-976. A time guard of 1 second delay shall prevent intermittent alarm indications. Alarm conditions shall be removed automatically when proper operation is restored. Audible alarms shall be provided and conform to human engineering criteria of MIL-STD-1472, paragraphs 5.3 through 5.3.6. This audible alarm shall sound different than the OW call signal. Visual alarm indicator lights shall conform to criteria of MIL-STD-1472 paragraphs 5.2.5.19 for color coding, 5.2.5.10 for brightness level and 5.2.72 for spacing.

3.4.15.1 Supergroup frame identification alarm. Loss of frame sync in the received supergroup signal shall activate a front panel lamp and an audible signal. The capability of remoting this alarm through an external wire pair shall be provided. The open circuit (no alarm) resistance shall be 5 k ohms or greater and an operated (alarm on) resistance of 50 ohms or less. This alarm shall deactivate the group alarms.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:58Z
Check the source to verify that this is the current version before use.

3.4.15.2 Supergroup transmit signal alarm. Loss of transmit signals shall activate a front panel lamp and an audible signal. Capability of remoting this alarm through an external wire pair shall be provided by a closure of 50 ohms or less. The open circuit resistance shall be 5 k ohms or greater.

3.4.15.3 Group input activity alarm. The absence of data transitions within each PCM group entering the TD-976 shall activate visual and audible indication for each specific group in which this failure is detected and the unique dummy pattern shall be inserted into the failed groups to prevent tandem failure indications of the group alarms. Less than 2,000 transitions in a second shall be considered a failure and activate the alarms. Capability to remote the alarm status of each group shall be provided. Activation of the alarm detection circuits shall not affect start-up operation of the multiplexer or any terminal equipment. Receipt of the unique dummy pattern shall inhibit this alarm.

3.4.15.4 Audible signal pushbutton switch. A push-button switch on the front of the TD-976 shall turn off the audible signal for each alarm occurrence. Even parity shall be used so that an even number of failure indications results in no audible alarm, and odd number shall cause an audible alarm which can be reset. Removing the cause of the alarm subsequent to being "reset", shall reactivate the audible alarm until pushbutton is again "reset". Operation of the pushbutton switch shall have no delay.

Source: https://assist.dla.mil -- Downloaded: 202640-26T03:58Z
Check the source to verify that this is the current version before use.

3.4.15.5 Dummy PCM signal. The absence of a PCM input signal or dummy shall be detected and shall automatically cause a dummy pattern to be inserted into the SG output stream. The unique dummy pattern shall not duplicate the 2 kHz or 4 kHz pattern used in the TD-660. Detection of the dummy pattern in the SG receive stream will cause a visual indication that the dummy pattern is present in one or more of the receive groups. Capability to remote this alarm status shall be provided.

3.4.15.6 Group output activity alarm. The absence of PCM activity within each group leaving the TD-976 shall be detected. A visual indication shall be activated for each specific group in which this failure is detected. Capability to remote the alarm status of each group shall be provided. Activation of this receive alarm shall not activate the audible alarms. Action of the alarm detection circuits shall not affect start-up operation of the TD-976 or any terminal equipment. Receipt of the unique dummy pattern shall inhibit this alarm.

3.4.15.7 Alarm operation check. A push button switch shall be provided on the TD-976 front panel and also on the remote unit to check the operation of the visual alarms without interrupting the equipment operation (see 3.4.14.5).

3.4.15.8 Power loss indicator. A front panel indication shall be provided to indicate loss of primary power to the active circuits of the equipment, i.e., the light indicating power 'on' shall go 'off' to indicate a power failure.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:58Z
Check the source to verify that this is the current version before use.

3.4.16 Cable system fault location (See 4.17.30 and 4.17.33). Means shall be provided in the TD-976 to locate faulty restorers. The detection shall be accomplished by a comparison of the cable power supply output voltage under the conditions of no bipolar signal and proper bipolar signal. Means to display the voltage reading shall be provided. Accuracy shall be such that a faulty restorer half can be located in a 1 to 5 mile system.

3.4.17 Combination case. The TD-976 shall have the chassis and case integrated as a combination case. The case shall be equipped with 2 handles for 2 man carry.

3.4.18 Size. The TD-976, including the RAU, shall be less than 16 inches high and 15 inches deep. Width shall be suitable for relay rack mounting. These outside dimensions shall conform to the engineering sketch furnished in the bid request and contract.

3.4.19 Weight. The combined weight of the TD-976 and RAU shall not exceed 75 pounds. A warning label shall be affixed to any unit weighing more than one man lift capability in conformance with paragraph 5.9.11.3.3 of MIL-STD-1472.

3.4.20 Power supply. Removal and replacement of the plug in power supply shall be from the front of the equipment.

3.4.21 Power dissipation (See 4.17.31). The power dissipation shall not exceed 150 watts and shall not use rotary cooling devices to cool equipment.

Source: https://assist.dla.mil -- Downloaded: 202640-26T03:58Z
Check the source to verify that this is the current version before use.

3.4.22 Prime power (See 4.17.31). The equipment shall operate satisfactorily when supplied power having a nominal 120V, single phase, with equipment ground, 50 and 60 hertz having characteristics as contained in MIL-STD-1332 for Type I, Class 2, Model 1 generator sets. In addition, the equipment shall operate satisfactorily when supplied power having the following steady-state voltage and frequency:

Voltage	-	115 ±10%
Frequency	-	50 ±5% hertz 60 ±5% hertz

A positive front panel indication shall be provided to indicate that the power supply has overloaded. Line fuses or circuit breakers shall be incorporated to provide overload protection for the line.

3.4.23 Test points. Voltages and waveforms not metered or monitored by the test facilities and which are useful in maintenance of the equipment shall be available at suitably located test jacks readily accessible, but not on the front panel. Additional test jacks as required shall be located on the plug-in subassembly panels so as to be accessible with the use of an extension panel. A test jack for synchronization of an oscilloscope shall be provided on the front panel. All test jacks shall be short circuit-protected. Only operator type adjustments shall be accessible without the extender panel. Test points shall be in accordance with MIL-STD-415.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:58Z
Check the source to verify that this is the current version before use.

3.4.24 Extender panel for TD-976 (See 4.14). The extender panel shall consist of a printed wiring panel with a multipin connector at each end mounted in a metal frame. It shall be constructed for insertion between the chassis and any plug-in circuit panel and shall contain test points for all connector pins. The extender panel shall be a separate item to be furnished in the quantities specified in the bid request and contract. All circuit paths shall be continuous when tested in accordance with 4.14.

3.4.25 Defective panels. The TD-976 shall have the facility to permit the operator to locate defective panels and replace them without the use of external test equipment. It shall be possible to replace panels in the TD-976 associated with PCM groups, without interrupting traffic. Major output failures such as those indicating failure of the Master Oscillator/Overhead Generator, Timing and Control, Supergroup Driver/Receiver, or Frame Sync Detector/Maintenance circuits shall be apparent by a visual indication on the card edge. A front panel visual indicator shall light to indicate that one of the above failures exists. Operation of diagnostics shall not cause an audible alarm.

3.4.26 Spare panel assemblies storage case for TD-976. The contractor shall supply spare panel assemblies storage cases for the TD-976. Each storage case shall contain one of each type of panel assembly printed circuit card used in the TD-976. The case holding these eleven (11) panel assemblies shall be of aluminum welded construction, suitable for rack mounting in 19 inch racks using mounting brackets in accordance with SC-B-693994. The overall dimensions of the case shall be 17 $\pm$1/32 inches wide, 8 1/2 inches high, maximum, and 15 inches deep, maximum. Two 10-32 UNF mounting inserts shall be installed in each side in a vertical row located 3 3/16 inch back from the front face. Vertical spacing shall be in accordance with

Source: https://assist.dla.mil -- Downloaded: 202640-26T03:58Z
Check the source to verify that this is the current version before use.

the 5 27/32 inches dimension, in SC-B-693994, with the bottom insert centered 9/16 inches above the external bottom surface. The internal top and bottom surfaces shall have guides to hold the panel assemblies in the vertical position, which shall protect the cards from damage, dislocation or contact when rack mounted. The front cover shall be hinged at the bottom and be latched at the upper sides. The contractor shall supply two mounting brackets per case in accordance with SC-B-693994.

3.5 TD-982 power supply (see 4.17.5 and 4.17.32). Power for cable system operation shall be supplied by the TD-982 power supply in the TD-976. It shall be operated by a separate front panel ON/OFF switch and shall have the capacity to provide constant current of 45 milliamperes DC ±3% to power up to 18 restorer halves with the distant terminal in the power loop back mode so that only one power supply is required for the 5 mile system. The TD-976 shall be capable of operating with the cable power supply switch in the OFF position.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:58Z
Check the source to verify that this is the current version before use.

3.5.1 Protection (See 4.17.32). Protection of the power supply from short circuits shall be provided. There shall be both over-current and under-current protection. The supply shall automatically shut off but not affect operation of the TD-976 when the cable current changes more than 10% from the nominal. After automatic shut off, a cable power switch shall be operated to the OFF position and back to the ON position before the cable current can be restored. The equipment shall discharge the cable input voltage to a value of 30 volts or less in less than 1.0 second after the cable current is cut off or turned off. The supply voltage shall never exceed 500 VDC.

3.6 TD-982. The TD-982 shall be an unattended full duplex pulse restorer. It shall include circuitry to detect and regenerate received bi-polar signals, circuitry to detect faults at the TD-982 outputs, independent orderwire circuitry and circuitry to develop the required internal power supplies from the constant current supplied by the TD-982 power supply in the TD-976. The circuitry of the TD-982 shall be protected from high voltage surges caused by lightning or static discharge. A connector with cap shall be provided on the TD-982 for connection of PCM test equipment. This connector shall be isolated from the cable DC voltage.

3.6.1 Operating characteristics (See 4.17.34 and 4.18.1).

3.6.1.1 Signal regeneration. The regenerated output signal shall be specified as follows when the output is terminated into 56 ohms. The digital "1" signals shall be alternate positive and negative half baud pulses as shown in Figure 2. Digital "0" shall be at the voltage developed by the constant restorer power supply. The output error rate of the TD-982 shall not exceed 1 per minute when compared to the original supergroup transmission. This error rate shall also be realized with the analog orderwire signal, 1600 Hz ring, intermittently superimposed on the supergroup cable signal from either a TD-976 or a AN/PTM-7 Test Set.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:58Z
Check the source to verify that this is the current version before use.

3.6.1.2 Signal detection. The received signal shall be the bipolar signal described in Figure 2, attenuated by one-half mile of Cable Assembly, Special Purpose, Electrical CX-11230( ).

3.6.1.3 Timing recovery (See 4.18.2). Timing for operation of the TD-982 shall be recovered from each of the input signals. The stability of the timing circuits shall be such that the relative phase shift through the TD-982 shall be less than ±20 nanoseconds for each of the following listed conditions:

a. Pattern density variation of from 10% to 90% digital "1's".

b. Transmitted timing frequency variation of ±0.004 percent of nominal.

3.6.1.4 Orderwire (See 4.18.3). The TD-982 shall provide proper loading to the cable system to maintain an impedance of 880 ohms ±10% for all frequencies between 300 and 1700 Hz. The orderwire circuits shall bypass the digital regeneration circuits and be operational even though TD-982 power and the regeneration circuitry should fail. Orderwire signals shall not produce digital errors in the regenerated output signal. The orderwire signals shall be brought to a connector on the TD-982 shell to provide quick access to the cable orderwire.

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:58Z
Check the source to verify that this is the currenversion before use.

3.6.1.5 Power supply (See 3.5). Power for operation of the active circuits of the TD-982 shall be derived from power supplied by the TD-976.

3.6.1.6 Fault location (See 4.17.30). Circuitry in the TD-982 shall detect loss of supergroup signal and insert a known resistance in series with the constant current power. This will produce a known change in cable voltage for each TD-982 half having no data signal output. The fault detection circuitry located in TD-976 shall be capable of using this resistance to pinpoint the faulty TD-982 section.

3.6.1.6.1 Cable System Test Set AN/PTM-7 (See 4.17.30). The AN/PTM-7, in accordance with MIL-T-55603EL, will be utilized for maintenance of the cable system. The orderwire facility of Test Set AN/PTM-7 shall be used to service the cable system. A special adapter cable shall be provided to make the PTM-7 and TD-982 interoperate properly. This assembly shall meet the requirements of MIL-P-11268.

3.6.2 Size. The dimensions of the TD-982 shall not exceed 4 inches in diameter or 9 inches in length.

3.6.3 Weight. The weight of the TD-982 shall not exceed 5 lbs.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:58Z
Check the source to verify that this is the current version before use.

3.7 Short circuit proofing all equipments and interfaces (See 4.21). All inputs and outputs of all interfaces between the TD-976, TD-982, and Cable CX-11230 shall be short-circuit proof. This shall also apply to all orderwire interfaces, alarms, and extensions.

3.8 Service conditions (See 4.7). The equipment shall meet the following service conditions. Where a test is referenced, meeting the test shall be considered as compliance with the requirement.

3.8.1 Temperature (See 4.7.1).

a. Low temperature (See 4.7.1.1 and 4.7.1.2).

(1) Operating - The TD-976 shall meet the specified performance of 4.16.1 and 4.17.13, at ambient temperatues as low as -25°F. The TD-982 shall meet the specified performance of 4.16.2, at ambient temperatures as low as -50°F.

(2) Storage and transportation - The equipment shall withstand exposure to ambient air temperatures as low as -70°F.

b. High temperature (See 4.7.1.3 and 4.7.1.4).

(1) Operating - The TD-976 shall meet the specified performance of 4.16.1 and 4.17.30, at ambient air temperatures as high as +125°F. The TD-982 shall meet the specified performance of 4.16.2, at ambient air temperatures as high as +160°F.

(2) Storage and transportation. The equipment shall withstand exposure to ambient air temperatures as high as +160°F.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:58Z
Check the source to verify that this is the current version before use.

3.8.2 Vibration (See 4.7.4). The equipment shall meet the specified performance of 4.16.1 for the TD-976 and 4.16.2 for the TD-982, after being tested in accordance with 4.7.4, frequency range 5.5-200 Hz.

3.8.3 Bounce vehicular, TD-976 (See 4.7.5). The TD-976 shall meet the specified performance of 4.16.1 after being tested in accordance with 4.7.5 with a 10g input load.

3.8.4 Bounce (loose cargo), TD-982 (See 4.7.6). The TD-982 shall meet the specified performance of 4.16.2 after being tested in accordance with 4.7.6.

3.8.5 Shock (bench handling), TD-976 (See 4.7.7). The TD-976 in its servicing condition shall withstand shock encountered during servicing and shall meet the specified performance of 4.16.1 after being tested in accordance with 4.7.7.

3.8.6 Shock (drop), TD-982 (See 4.7.8). The TD-982 shall meet the specified performance of 4.16.2 after being tested in accordance with 4.7.8.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:58Z
Check the source to verify that this is the current version before use.

3.8.7 Humidity (See 4.7.2). Each of the equipments shall exhibit no physical damage, such as corrosion, rust, blistering, swelling or deterioration of parts and materials, and shall meet the specific performance of 4.16.1 for the TD-976 and 4.16.2 for the TD-982.

3.8.8 Altitude (See 4.7.3). Each of the equipments shall meet the specific performance of 4.16.1 for the TD-976 and 4.16.2 for the TD-982 at altitudes up to 15,000 feet above sea level and shall withstand air transportation at altitudes up to 40,000 feet above sea level.

3.8.9 Dust (See 4.7.9). After testing as specified in 4.7.9, the TD-976 and TD-982 shall meet full specification performance when tested in accordance with 4.16.1 and 4.17.30 for the TD-976 and 4.16.2 for the TD-982.

3.8.10 Salt fog (See 4.7.10). After the salt fog test of 4.7.10, the TD-976 and TD-982, when examined visually with the aid of a 10-power magnifier, shall show no evidence of degradation, such as flaking, pitting, blistering or loosening of finish or metal surface; or exfoliation (see 6.8) of metal. Electrical operation is not required.

3.8.11 Fungus (See 4.7.11). After the fungus test of 4.7.11, the equipment shall be visually examined, using a 10-power magnifier. The equipment shall show no more than six minute unrelated spots, each no greater than 0.015 square inch in area, of sparse microbial growth (see 6.8) as evidenced by growth colonization (see 6.8) (which includes branching (see 6.8) and sporulation (see 6.8) on or within each cubic foot, or fraction thereof, of equipment assembly volume. Isolated instance of partial tubular germination (see 6.8) shall not be included in this evaluation.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:58Z
Check the source to verify that this is the current version before use.

3.8.12 Immersion, TD-982 (See 4.7.12). The restorer shall show no evidence of leakage after it is tested as specified in 4.7.12.

3.8.13 Rain, TD-976 (See 4.7.13). The multiplexer when tested in accordance with 4.7.13 shall not leak or show evidence of leakage and shall meet full specification performance when tested in accordance with 4.16.1.

3.9 Electromagnetic compatibility (See 4.10). The TD-976 and the TD-982 shall comply with the requirements of Table VI of MIL-STD-461A, Notice 4 as defined hereinafter:

CE01	CS01	RE02	RS03
CE02	CS02	(1)RE02.1	(1)RS03.1
CE03	CS06		
CE04			
CE05			

(1) The upper frequency limit of test of RE02.1 and RS03.1 shall be 1000 MHz.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:58Z
Check the source to verify that this is the current version before use.

3.9.1 Bonds, grounds and connectors. Bonding for equipment current path return, RF potentials and shock hazard shall be so installed that expansion, contraction or movement incident to normal service use will not break or loosen the connection. Surface preparation for bonds and grounds shall be accomplished by removing all anodic film, grease, paint, lacquer or other high resistance properties from the immediate area of contact. Bonding impedance shall not exceed 2.5 milliohms. Case or chassis mounted connectors shall have an approved RF conductive finish and shall be installed as above. Bonding impedance of the connector shell to the case shall not exceed 2.5 milliohms. Interconnecting cables requiring overall shields shall have a minimum of 40 dB attenuation to electric fields over the frequency range of 10 kHz to 400 MHz. The shield shall be constructed of tinned copper braid. Connectors used with cables having an overall shield shall have an approved RF conductive finish and provided with EMI backshells for peripheral bonding of the cable shield.

3.10 Nuclear survivability (See 4.11). The TD-976 and the TD-982 shall withstand the nuclear environment specified in EL-CP0076-0001A, Appendix I, when tested in accordance with 4.11. The equipment shall be energized during exposure. Subsequent to the tests, the equipment shall be operable as demonstrated by the test of 4.16.1 for the TD-976 and the test of 4.16.2 for the TD-982.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:58Z
Check the source to verify that this is the current version before use.

3.11 Equipment reliability (mean-time-between-failure (MTBF)) (See 4.8). The equipment MTBF's listed below shall be provided while operating at a duty cycle of 100% in a temperature environment of 40°C for the multiplexer and 50°C for the restorer and considering a fixed ground application:

a. TD-976 - MTBF (θo) = 2000 hours

b. TD-982 - MTBF (θo) = 3500 hours

3.12 Marking (See 4.4).

3.12.1 General. Marking shall conform to MIL-M-13231. Front panel marking shall be Group I as described in MIL-M-13231.

3.12.2 Visibility. Wherever practicable, parts and assemblies shall be so mounted that their identification markings will be readily visible with minimum disassembly of the equipment.

3.12.3 Serial numbers. Each TD-976 and TD-982 shall be serial numbered.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:58Z
Check the source to verify that this is the current version before use.

3.13 Finish, protective (See 4.4). The equipment shall be given protective finish in accordance with MIL-F-14072. This includes finish of hardware such as hinges, screws, etc., and necessary touch up after mounting. The top coats of paint finish on front panels shall be semi-gloss light gray enamel in accordance with MIL-E-15090. The outside of the case shall be semi-gloss olive drab, Color No. X-24087 in accordance with Federal Standard No. 595. Front panel markings shall be in black, Color No. 17038 per FED-STD-595.

3.14 Connectors (See 4.4). Power and general purpose connectors shall be of the quck disconnect type. Coaxial connectors for the PCM group inputs and outputs shall be type BNC. Supergroup input and output connectors shall be the RF high voltage type in accordance with SM-B-529174-1 and SM-B-529174-2.

3.14.1 Bulkhead connector. The TD-982 shall utilize bulkhead Connector UG-1837/U in accordance with MIL-C-55661.

3.15 Plug-in printed circuit cards (See 4.11). Plug-in printed circuit cards shall utilize separate guide and keying systems, i.e., connector guide pins shall not be used for keying purposes. The use of adhesives to affix or mount keying pins is prohibited. Cards shall provide an adequate surface other than the card edge, for use in pressing cards into place. The use of special tools for card removal is prohibited. Cards shall be capable of rapid and easy removal and insertion without the need for excessive force to perform a replacement or removal function. A wide tapered opening shall be provided on the front edge of the slides to facilitate card insertion. A slotted flange marked with applicable card identification symbol shall be mounted in front of the tapered slide in order to preclude the necessity for insertion at eye level only.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:58Z
Check the source to verify that this is the current version before use.

3.16 Panel assemblies and power supply (electrical test) (See 4.19). All TD-982 and TD-976 panel assemblies and the TD-976 plug-in power supply shall meet the requirements of their respective test requirement drawings when tested in accordance with 4.19.

3.17 Microelectronic integrated circuits (See 4.4). All microelectronic integrated circuit devices used in this equipment shall meet the requirements of MIL-M-38510 for Class B devices and their individual drawing requirements.

3.18 Mean-time-to-repair (MTTR or MCT) (See 4.9). The inherent equipment MTTR shall not exceed 30 minutes at organizational level. In addition, the maximum time-to-repair shall not exceed 60 minutes for 90% of all maintenance actions.

3.19 Interchangeability (See 4.12). Like units, assemblies, subassemblies and replaceable parts shall conform to requirement 7 of MIL-STD-454.

3.20 Systems safety engineering.

3.20.1 Personnel hazards (See 4.13). Personnel hazards shall be kept to a minimum. The criteria of MIL-STD-454, Requirement 1, shall be made applicable for tasks and efforts such as the selection of parts, the complete manufacturing and assembly processes, and any product baseline configuration changes that may be implemented during the course of the contract. Verification that compliance with this requirement has been achieved shall be through a visual examination and inspection.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:58Z
Check the source to verify that this is the current version before use.

3.20.2 Radioactive materials. Radioactive materials shall not be used (e.g., luminous dials/markings, electron tubes, surge arrestors and lenses).

3.20.3 Warning labels. Warning labels shall be provided on the equipment in accordance with ANSI Z35.1-1972.

3.21 Preconditioning (See 4.6).

3.21.1 Bounce preconditioning (See 4.6.1). Each TD-976 and TD-982 shall meet the requirements herein, without subsequent processing, after subjection to the bounce preconditioning of 4.6.1.

3.21.2 Electrical preconditioning (burn-in) (See 4.8.2). Each equipment including plug-in assemblies and spare panel assemblies shall be subjected to a 48 hour burn-in period with the final 24 hours failure free in accordance with 4.8.2

3.22 Air seal, provision for (See 4.20). The TD-982 shall provide means for enabling performance of the air seal test of 4.20.

3.23 Operational test.

3.23.1 Multiplexer (See 4.17.33). Each fabricated and tested Mutliplexer TD-976 shall be subjected to a final operational test in accordance with 4.17.33 after all other inspections and tests are completed to determine that the TD-976 is operational prior to packaging.

3.23.2 Restorer, pulse form (See 4.17.34). Each fabricated and tested TD-982 shall be subjected to a final operational test in accordance with 4.17.34 after all other inspections and tests are completed to determine that the TD-982 is operational prior to packaging.

3.24 Workmanship (See 4.4). The equipment shall be manufactured and assembled in accordance with the requirements of MIL-P-11268 and requirement 9 of MIL-STD-454.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:58Z
Check the source to verify that this is the current version before use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary. Test equipment and facilities for Government inspection shall be made available by the contractor. The Government reserves the right for authorized technical representatives to witness any contractor performed inspections and tests.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3). Does not include preparation for delivery.

b. Inspections covered by subsidiary documents. (See 4.4).

c. Quality conformance inspections.

(1) Quality conformance inspection of equipment before preparation for delivery. (See 4.5).

(2) Quality conformance inspection of preparation for delivery. (See 4.23).

Source: https://assist.dla.mil -- Downloaded: 202640-26T03:58Z
Check the source to verify that this is the current version before use.

4.3 First article. Unless otherwise specified in the contract or purchase order, the first article inspection shall be performed by the contractor.

4.3.1 First article units. The contractor shall furnish the quantity of first article units in accordance with the following:

a.	TD-976	32 each
b.	TD-982	20 each
c.	TD-976 extender panel	10 each
d.	AN/PTM-7 to TD-982 special adapter cable assembly	10 each
e.	Spare panel assemblies storage case (including spare panel assemblies)	12 each

4.3.2 First article inspection. The first article inspection shall consist of inspections specified in Table I and shall be performed in the order specified therein.

4.3.3 First article data. The first article test plan and test report(s) shall be as required in the contract.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:58Z
Check the source to verify that this is the current version before use.

Table I - First Article Inspection

Inspections	Req Para	Test Para.	Unit 1R 1M	2R 2M	3R 3M	4R 4M	5R 5M	6R 6M	7-15R 7-11M
1. Inspections covered by subsidiary documents		4.4	Inspection to be performed on all units						
2. Bounce preconditioning	3.21.1	4.6.1	"	"	"	"	"	"	"
3. Electric preconditioning (Burn-in)	3.21.2	4.6.2	"	"	"	"	"	"	"
4. Group A inspection	See Table II		"	"	"	"	"	"	"
5. Group B inspection	See Table III		"	"	"	"	"	"	"
6. Group C inspection (Electrical)	See Table IV		All electrical inspections lsited in subgroup IV of the Table to be performed on all units						
7. Environmental tests									
a. Temperature tests	3.8.1	4.7.1							
(1) Low Temp (TD-976 only)	3.8.1a	4.7.1.1			1M				
(2) Low Temp (TD-982 only)	3.8.1a	4.7.1.2			1R				
(3) High Temp (TD-976 only)	3.8.1b	4.7.1.3			2M				
(4) High Temp (TD-982 only)	3.8.1b	4.7.1.4			2R				
b. Humidity	3.8.7	4.7.2					3M, 2R		
c. Altitude (elevation)	3.8.8	4.7.3				1 M,R			

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:58Z
Check the source to verify that this is the current version before use.

Table I - First Article Inspection (Cont'd)

Inspections	Req. Para.	Test Para.	Unit 1R 1M	2R 2M	3R 3M	4R 4M	5R 5M	6R 6M	7-15R 7-11M
d. Vibration	3.8.2	4.7.4				2 M,R			
e. Bounce [TD-976 and storage case (see note f.) only]	3.8.3	4.7.5				3M			
f. Bounce (TD-982 only)	3.8.4	4.7.6				3R			
g. Shock Drop (TD-982 only)	3.8.6	4.7.8				4R			
h. Shock Bench Handling [TD-976 and storage case (see note f)only]	3.8.5	4.7.7				4M			
i. Fungus (See Note c.)	3.8.11	4.7.11	1 M,R						
j. Rain (TD-976 only)	3.8.13	4.7.13					1M		
k. Immersion (TD-982 only)	3.8.12	4.7.12					1R		
l. Salt Fog (See Note c.)	3.8.10	4.7.10			3 M,R				
m. Dust	3.8.9	4.7.9			4 M,R				
8.Nuclear Survivability	3.10	4.11						2 M,R	
9.Group D Reliability	3.11	4.8							1 M,R
10.EMI	3.9	4.10		1 M,R					

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:58Z
Check the source to verify that this is the current version before use.

Table I - First Article Inspection (Cont'd)

Inspections	Req. Para.	Test Para.	Unit 1R 1M	2R 2M	3R 3M	4R 4M	5R 5M	6R 6M	7-15R 7-11M
11. Maintainability (TD-976 only)	3.18	4.9					2M		
12. Safety	3.20	4.13						1 M,R	
13. 10 Tandem hop test (20 TD-976) (See note e.)	3.4.8	4.17.7							

NOTES:

a. The inspections 1 through 6, in the order shown, shall be performed on all first article units before subjecting these units to any other inspection requirements specified in the Table.

b. The numbers in the unit columns in the table covered by inspection paragraph 7 thru 12 specify the order of inspections for the indicated unit. Designations are "M" for the TD-976 and "R" for the TD-982.

c. The equipment shall be thoroughly washed, cleaned, dried and refurbished after this inspection before proceeding with subsequent inspection.

d. Units (7-11)M and (7-15)R. These units are to undergo inspections 1, 2, 3, 4, 5, 6 and 9.

e. This test may be conducted in any period during First Article testing, whenever sufficient units are available.

f. The spare panel assemblies storage case for the TD-976 shall be tested with a full complement of spare panels. The spare panels shall be inserted in a TD-976 and tested in accordance with 4.16.1 and 4.17.30 as stated in 4.7.5 and 4.7.7. (See 4.22).

g. A minimum of six each TD-976 units will be required for the conduct of an in-plant training course of up to two (2) weeks prior to submission of the First Article Test Report. The units used for training shall have successfully completed inspections 1 through 5. The training, as specified in the RFP and contract (see 6.2e.), is required to be programmed and performed without impacting the first article testing schedule.

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:58Z
Check the source to verify that this is the current version before use.

4.4 Inspections covered by subsidiary documents. The equipment shall be inspected under the following applicable subsidiary documents as part of the inspection required by this specification, and the inspection requirement specified in the contract.

Item	Where required
Parts, materials, and processes-general	3.3
Display lamp assembly	3.4.14.5
Finish, protective	3.13
Marking	3.12
Microelectronic integrated circuits	3.17
Connectors	3.14, 3.4.14.3, 3.4.14.4.1
Workmanship	3.24

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:58Z
Check the source to verify that this is the current version before use.

4.5 Quality conformance inspection of equipment before preparation for delivery. The contractor shall perform the inspections specified in 4.4, 4.6, 4.5.1 through 4.5.4. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's inspection records. In addition, the Government, at its discretion, may perform all or any part of the specified inspection, to verify the contractor's compliance with specified requirements.

4.5.1 Group A inspection. Subassemblies and completed equipments shall be inspected for conformance to the inspections specified in Table II. Each subassembly, after 100% inspection, shall be formed into discrete lots for quality assurance verification. The equipments assembled from lots accepted under subassembly verification shall be inspected 100% and formed into discrete lots for final QA verification. Factors of lot composition not defined herein, or in the contract or purchase order, shall be in accordance with MIL-STD-105. Each lot shall be submitted to sampling inspection, utilizing the procedures of MIL-STD-105 using general inspection levels and AQL's indicated in Table II (Quality Assurance Verification). Group A inspection shall be performed in any order which is satisfactory to the government, except that the operational test 4.17.33 shall be last for the TD-976. For the TD-982, the operational test 4.17.34 shall be next to last and the air seal test in 4.7.12.1 shall be last.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:58Z
Check the source to verify that this is the current version before use.

Table II - Group A Inspection

Inspection	Req. Para.	Ref. Fig.	Test Para.	AQL Major	AQL Minor
Visual-Mechanical TD-976					
Case and Covers	3.24		4.13	1%	4%
Remote Access Unit	3.24		4.13	1%	4%
Power Supply	3.24		4.13	1%	4%
Panel Assemblies (Including spares)	3.24		4.13	1%	4%
Assembled TD-976	3.24		4.13	1%	4%
TD-982					
Case	3.24		4.13	1%	4%
Panel Assemblies	3.24		4.13	1%	4%
Assembled TD-982	3.24		4.13	1%	4%
Ancillary Items					
Spare Panel Assembly Storage Case	3.24		4.13	1%	4%
Adapter Cable (PTM-7 to TD-982)	3.24		4.13	1%	4%
Extender panel	3.24		4.13	1%	4%

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:58Z
Check the source to verify that this is the current version before use.

Table II - Group A Inspection (Cont'd)

Inspection	Req. Para.	Ref. Fig.	Test Para.	AQL Major	AQL Minor
Electrical Test TD-976					
Assemblies, including spares (covered by TRS drawings plus all ancillary items)	3.16		4.19	1%	*
SG Input/Output Signals	3.4.9	11	4.17.11		
SG Loop Back Switch	3.4.9.5	11	4.17.12	1%	
Master OSC. Freq. Accur.	3.4.10	15	4.17.15	For	*
Cable Fault Location	3.4.16	31	4.17.30	The	
Input Power Supply (60 Hz, Nominal Voltage Only)	3.4.20	32	4.17.31	Group	
Cable Power Supply	3.5	33	4.17.32		
Cable OW Level	3.4.9.1.2	13	4.17.10		
OW Appearances	3.4.14	17	4.17.17	1%	
AVOW Output Level	3.4.14.1.3	19	4.17.19	For	*
DVOW Output Level	3.4.14.2.1	21	4.17.21	The	
Dig. Data/TTY OW Sig. Level	3.4.14.2.2	28	4.17.28	Group	
Group ON/OFF Switch	3.4.12	16	4.17.16	1%	
Alarms	3.4.15	30	4.17.29	For The Group	*

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:58Z
Check the source to verify that this is the current version before use.

Table II - Group A Inspection (Cont'd)

Inspection	Req. Para.	Ref. Fig.	Test Para.	AQL Major	AQL Minor
TD-976					
Operational Test	3.4.8.1 3.4.9.1 3.4.15 3.4.16	3a	4.17.3.3	1% For The Group	*
TD-982					
Output Signal	3.6.1.1	34	4.18.1	1%	*
Operational Test	3.6.1.1	3b	4.17.34	1%	*
Air Seal Test	3.22		4.20	1%	*

*Defects in all characteristics other than visual-mechanical are considered major.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:58Z
Check the source to verify that this is the current version before use.

4.5.2 Group B inspection. Group B inspection shall normally be performed on inspection lots that have passed Group A inspection and on samples selected from units that have been subjected to and met the Group A inspection. This inspection shall conform to Table III and to the special inspection levels of Table I of MIL-STD-105.

4.5.2.1 Group B sampling plans. The Group B AQL shall be as shown in Table III, and the inspection level shall be S4.

4.5.2.2 Order of inspection within Group B. Group B inspection shall be performed in any order satisfactory to the Government.

4.5.3 Group C inspection. Group C inspection shall be performed on units that have passed Group A and Group B inspection. The inspection shall consist of the inspections specified in Table IV. Samples shall be selected in accordance with 4.5.3.1.

4.5.3.1 Sampling for Group C inspection.

4.5.3.1.1 Sub-Group I. For this sub-group, one unit each (1 TD-976, 1 TD-982) shall be selected at random from the first production lot. For subsequent Group C inspection, one unit shall be selected every month. Units shall be subjected to tests in the order listed in Table IV.

4.5.3.1.2 Sub-Group II. For this sub-group, one unit each (1 TD-976, 1 TD-982) shall be selected at random from the first production lot. For subsequent Group C inspection, one unit shall be selected every two months. Units shall be subjected to tests in the order listed in Table IV.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:58Z
Check the source to verify that this is the current version before use.

Table III - Group B Inspection

Inspection	Req. Para.	Ref. Fig.	Test Para.	AQL
Interchangeability, Mechanical	3.19		4.12	6.5%
Electrical tests TD-976				
Input signals	3.4.5	5	4.17.1	
Group output signal	3.4.6	5	4.17.3	
Input impedance	3.4.5.1	6	4.17.2	4%
Input jitter test	3.4.6.4	7	4.17.4	For
Group BCI and power switch operation	3.4.7	8	4.17.5	The
Time to BCI	3.4.7.1	9	4.17.6	Group
SG output	3.4.9.1	12	4.17.9	
SG sync time	3.4.11	10	4.17.14	
AVOW, DVOW impedance	3.4.14.1.3 3.4.14.1.4	18	4.17.18	
AVOW freq. response	3.4.14.1.3	20	4.17.20	
DVOW freq. response	3.4.14.1.4	22	4.17.22	4%
DVOW 4-wire freq. response and linearity	3.4.14.2.1	23	4.17.23	For
DVOW S/N	3.4.14.2.1	24	4.17.24	The
DVOW distortion	3.4.14.2.1	25	4.17.25	Group
Dig. Data/TTY OW impedance	3.4.14.2.2	26 & 27	4.17.26	
Dig. Data/TTY OW bias distortion	3.4.14.2.2	29	4.17.28	

Source: https://assist.dla.mil -- Downloaded: 202640-26T03:58Z
Check the source to verify that this is the currentversion before use.

Table III - Group B Inspection (Cont'd)

Inspection	Req. Para.	Ref. Fig.	Test Para.	AQL
TD-982				
TD-982 OW characteristics	3.6.1.4	36	4.18.3	2.5% For The Group
TD-982 timing recovery	3.6.1.3	35	4.18.2	

Source: https://assist.dla.mil -- Downloaded: 202640-26T03:58Z
Check the source to verify that this is the current version before use.

Table IV - Group C Inspection

Inspection	Req. Para.	Ref. Para.	Test Para.
Subgroup I			
Temperature	3.8.1	4a,4b,31	4.7.1;4.17.30
Immersion (TD-982)	3.8.12	4b	4.7.12
Subgroup II			
Humidity	3.8.7	4a,4b	4.7.2
Drop (TD-982 only)	3.8.6	4b	4.7.8
Subgroup III			
Rain (TD-976 only)	3.8.13	4a	4.7.13
Altitude (elevation)	3.8.8	4a,4b	4.7.3
Bench Handling (TD-976)	3.8.5	4a	4.7.7
Bounce (vehicular) (TD-976 only)	3.8.3	4a	4.7.5
Vibration	3.8.2	4a,4b	4.7.4
Subgroup IV			
System operation	3.4.8	8b	4.17.8
Dynamic range	3.4.9.3	14	4.17.13
Dust	3.8.9		4.7.9
Fungus	3.8.11		4.7.11 (see note a)
EMI	3.9		4.10
Salt fog	3.8.10		4.7.10 (see note a)

NOTE: The equipment shall be washed, cleaned, dried, and refurbished if necessary before proceeding with subsequent tests.

Source: https://assist.dla.mil -- Downloaded: 202640-26T03:58Z
Check the source to verify that this is the current version before use.

4.5.3.1.3 Sub-Group III. For this sub-group, two TD-976's and one TD-982 shall be selected at random from the first production lot. For subsequent Group C inspection, two units each shall be selected every third month. One TD-976 shall be subjected to vibration and bounce, one TD-976 shall be subjected to rain, altitude and bench handling, and one TD-982 shall be subjected to altitude and vibration in that order.

4.5.3.1.4 Sub-Group IV. For this sub-group, two units each (2 TD-976, 2 TD-982) shall be selected at random from the first production lot. For subsequent Group C inspection, two units each shall be selected every six months. The tests in this sub-group may be divided between the two units, with the last test on the first unit being dust and on the second unit salt fog. An additional unit shall be selected for fungus tests only. Units shall be subjected to tests in the order listed in Table IV.

4.5.3.2 Group C failures. Actions required relative to Group C failures shall be as specified in the contract.

4.5.3.3 Reinspection of conforming Group B and Group C sample units. Unless otherwise specified, sample units which have been subjected to and passed both Group B and Group C inspection may be accepted on contract, provided all damage is repaired and the sample units are resubjected to and pass Group A inspection.

Source: https://assist.dla.mil -- Downloaded: 202640-26T03:58Z
Check the source to verify that this is the currentversion before use.

4.5.4 Group D inspection. This inspection shall consist of the test specified in Table V and shall be performed on units from lots which have been subjected to and met Group A and Group B inspection.

4.5.4.1 Group D failure. In case of a Group D failure occurrence, procedures and requirements specified elsewhere in this contract shall be followed.

4.5.4.2 Reinspection of conforming Group D sample units. Unless otherwise specified, sample units which have been subjected to and passed Group D inspection, may be accepted on the contract provided all damage is repaired and the sample units are re-subjected to and pass Group A inspection.

TABLE V - GROUP D INSPECTION

INSPECTION	REQ. PARA.	TEST PARA.
Reliability	3.11	4.8

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:58Z
Check the source to verify that this is the current version before use.

4.6 Preconditioning (See 3.21).

4.6.1 Bounce preconditioning (See 3.21.1). The unit, with shock mounts (if any) removed or blocked, shall be placed in its normal operation position on the table of the Package Tester, Type 1000-SC, as made by L.A.B. Corporation, Skaneateles, New York, or equal. The package tester, shafts in phase, shall have a speed such that it is just possible to insert a 1/32 inch strip of material under one corner or edge of the unit to a distance of 3 inches as the unit bounces. Each unit of each lot of equipments shall be subjected to bounce preconditioning for 1 minute, prior to selection of units for inspection. After bounce preconditioning, the units shall not be repaired, aligned, cleaned, or otherwise changed prior to subjection to inspection.

4.6.2 Electrical preconditioning (burn-in) (See 3.21.2). All units submitted for Government acceptance shall be subjected to and pass a burn-in period. The burn-in period will consist of 48 hours of "ON" time with the last 24 hours of "ON" time failure free. Equipments shall be operated in the same manner and under the same environmental conditions as during reliability testing. At the end of each 24 hour period the measurements of 4.8.2 shall be performed. Noncompliance will be considered a failure. If a unit fails testing during the final 24 hours of "ON" time it shall be repaired and returned to testing until it has accumulated 24 hours of failure free "ON" time. Failures shall be recorded and analyzed, but neither burn-in time nor burn-in failures shall be used to calculate equipment MTBF. Burn-in testing shall be performed prior to Group A testing. The equipment, including plug-in assemblies and spares shall have no less than 48 hours nor more than 144 hours of burn-in (exclusive of production test time).

Source: https://assist.dla.mil -- Downloaded: 202640-26T03:58Z
Check the source to verify that this is the current version before use.

4.7 Service conditions tests (See 3.8). The following service condition tests shall be performed on the equipment.

4.7.1 Temperature tests (See 3.8.1).

4.7.1.1 Low temperature, TD-976. The equipment shall be subjected to the test of Method 502.1, Procedure I, of MIL-STD-810. The storage temperature (Step 2) shall be -57°C (-70°F), and shall be maintained for a period of not less than 2 hours following stabilization of the test item. Low operating temperature (Step 4) shall be -32°C (-25°F). At operation temperature, tests shall start within 5 minutes after a 6 minute power application. Failure of the equipment to meet the specification requirements after testing in accordance with 4.16.1 and 4.17.30 shall constitute a failure of this test.

4.7.1.2 Low temperature, TD-982. Procedure shall be as described in 4.7.1.1 except operating temperature shall be -45.6°C (-50°F). Failure of the equipment to meet the specification requirements after testing in accordance with 4.16.2 shall constitute a failure of this test.

4.7.1.3 High temperature, TD-976. The equipment shall be subjected to the test of Method 501.1, Procedure II, of MIL-STD-810. For Step 1 the chamber temperature shall be 71°C (+160°F). High operating temperature (Step 7) shall be 52°C (125°F). Failure of the equipment to meet the specification requirements after testing in accordance with 4.16.1 and 4.17.30 shall constitute a failure of this test.

4.7.1.4 High temperature, TD-982. The equipment shall be subjected to the test of Method 501.1, Procedure II, of MIL-STD-810. For Step 4 the chamber temperature shall be 71°C (+160°F). For Step 7, the chamber temperature shall be 71°C (+160°F) for test item operation (this includes effects of solar radiation). Failure of the equipment to meet the specification requirements after testing in accordance with 4.16.2 shall constitute a failure of this test.

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:58Z
Check the source to verify that this is the current version before use.

4.7.2 Humidity (See 3.8.7).

a. TD-976. The equipment shall be subjected to the test of Method 507.1, Procedure II of MIL-STD-810. In Step 6 measurements in accordance with 4.16.1 shall be taken during the last 5 hours of the last cycle.

b. TD-982. The equipment shall be subjected to the test of Method 507.1, Procedure III of MIL-STD-810. In Step 7, the chassis shall be removed from its enclosure or the test item shall be otherwise opened so as to expose the normally sealed areas to the chamber environment. In Step 6, measurements in accordance with 4.16.2 shall be taken during the last 5 hours of the last cycle.

4.7.2.1 Failure. If the equipment fails to meet the specification requirements after testing in accordance with 4.16.1 and 4.16.2 and after conditioning and adjustment, it does not pass the test.

4.7.3 Altitude (See 3.8.8). The equipment shall be subjected to the test of Method 500.1, Procedure I, of MIL-STD-810. The test procedure shall incldue a 40,000 feet altitude non-operational exposure for not less than 1 hour, previous to the 15,000 feet altitude operational portion of Step 2. Altitude changes should not exceed 2,000 feet per minute. Failure of the equipment to meet the specification requirements after testing in accordance with 4.16.1, 4.16.2, and 4.17.30 shall constitute a failure of this test.

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:58Z
Check the source to verify that this is the current version before use.

4.7.4 Vibration (See 3.8.2). The equipment shall be subjected to Method 514.2, Procedure VIII, Curve V, Figure 514.2-6 of MIL-STD-810. Test time shall be 3 hours in each of 3 mutually perpendicular directions. Failure of the equipment to meet the specification requirements after testing in accordance with 4.16.1, 4.16.2, and 4.17.30 shall constitute a failure of this test.

4.7.5 Bounce (vehicular) TD-976 (See 3.8.3). The equipment shall be subjected to Test Method 514.2, Procedure IX, Part 2 of MIL-STD-810. Failure of the equipment to meet the specification requirements after testing in accordance with 4.16.1 and 4.17.30 shall constitute a failure of this test.

4.7.6 Bounce (loose cargo) TD-982 (See 3.8.4). The equipment shall be subjected to Test Method 514.2, Procedure XI, Part 2 of MIL-STD-810. Failure of the equipment to meet the specification requirements after testing in accordance with 4.16.2 shall constitute a failure of this test.

4.7.7 Shock (bench handling) TD-976 (See 3.8.5). The equipment shall be subjected to Test Method 516.2, Procedure V, of MIL-STD-810. Failure of the equipment to meet the specification requirements after testing in accordance with 4.16.1 and 4.17.30 shall constitute a failure of this test.

Source: https://assist.dla.mil -- Downloaded: 20260-26T03:58Z
Check the source to verify that this is the current version before use.

4.7.8 Shock (drop) TD-982 (See 3.8.6). The equipment shall be subjected to Test Method 516.2, Procedure II, of MIL-STD-810. Height of drop shall be 20 feet. Number of drops shall be six. Failure of equipment to meet the specification requirements after testing in accordance with 4.16.2 shall constitute a failure of this test.

4.7.9 Dust (See 3.8.9).

a. TD-976. The equipment shall be subjected to the test of Method 510.1, Procedure I, with the exception that the air velocity in Steps 1, 2 and 3 shall be 200 ±100 FPM. Step 3 shall proceed immediately after stabilization in Step 2. Failure of the equipment to meet the specification requirements after testing in accordance with 4.16.1 and 4.17.30 shall constitute a failure of this test.

b. TD-982. The equipment, in its operating configuration, shall be subjected to Method 510.1, Procedure I, of MIL-STD-810. Failure of the equipment to meet the specification requirements after testing in accordance with 4.16.2 shall constitute a failure of this test.

4.7.10 Salt fog (See 3.8.10). The TD-976 and TD-982 shall be subjected to the salt fog test specified in Test Method 509.1, Procedure I, of MIL-STD-810. No corrosion products shall be deposited on the test item by the facility. Inability of the equipment to meet the requirements of 3.8.10 shall constitute failure of this test.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:58Z
Check the source to verify that this is the current version before use.

4.7.11 Fungus (See 3.8.11). The equipment shall be subjected to the fungus test specified in Method 508.1, Procedure I, of MIL-STD-810. There shall be abundant growth colonization (See 6.8) on 50 percent or more of the area of the control item after 14 and 28 days. No cleaning of the equipment is permitted for 72 hours prior to the fungus test. Handling, prior to and during testing, shall be accomplished without contamination of the equipment. Inability of the equipment to meet the requirements of 3.8.11 shall constitute failure of this test. The equipment shall be exposed with the chassis removed and opened from the case.

4.7.12 Immersion, TD-982 (See 3.8.12). The equipment shall be subjected to the test of Method 512.1, Procedure I of MIL-STD-810. Failure of the equipment to meet the requirements of 3.8.12 shall constitute a failure.

4.7.13 Rain, TD-976 (See 3.8.13). The equipment shall be subjected to the following:

a. Condition for 4 hours at 77°F ±5°F and 40 to 60 percent relative humidity.

b. Take pretest data.

c. Test the TD-976 in a non-operating configuration with cover in place in accordance with Method 506.1, Procedure I, of MIL-STD-810. The wind source shall be maintained at the 40 miles per hour velocity for 15 minutes after which the wind source shall be turned off. Complete final measurements in accordance with 4.16.1 within 1 hour after the rain has been discontinued. Failure of the equipment to meet the requirements of 3.8.13 shall constitute a failure.

4.8 Reliability (See 3.11). The contractor shall conduct all reliability testing in accordance with the following:

Source: https://assist.dla.mil -- Downloaded: 202640-26T03:58Z
Check the source to verify that this is the currentversion before use.

4.8.1 Reliability test conditions. All reliability testing shall be conducted under the environment of Test Level B of MIL-STD-781 for the TD-976 and Test Level C of MIL-STD-781 for the TD-982. Vibration on the TD-976 shall only be applied during the OFF cycle. The equipment shall be vibrated at least 10 minutes for every hour of ON time. However, for the TD-976, the total vibration time per ON-OFF cycle shall not be greater than the normal OFF time per cycle. Equipments shall be operated at a 100% duty cycle.

4.8.2 Failure definition. During reliability testing the following parameters shall be performed at least once every 24 hours:

TEST	REQ. PARA.	TEST PARA.
TD-976 error rate - all eight channels	3.4.8.1	4.16.1 (only at nominal, i.e., 576 kb)
Alarms	3.4.15	4.16.1
OW talk test & signal	3.4.14.1.1.1	4.17.17
RAU (alarms, OW indications	3.4.14.2	
TD-982 error rate -		4.16.2
PTM signalling to 976 (& vice versa)		

A failure for reliability purposes shall be defined as the inability of the TD-976 or TD-982 to meet the specification requirements for any of the parameters listed in this paragraph. A failure shall be considered to have occurred at the time the equipment was last known to be operating properly. Total test time per equipment shall be in accordance with paragraph 4.2.2.4 of MIL-STD-781. Paragraphs 5.5 through 5.9 of MIL-STD-781 shall also apply. The presence of a pattern failure as defined by MIL-STD-781, shall result in a reject.

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:58Z
Check the source to verify that this is the current version before use.

4.8.3 First article reliability acceptance testing. A sample of five TD-976's and nine TD-982's First Article units shall be subjected to reliability testing. Accept-reject criteria shall be as follows:

EQUIPMENT	MIL-STD-781 TEST PLAN	SPECIFIED MTBF (θ_o)
TD-976	XX	2000 hrs.
TD-982	XX	3500 hrs.

4.8.4 Sampling for Group D inspection. For reliability purposes, a lot shall be defined as one month's production.

4.8.4.1 TD-976. A random sample of 5 TD-976's shall be selected from the first month's production and subjected to reliability testing in accordance with the accept-reject criteria of Test Plan IV of MIL-STD-781. Every fourth lot thereafter, a random sample of 5 TD-976's shall be selected and subjected to reliability testing in accordance with the accept-reject criteria of Test Plan IV of MIL-STD-781.

4.8.4.2 TD-982. A random sample of 7 TD-982's shall be selected from the first month's production and subjected to reliability testing in accordance with the accept-reject criteria of Test Plan IVA of MIL-STD-781. Every sixth lot thereafter, a random sample of 7 TD-982's shall be selected and subjected to reliability testing in accordance with the accept-reject criteria of Test Plan IVA of MIL-STD-781.

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:58Z
Check the source to verify that this is the current version before use.

4.9 Maintainability (See 3.18). The mean corrective maintenance time of 30 minutes of the organizational level and the maximum corrective maintenance time of 60 minues of the 90th percentile of the organizational level of maintenance shall be demonstrated using Test Method 9 of MIL-STD-471. The consumer's risk, β, shall be 10%.

4.10 Electromagnetic compatibility (See 3.9). The equipment shall be tested for compliance with the requirements of 3.9. Test setups and measurement techniques shall be in accordance with MIL-STD-462, Notice 3, as detailed in a contractor prepared government approved test plan. Interconnecting cables used during test shall be similar in type, shielding and length to those to be used in an operational installation. Test shall be performed with the equipment simulating normal operation.

4.10.1 Bonds, grounds and connectors (See 3.9.1). Prior to the performance of EMI tests the bonding impedance requirements of 3.9.1 shall be verified. Bonding impedance of the following items shall be measured with a Shallcross Meter Model 670A or equal:

a. All connector shells to chassis frame.

b. Power line filter body to ground and chassis frame.

c. All surfaces requiring EMI gaskets.

d. All switches, meters, hinge fasteners, transformers, iron core inductors and all mechanical mating surfaces which become exposed during maintenance and repair.

Source: https://assist.dla.mil -- Downloaded: 202640-26T03:58Z
Check the source to verify that this is the currentversion before use.

4.11 Nuclear tests (See 3.10). Nuclear tests (blast, thermal, E.M.P. initial nuclear radiation) shall be performed on the equipment in accordance with 4.10.1 through 4.10.4 inclusive.

4.11.1 Nuclear air blast test. The equipment shall be placed in a nuclear air blast simulation facility and exposed to the air blast environment as specified in 3.10. The equipment shall meet the operational requirements of Section 3 herein after being subjected to this test. Test facilities for this test are available at:

a. BRL, Aberdeen, Maryland.

b. Blast load generator, Waterways Experimental Station, Vicksburg, Mississippi.

c. U.S. Naval Civil Engineering Laboratory, Port Hueneme, California.

d. Sandia Corp., Albuquerque, New Mexico.

e. Air Force Weapons Laboratory, Albuquerque, New Mexico.

Calibration certification of listed test facilities or their equivalents are required by the Government, 45 days prior to nuclear test. Verification of the wave shape of the test simulator is required. Calibration and wave shape verification shall be supplied to the Government by the contractor.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:58Z
Check the source to verify that this is the current version before use.

4.11.2 Thermal radiation. One complete equipment shall be exposed to thermal radiation levels as specified in 3.10. Since the available simulation facilities provide these thermal radiation levels only over relatively small areas, many individual exposures will be required so that each of the exposed surfaces (case, meters, knobs, antennas, cables, and any other outboarded components) will receive the required thermal exposure. The equipment, following exposures, shall show no degradation that will affect the operation of the equipment as required in Section 3. Sample sections may be used for this test. Test facilities for this test are available at:

a. AVCO, Wilmington, Mass.

b. NASA, Moffet Field, California.

c. Spectro Lab., Division of Textron Inc., Sylmar, California.

d. Boeing Co., Seattle, Washington.

e. Tamarack Scientific Co., Orange, California.

f. Southwest Research Institute, San Antonio, Texas.

g. Thermal Radiation Facility, White Sands, New Mexico.

Calibration certification of listed test facilities or their equivalents are required by the Government, 45 days prior to nuclear test. Verification of the wave shape of the test simulator is required. Calibration and wave shape verification shall be supplied to the Government by the contractor.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:58Z
Check the source to verify that this is the current version before use.

4.11.3 Electromagnetic pulse. The equipment shall be pulsed at threat level in accordance with the procedures of (U) Nuclear Hardening EMP Final Report for Multiplexer TD-976( )/G and Pulse Form Restorer TD-982( )/G, OR12, 991-1, revised May 1975, or equivalent.

4.11.4 Initial nuclear radiation test. One complete set of equipment will be exposed to a LINAC and a pulsed or fast burst reactor in such a manner that all electronic piece parts and circuits in the set receive the neutron and gamma doses and rates specified in 3.10, Nuclear Survivability. The equipment shall then perform as specified in Section 3. For the purpose of design (not acceptance testing), electronic piece parts and circuits which have nuclear test data available or whose mathematical analysis has indicated that a level of confidence has been established, need not be subjected to further nuclear design testing. Acceptance of these electronic piece parts, circuits and mathematical analysis will be determined by the procuring agency from data supplied to the Government by the contractor. Typical test facilities for initial nuclear radiation tests are located at:

a. Fast Burst Reactors, White Sands, New Mexico.

b. Sandia Pulsed Reactor, Albuquerque, New Mexico.

c. Diamond Ordnance Radiation Facility, Washington, D.C.

d. Gulf General Atomic, La Jolla, California.

e. Northrop Corp., Hawthorne, California.

f. LINAC, WSMR, White Sands, New Mexico.

g. Aurora, HDL, Adelphi, Maryland.

h. HERMES II, Sandia Labs., Albuquerque, New Mexico.

Calibration certification of listed test facilities or their equivalents are required by the Government 45 days prior to nuclear test. Calibration and wave shape verification shall be supplied to the Government by the contractor.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:58Z
Check the source to verify that this is the current version before use.

4.12 Inspection for dimensional interchangeability (See 3.19). The dimensions listed below shall be gaged or measured to determine conformance to the physical interchageability requirement of 3.19. When a listed dimension is not within specified or design limits, it shall be considered a major defect as follows:

a. External and internal dimensions of cases, covers, and insertable assemblies, when such dimensions affect mating of parts.

b. Dimensions of cavities, when such dimensions affect insertion of items.

c. Location of hinges and fasteners on separable parts or assemblies which must mate, such as cases, covers, and mountings.

d. Location of connectors, locking pins, fasteners, slides, and mountings which receive mating parts of plug-in assemblies and major units; and location of the mating parts on the plug-in assembly or major unit.

e. Size and form of special threads.

4.13 Visual and mechanical inspection. Equipment shall be examined for the defects listed in MIL-STD-252. An inspection shall be performed to verify compliance with those portions of 3.20 which can be determined visually.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:58Z
Check the source to verify that this is the current version before use.

4.14 Continuity test (See 3.4.24). A dc continuity check shall be made between each end connector and the test points.

4.15 Electrical tests.

4.15.1 Standard conditions. Unless otherwise specified, all tests shall be performed at 115 volts ±5 percent from a single-phase source operating between 57.0 and 63.0 Hz. Tests shall be conducted at room ambient temperatures, pressure and humidity per MIL-STD-810 unless otherwise specified.

4.15.2 Standard test equipment. Standard test equipment is indicated in test figures by manufacturer and model number. In all cases, equivalent substitutes may be used. The equipment indicated in the figures is listed in Table VII. Standard test equipment shall be provided by the contractor. Government Technical Representative approval is required prior to use of proposed substitute items.

4.15.3 Special test equipment. Special test equipment is identified by name in test figures. Unless otherwise specified, special test equipment shall be provided by the contractor subject to approval of the Government. Detailed equipment description is given in Martin Marietta Equipment Compliance Report OR12, 397-2, dated December 1975. A brief description of this equipment is as follows:

a. PRSG/C (Pseudo-Random Pulse Generator and Correlator). Provides PCM output on up to eight channels to simulate a TD-660. The correlator can be switch selected to any of the eight channels. BCI acquisition indication and an error count indicator is provided on front panel. Both may be remoted to provide total counts.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:58Z
Check the source to verify that this is the current version before use.

b. Variable attenuator test box. Provides either termination or connection for PCM and supergroup signals.

c. Link error inserter. Has switch selected error rates. These are random error rates between .001 and 250 errors per 1000 bits.

d. Cable fault location test box. Provides switches and jumpers to simulate TD-982's in a five mile cable system.

e. Special purpose cable assembly to permit interconnection of the TD-976 to the RAU shall be provided and wired in accordance with the engineering sketch as furnished in the bid request and contract

f. Display power cable assembly to permit interconnection between the RAU and a remote 24 Vdc power source. A 6 foot cable wired to the mating connector of that specified in 3.4.14.4.1 shall be supplied.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:58Z
Check the source to verify that this is the current version before use.

Table VII - Standard Test Equipment

Designation	Function	Manufacturer
HP 5245L	Frequency Counter	Hewlett Packard
HP 5100B/HP5110B	Frequency Synthesizer	"
HP 183	Oscilloscope	"
HP 1402A (HP 141 Plug In)	Oscilloscope	"
HP 651B	Oscillator	"
HP 205AG	Oscillator	"
HP 400E	Voltmeter	"
HP 3400A	Voltmeter	"
HP 302A	Wave Analyzer	"
HP 334A	Distortion Analyzer	"
HP 3550A	Test Set	"
HP 11005A	Matching Transformer	"
DP 106A	Pulse Generator	Data Pulse
TEK 503	Oscilloscope	Tektronix
TEK 454	Oscilloscope	"
GR 1432N	Decade Box	General Radio
Digitech 2102A	Distortion Generator	Digitech
DQA-10	Data Quality Analyzer	"

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:58Z
Check the source to verify that this is the current version before use.

4.16 Service condition performance verification test.

4.16.1 TD-976.

a. Connect equipment per Figure 4a.

b. Check operation of alarms on the TD-976 and its' RAU and also the cable fault location indicator of TD-976 Unit 2 by making suitable switch changes to TD-976 Unit 1 in 96 channel mode. (Note: PRSG is set to supply input to all 8 channels; correlator checks one channel at a time).

c. Repeat b in 48 channel mode in reverse direction. (Note: PRSG is set to supply input to 4 channels).

d. With all 8 inputs set to 576,000 +.0045% measure error rate on two even channels for 3 minutes each.

e. Repeat d for 576,000 -.0045% on two odd channels.

f. With TD-976's set to 48 channels, one group input switch on, check error rate for 3 minutes. Repeat at 96 channel position using different input channel.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:58Z
Check the source to verify that this is the current version before use.

g. During above intermittently depress Analog OW call switch on both the TD-976 and its RAU. No errors should be introduced.

h. Check master oscillator frequency of TD-976 under test.

i. Check for proper operation of AVOW and DVOW circuits (ring and talk/listen operation).

j. The requirements of 3.4.5, 3.4.8.1, 3.4.10, 3.4.14, 3.4.15, 3.4.16 shall be met.

4.16.2 TD-982.

a. Connect equipment per Figure 4b.

b. Check error rate for 5 minutes on any one channel in one direction; another channel in other direction. This test should be performed with TD-976 in 96 channel mode, all 8 group input switches set to "ON" position, only one input provided.

c. Check for proper PTM-7 operation (both calling and talk/listen) from two TD-982's including the one under test to each of the TD-976's. (Also check AVOW operation with cable power "OFF").

d. The requirements of 3.6.1.1, 3.6.1.6.1, 3.6.1.4 shall be met.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:58Z
Check the source to verify that this is the current version before use.

4.17 Electrical performance for TD-976.

4.17.1 PCM input signal test.

a. Connect equipment per Figure 5.

b. Set TD-976's for 48 channel operation, PRSG to PCM input channels 1-4.

c. Set PRSG to 576 kb +.0045%, volt settings to provide minimum 1,0 separation of .4V. Correlator shall successively monitor each channel's error rate performance for 5 minutes.

d. Repeat c for 576 kb -.0045%.

e. Set TD-976 for 96 channel operation. Seven channels in "OFF" switch position.

f. Apply 576 kb +.0045% to remaining channel one and monitor error rate five minutes.

g. Repeat, successively for three more channels, five minutes each (channels 2,3,4).

h. Repeat on channels 5-8 successively, with PRSG set to 576 kb -.0045%.

i. The requirements of 3.4.8.1 shall be met under condition of 3.4.5.

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:58Z
Check the source to verify that this is the current version before use.

4.17.2 PCM input impedance.

a. Connect equipment per Figure 6.

b. Measure input impedance of PCM IN for each channel at 50, 1000, 100,000 and 5,000,000 Hz with power applied to TD-976.

c. The requirement of 3.4.5.1 shall be met.

4.17.3 Output signal test.

a. Connect equipment per Figure 5, with 91Ω load connected as shown.

b. Using the oscilloscope, measure the output signals.

c. The requirements of 3.4.6, shall be met.

4.17.4 PCM output jitter test.

a. Connect equipment per Figure 7 with PRSG/C connected directly to the 40 mile cable system using TD-204's.

b. Check for error free operation over range of 576 kb ±40 Hz. (Note: If this range cannot be met, replace TD-204.)

c. Reconnect equipment so PRSG output is first applied through TD-976 and then to the cable system. The output of the TD-204 remains connected to correlator.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:58Z
Check the source to verify that this is the current version before use.

d. Successively check each of the eight channels of the TD-976 for error free operation over at least ±17 Hz from 576,000 Hz.

e. Repeat steps a thru d using TD-754's.

f. The requirements of 3.4.6.4 shall be met.

4.17.5 Group bit count integrity (BCI) test (also cable power switch operation test).

a. Connect the equipment per Figure 8.

b. Set TD-976's for 48 channel operation, link error inserter set to provide 'random' errors of 1 per 1000.

c. After SG sync acquisition, let equipment run for 24 hours, monitor for loss of BCI.

d. Check for proper operation of cable power switch on each TD-976.

e. Repeat b and c with TD-976 in 96 channel mode.

f. The requirement of 3.4.7 and cable power requirement of 3.5 shall be met.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:58Z
Check the source to verify that this is the current version before use.

4.17.6 Time to group bit count integrity test.

a. Connect the equipment per Figure 9.

b. Set TD-976 for 48 channel operation with switch between PRSG and PCM input terminal to 'open'. .

c. Turn power switches to "ON".

d. Close switch (apply PCM to TD-976) and read time for BCI.

e. Repeat 4 more times.

f. Repeat for 96 channel operation (5 readings).

g. The requirement of 3.4.7.1 shall be met.

4.17.7 10 TANDEM hop test.

a. Connect equipment per Figure 8a. In all, 20 TD-976's are interconnected. Short interconnecting cables and the appropriate LBO settings are required.

b. The requirements of 3.4.8.1 shall be met for an 8 hour period under the conditions of 3.4.8.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:58Z
Check the source to verify that this is the current version before use.

4.17.8 System operation test.

a. Connect equipment per Figure 8b.

b. The requirements of 3.4.8.1 shall be met under the conditions of 3.4.8 for an 8 hour period. Also, there shall be no loss of BCI during 8 hours.

4.17.9 SG output signal test and data output pulse shape test.

a. Connect equipment per Figure 12.

b. Set TD-976 for 96 channel operation.

c. Observe 1/2 BAUD bipolar output on scope.

d. Observe data pulse levels, rise and fall time, etc.

e. Repeat c and d for 48 channel operation.

f. The requirements of 3.4.9.1 and 3.4.9.1.1 shall be met.

4.17.10 Cable orderwire signal level test.

a. Connect equipment per Figure 13.

b. Apply 0 dBm input at 1000 Hz into the handset terminal and measure voltage out across 880Ω load.

c. Repeat b with 0 input.

d. Repeat b and c using the handset terminal on Remote Access Unit.

e. The requirements of 3.4.9.1.2 shall be met.

Source: https://assist.dla.mil -- Downloaded: 202640-26T03:58Z
Check the source to verify that this is the current version before use.

4.17.11 SG output-input signals.

a. Connect equipment per Figure 11.

b. Set LBO setting on each TD-976 for 0 miles of cable between TD-976's.

c. Monitor error rate for 3 minutes.

d. Repeat b and c for 1/4 mile of cable.

e. Repeat b and c for 1/2 mile of cable.

f. The error rate requirement of 3.4.8.1 shall be met under conditions of 3.4.9.

4.17.12 SG loop-back switch test.

a. Connect equipment per Figure 11.

b. Apply power and notice 0 error rate on correlator.

c. Remove power on unit #2, TD-976.

d. Correlator should indicate loss of BCI.

e. Operate SG loop-back switch on TD-976, unit #1. Correlator should regain BCI, 0 errors.

f. The requirements of 3.4.9.5 shall be met.

Source: https://assist.dla.mil -- Downloaded: 202640-26T03:58Z
Check the source to verify that this is the currentversion before use.

4.17.13 Dynamic range test.

a. Select lengths of cable so that their attenuation when measured at 2.3 MHz is 22 dB.

b. Repeat a to have another calibrated cable (reading at 2.3 MHz) of 14 dB attenuation.

c. Connect the equipment as shown in Figure 14.

d. Perform 5 minute PCM error rate test using each calibrated cable to simulate the extreme variations expected for 1/2 mile cable lengths.

e. The error rate requirements of 3.4.8.1 shall be met under conditions of 3.4.9.3.

4.17.14 Super group synchronization time test.

a. Connect equipment per Figure 10.

b. Set random error inserter to 0 errors.

c. Measure super group synchronization time. Repeat 20 times.

d. Set random error inserter to one error per 1000 bits.

e. Repeat c.

f. The super group synchronization time requirements of 3.4.11 shall be met.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:58Z
Check the source to verify that this is the current version before use.

4.17.15 Master oscillator frequency accuracy test.

a. Connect equipment per Figure 15.

b. Read oscillator frequency on counter.

c. The requirements of 3.4.10 shall be met.

4.17.16 PCM group switch, On/Off test.

a. Connect equipment per Figure 16.

b. With power on both units, PCM group switches "ON", both units should indicate receipt of dummy per 3.4.15.5.

c. Remove power from TD-976, Unit 2. Unit 1 should indicate super group frame identification alarm per 3.4.15.1.

d. Reapply power to Unit #2. PCM group switches on Unit 2 to "OFF", Unit 1 left in "ON" position.

e. Unit #1 should indicate PCM input alarms per 3.4.15.3 and Unit #2 dummy alarm per 3.4.15.5.

f. Repeat Step c.

g. The requirements of 3.4.12 shall be met.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:58Z
Check the source to verify that this is the current version before use.

4.17.17 Orderwire appearance test.

a. Connect equipment per Figure 17.

b. Ring analog voice OW from Unit #1. A visual and audible signal on Unit #2 should indicate presence of incoming call in accordance with 3.4.14.1.1 and 3.4.14.1.1.1.

c. Respond on Unit #2 using talk/listen switch. Call indications should clear on both Unit #2 and its RAU in accordance with 3.4.14.1.1.1.

d. Repeat b and c above to demonstrate operation of switches on RAU in accordance with 3.4.14.2.

e. Repeat b, c, d originating ring analog voice OW signal from RAU of TD-976 Unit 1.

f. Talk/listen from Unit #1 to Unit #2 and simultaneously on the RAU of Unit #2. Parallel operation in accordance with 3.4.14 should be noted.

g. With Unit #1 switched to DVOW position note that interconnection between DVOW and AVOW not possible in accordance with 3.4.14.1.

h. With both units connected in AVOW position, connect counter to earphone output of Unit #2. Ring AVOW from Unit #1. Frequency at Unit #2 should be 1600 $\pm$30 Hz in accordance with 3.4.14.1.1.1.

i. Repeat h in other direction.

j. Repeat f and note that no talk down possible in accordance with 3.4.14.1.1.1.

k. Check DVOW operation by repeating steps b through g above for DVOW switch positions.

l. The requirements of 3.4.14 shall be met.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:58Z
Check the source to verify that this is the current version before use.

4.17.18 AVOW and DVOW circuit impedance test.

a. Connect the equipment per figure 18.

b. Set OW select switch to "Cable". Adjust HP 205AG #1 and decade box to provide 0 dBm into TD-976 Unit #1, with HP 400E units 1 and 2 reading equally. Read input impedance on decade box. Use same 1/2 voltage reading to read output impedance.

c. Repeat b with select switch set to "System".

d. Repeat b, c for remote panel readings.

e. Repeat above for TD-976 Unit #2.

f. The requirements of 3.4.14.1.3a, b and 3.4.14.1.4a, b shall be met.

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:58Z
Check the source to verify that this is the current version before use.

4.17.19 AVOW output level test.

a. Connect the equipment per Figure 19.

b. Adjust oscillator for 0 dBm at 1 kHz.

c. Measure output level at handset connector of TD-976 Unit #2 and side tone level at Unit #1.

d. Repeat with TD-976 Unit #1 and Unit #2 interchanged.

e. The requirements of 3.4.14.1.3d shall be met.

4.17.20 AVOW frequency response test.

a. Connect the equipment per Figure 20.

b. Set input level of 0 dBm at 1 kHz on TD-976 Unit #1.

c. Measure output level, in dBm, at Unit #2 and its RAU.

d. Repeat steps b and c at frequencies of 200, 300, 500, 700, 900, 1200, 1400, 1600 and 1700 Hz.

e. Repeat steps b, c, d with cable power "OFF".

f. Repeat with input to the RAU of TD-976 Unit #1.

g. The requirements of 3.4.14.1.3e, f shall be met.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:58Z
Check the source to verify that this is the current version before use.

4.17.21 DVOW output signal level test.

a. Connect the equipment per Figure 21.

b. The signal generator is set to provide 0 dBm at 1000 Hz to Unit #1.

c. The output level is measured on TD-976 Unit #2 and the sidetone at Unit #1 and also on their respective RAU's.

d. The signal generator is set to provide -4 dBm into the 4 wire input on the RAU, at 600 ohm output and input impedance.

e. The requirements 3.4.14.1.4c, d and 3.4.14.2.1d shall be met.

4.17.22 DVOW frequency response test.

a. Connect the equipment per Figure 22.

b. With data rate switch in 96 channel position adjust input signal level to 0 dBm at 1000 Hz to TD-976 Unit #1. The output level on TD-976 Unit #2 shall be measured. It shall be -15 dBm in accordance with 3.4.14.1.4e.

c. Input frequencies of 200, 300, 500, 2000, 3000, 3400, 3500 and 3640 Hz are used at 0 dBm. The dB output relative to the 1000 Hz reading is recorded.

d. The requirements of 3.4.14.1.4e and 3.4.14.1.4f shall be met.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:58Z
Check the source to verify that this is the current version before use.

4.17.23 DVOW 4 wire frequency response and linearity test.

a. Connect the equipment per Figure 23.

b. The oscillator is set to provide a signal level of -4 dBm at 1000 Hz into the 600 Ω load. The output level shall be in accordance with 3.4.14.2.1d.

c. The input level is maintained at -4 dBm and the frequency changed to 200, 300, 500, 2000, 3000, 3400, 3500, and 3640 Hz. The output level is recorded in dB relative to the 1000 Hz.

d. The requirement of 3.4.14.2.1e shall be met.

e. Step d will be repeated for input levels of -8, -12, -16, -20, -24, -28, and -30 dBm.

f. The requirements of 3.4.14.2.1d, e, f shall be met.

4.17.24 Digital voice orderwire signal-to-noise ratio test.

a. Connect the equipment per Figure 24.

b. Set data rate switch to 48 channels.

c. Apply input signal (to RAU of TD-976 Unit #1) of -4 dBm at 1000 Hz. Determine output signal at TD-976 Unit #2.

d. Replace input signal with a 600 ohm resistor and measure output signal.

e. Repeat steps b through d at the 96 channel rate.

f. The requirement of 3.4.14.2.1g shall be met.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:58Z
Check the source to verify that this is the current version before use.

4.17.25 Digital voice orderwire distortion test.

a. Connect the equipment per Figure 25.

b. Set data rate switch to 96 channel rate.

c. Apply input signal of -4 dBm at 1000 Hz.

d. Measure output signal at 1000 Hz and it's harmonics.

e. Repeat b through d at 48 channel rate.

f. The requirement of 3.4.14.2.1h shall be met.

4.17.26 Digital data/TTY order wire circuit impedance.

a. The test item will be connected as shown in Figure 26 and 27.

b. The TD-976 and remote panel baud rate select straps will be set for 75, the RECEIVE-READY switch will be set to RECEIVE, the SEND-RING switch to SEND, the DATA RATE switch to 96, the MILE switch to 0, and the LOOP BACK switch to ON. The test message generator will be set for 75 baud, 20 mA output current, 5-level internal message operation.

c. Switch S1 will be set to the 976 position and the timing peak output level will be determined with the oscilloscope. Switch S1 will be set to the DEC position and the decade resistance will be adjusted until the timing's peak output level is the same as determined previously. The decade resistance box setting will be recorded as the input impedance.

d. The teletype will be set up for 75 baud, 20 mA operation and switch S1 will be returned to the 976 position.

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:58Z
Check the source to verify that this is the current version before use.

e. While the timing is feeding a signal to the remote panel, the decade resistor will be adjusted until the signal or scope trace "B" is one half the amplitude of trace "A", both trace sensitivities being the same. The setting of the decade box will be recorded as the DD/TTY output impedance.

f. The test items will be re-interconnected as shown in Figure 27. The combiner baud select straps will be reconnected for 1200 baud. The distortion generator will be set for 1200 baud, +6 V polar operation. Switch S1 will be set to 976, switch S2 set to "RL".

g. Set c will be repeated and the decade resistance box setting recorded as the input impedance.

h. Switch S2 will be returned to the 976 position and switch S1 will be moved to the "OC" position.

i. The peak signal amplitude at the DD/TTY receive terminal will be determined and recorded as e_{OC}.

j. Switch S2 will be moved to "RL" position and the output signal peak amplitude will again be determined and recorded as e_{RL}.

k. The output impedance will be calculated as:

$$Z_o = \left(\frac{e_{OC}}{e_{RL}} - 1 \right) RL$$

from the data steps i and j.

l. The requirements of 3.4.14.2.2 shall be met.

Source: https://assist.dla.mil -- Downloaded: 202640-26T03:58Z
Check the source to verify that this is the currentversion before use.

4.17.27 Digital data/teletype orderwire signal levels test.

a. The test item will be connected as shown in Figure 28.

b. The digitech generator will be set for 75 baud, 5-level, internal message, 20 mA operation. The teletype will be set for 75 baud, 20 mA internal battery operation. The combiner and remote panel baud rate select straps will be set for 75, the RECEIVE-READY switch will be set to RECEIVE, the SEND-RING switch to SEND, the DATA RATE switch to 48, the MILE switch to 0, and the LOOP BACK switch to ON.

c. While the test message is being passed, the input and output waveforms will be monitored and the peak amplitudes recorded.

d. The digitech generator will be reset to 60 mA operation. Step c will be repeated.

e. The digitech generator will be set for 1200 baud, ±1 mA, ±6 V polar operation. The combiner and remote panel baud select straps will be reconnected for 1200 baud operation.

f. While the digitech generator is sending the test message, the input waveform will be monitored across the 1-ohm resistor and the peak amplitude measured.

g. The requirements of 3.4.14.2.2 shall be met.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:58Z
Check the source to verify that this is the current version before use.

4.17.28 Digital data/TTY orderwire bias distortion test.

a. The test items will be connected as shown in Figure 29.

b. The TD-976 and remote panel baud rate select straps will be connected for 75 baud operation. The MILE switch will be set to 0 mile, the LOOP BACK switch at OFF, the DATA RATE switch to 96, the SEND-RING switch to SEND, and the RECEIVE-READY switch to RECEIVE. The digitech generator will be set for 75 baud, 20 mA neutral operation, with 0 percent distortion. The teletype will be set for 20 mA, 75 baud operation.

c. With 0 percent distortion, the test items will be allowed to pass the test message for 1 minute to ensure the system is operating properly.

d. The digitech generator will be set for 35 percent mark bias distortion and the test items will pass the test message for 5 minutes. The number of errors in the output copy will be recorded.

e. The mark bias distortion will be reduced to 0 percent and space bias distortion will be set to 35 percent. The test items will pass the test message for 5 minutes. The number of errors in the output copy will be recorded.

f. The digitech generator will be set for 0 percent distortion and the teletype replaced with the data quality analyzer. The analyzer will be set for 75 baud, 5-level, 20 mA input, and set for bias distortion measurement.

g. With a distortion-less signal into the combiner, the bias distortion of the combiner output signal will be determined and recorded.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:58Z
Check the source to verify that this is the current version before use.

h. The digitech generator will be set to produce 35 percent mark bias distortion and for each distortion value, step g will be repeated.

i. The digitech generator will be set for 0 percent distortion, 1200 baud 8-level, ± 6 Vdc polar operation. The combiner baud rate select straps will be connected for 1200 baud operation. The data quality analyzer will be set for 1200 baud, 8-level, ±6 V polar operation. The combiners will be interconnected with a short cable. The combiner MILE switch will be set for 0 mile, and the DATA RATE switch to 96.

j. The digitech generator will pass data through the test items for 1 minute. During the time interval, the data quality analyzer will be monitored and the average bias distortion recorded.

k. The digitech generator will be set for 35 percent mark and space bias distortion. At each distortion setting, step g will be repeated.

l. The requirements of 3.4.14.2.2 shall be met.

Source: https://assist.dla.mil -- Downloaded: 20261O-26T03:58Z
Check the source to verify that this is the currentversion before use.

4.17.29 Alarm test.

a. Connect equipment per Figure 30.

b. Set switches for 48 channel operation.

c. Apply power to all equipment, including cable power.

d. Remove cable power and note supergroup frame identification alarm both audible and visual on both TD-976's and their RAU's in accordance with 3.4.15.1.

e. Use audible push button switch. The requirement of 3.4.15.4 shall be met.

f. Return to step c.

g. Remove power from TD-976 Unit #2. Note alarm operation on TD-976 Unit #1 as in step d.

h. Return to step c, above.

i. Remove power from the TD-660 connected to channel 1 of TD-976 Unit #1.

j. Note PCM input alarm, both visual and audible, on TD-976 Unit #1 and its RAU in accordance with 3.4.15.3 and also operation of audible pushbutton switch as in e, above. Note dummy indication of TD-976 Unit #2 per 3.4.15.5.

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:58Z
Check the source to verify that this is the currentversion before use.

k. Return to step c, above.

l. Turn off PCM switches on channel 1 on both TD-976's.

m. Repeat step i, above and note no alarm indications in accordance with 3.4.12.2.

n. Check operation of alarm check switch.

o. The requirements of 3.4.15.7 shall be met.

p. The requirements of 3.4.12 and 3.4.15 shall be met.

Source: https://assist.dla.mil -- Downloaded: 202640-26T03:58Z
Check the source to verify that this is the current version before use.

4.17.30 Cable fault location test.

a. The equipment will be connected in the five mile system as shown in Figure 31 and the CABLE POWER and CABLE TEST switches on ADC Unit #2 will be placed to OFF positions. The CABLE POWER and CABLE TEST switches on ADC Unit #1 will be placed to ON positions. Verify that no cable fault indication exists.

b. A fault will be inserted at the near end of TD-982 #1 and a verification will be made that a cable fault at TD-982 half #1 is indicated on the front panel.

c. Connect the AN/PTM-7 test set to the faulted TD-982 half. Verify call, talk and listen capability to and from Unit #1 with a fault inserted and with no fault inserted. Remove the AN/PTM-7 test set during all cable fault location measurements.

d. The CABLE POWER Switch on Unit #1 will be placed to OFF and the cable fault will be inserted in TD-982 half #18. The CABLE POWER switch will be placed to ON and verification made that half #18 is indicated on the front panel. Repeat step c into #18.

e. A fault will be inserted at TD-982 half #9 and a verification will be made that TD-982 half #9 is indicated on the front panel. Repeat step c into #9.

f. A fault will be inserted in TD-982 half #10 and a verification made that #10 is indicated on the front panel. Repeat step c into #10.

Source: https://assist.dla.mil -- Downloaded: 202640-26T03:58Z
Check the source to verify that this is the currentversion before use.

g. Remove four TD-982's from the system diagram of Figure 31.

h. Repeat steps a through f for a 3 mile system (5 TD-982's).

(1) Insert a fault in TD-982 half #1, #10, #5 and #6.

i. Repeat steps a through f for a one (1) mile system.

(1) Insert a fault in TD-982 half #1, and #2.

j. The requirements of 3.4.16, 3.6.1.6 and 3.6.1.6.1 shall be met.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:58Z
Check the source to verify that this is the current version before use.

4.17.31 Input power supply test.

a. The equipment will be connected as shown in Figure 32. The input voltage will be adjusted to 115 Vac, 60 Hz.

b. Perform an error rate test for 3 minutes per channel, also check master oscillator frequency.

c. Vary input voltage and frequency on TD-976 Unit #1 in accordance with table below and repeat step b, above for each.

Voltage	Hz
103.5	47.5
103.5	63
126.5	47.5
126.5	63

d. The input power and current to the test item will be measured at all the combinations listed above.

e. The requirements of 3.4.20 shall be met.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:58Z
Check the source to verify that this is the current version before use.

4.17.32 Cable power supply test.

a. The equipment will be connected as shown in Figure 33 and the cable power supply test box set to simulate 18 TD-982 halves with faults.

b. The ADC and the cable power supply will be placed to ON position, and the cable current and voltage at the supergroup output will be measured.

c. Step b will be repeated for loop back cable systems simulating the #1 through #18 TD-982 halves.

d. The supergroup cable circuit will be opened and the cable power voltage at the ADC will be measured on the oscilloscope.

e. The supergroup cable circuit at the ADC output will be shortcircuited and the cable current will be measured.

f. Step a above will be repeated. The cable power supply will be turned off and the voltage decay will be monitored on the oscilloscope to obtain a reading after 1 second of decay time.

g. The requirements of 3.5 and 3.5.1 shall be met.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:58Z
Check the source to verify that this is the current version before use.

4.17.33 Operational test for TD-976.

a. Connect equipment in accordance with Figure 3a.

b. Set LBO network of TD-976 under test for 1/4 mile of cable; use 1/4 mile CX-11230 cable.

c. In 96 channel mode, supply input to one group input channel; the other 7 in OFF position.

d. Measure error rate for one minute.

e. Repeat d for remaining seven channels.

f. Check for proper operation of alarm indicators and cable fault indicator.

g. Check operation of talk/listen-call for both DVOW and AVOW.

h. The requirements of 3.4.8.1, 3.4.9.1, 3.4.15 and 3.5.16 shall be met.

4.17.34 Operational test for TD-982.

a. Connect equipment in accordance with Figure 3b.

b. With TD-976 in 96 channel mode, apply PCM to one channel (other 7 in "OFF" position). Measure error rate for five minutes while intermittently signalling on AVOW. No errors should be introduced.

c. The error rate requirements of 3.6.1.1 shall be met.

Source: https://assist.dla.mil -- Downloaded: 202640-26T03:58Z
Check the source to verify that this is the current version before use.

4.18 Electrical performance for TD-982.

4.18.1 Output signal characteristics test.

a. The test setup is shown in Figure 34.

b. The equipment will be turned on. The output signal of TD-982 half A/B will be measured with an oscilloscope for the parameters listed below:

(1)	Pulse amplitude (logic 1)	(volt)
(2)	Pulse width	(nsec)
(3)	Pulse rise time	(nsec)
(4)	Pulse fall time	(nsec)
(5)	Pulse droop	(volt)
(6)	Back swing	(volt)
(7)	Ripple and noise	(volt)
(8)	Peak to level	(volt)

(Pulse widths will be measured at the 50 percent transition points. Rise and fall times will be measured between the 10 and 90 percent amplitude points).

c. The requirements of 3.6.1.1 shall be met.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:58Z
Check the source to verify that this is the current version before use.

4.18.2 Timing recovery rate test.

a. The test setup is shown in Figure 35.

b. The link error inserter shall be set for a pattern density of 10 percent ones.

c. The relative phase position of the test item's output will be observed on an oscilloscope.

d. The link error inserter output will be set for an output pattern density of 90 percent 1's and the phase position of the test items output will be observed on the oscilloscope.

e. The master oscillator clock frequency will be set to 9830400 Hz and the relative phase position of the test item output will be observed on the oscilloscope.

f. The master oscillator clock frequency will be set to 9830400 Hz $\pm$.004% Hz and the relative phase position of the test item output will be observed on the oscilloscope.

g. The test will be repeated for TD-982 half "B".

h. The requirements of 3.6.1.3 shall be met.

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:58Z
Check the source to verify that this is the current version before use.

4.18.3 Order wire characteristics test.

a. The test setup is shown in Figure 36.

b. The input impedance of TD-982 half "A" will be measured using the 6 dB loss method at the following frequencies: 300, 500, 700, 900, 1000, 1300, 1500, and 1700 Hz.

c. The output impedance of TD-982 half "A" will be measured as above.

d. The requirements of 3.6.1.4 shall be met.

4.19 Electrical subassembly test (See 3.16). TD-976, TD-982, and plug-in subassemblies including running spare panel assemblies and separate TD-976 power supply, shall be tested in accordance with their respective test requirement specification drawings.

4.20 Air seal test (See 3.22). The equipment shall be opened and closed again in such a manner as to break and remake the seal. Immediately thereafter, the equipment, as field transported, shall be subjected to a vacuum of 1 pound per square inch (1 pound per square inch less than the atmospheric pressure surrounding the equipment) applied to the interior of the equipment. The vacuum shall then be valved off and the interior pressure measured during the ensuing period of 1 minute. During this 1 minute period the decrease in vacuum shall not exceed 0.01 pound. The gage used for measurement of the vacuum shall be of such accuracy that a difference of 0.01 pound can be determined readily.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:58Z
Check the source to verify that this is the current version before use.

4.21 Short circuit proof test. The TD-976 shall be energized for normal TD-976 to TD-976 operation. A DC short shall be sequentially applied to each of the accessible inputs, outputs and interfaces. Failure of the equipment to meet the specification requirements after testing in accordance with 4.17.33 shall constitute a failure of this test.

4.22 Spare panel assemblies storage case test. The storage case shall be rack mounted and subjected to tests in accordance with 4.7.5 and 4.7.7. Subsequent to each test, the spare panel shall be removed from the case and inserted in the TD-976 which shall be subjected to the tests of 4.16.1 and 4.17.30.

4.23 Quality conformance inspection of packaging. Packaging inspection shall be in accordance with MIL-P-116 to determine conformance to the requirements of Section 5. The rough handling test shall be performed only when invoked in the bid request or contract (see 6.2d.).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:58Z
Check the source to verify that this is the current version before use.

5. PACKAGING

5.1 Preservation. Preservation shall be Level A, B or Commercial, as specified (See 6.2c.).

5.1.1 Level A.

5.1.1.1 Cleaning. Each TD-976 and TD-982 shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying. Each TD-976 and TD-982 shall be dried in accordance with the applicable procedure of MIL-P-116.

5.1.1.3 Preservation application. No preservative required.

5.1.1.4 Unit packing. Unit packing shall be in accordance with the methods prescribed in MIL-P-116 or as otherwise specified herein.

5.1.1.4.1 Technical literature. Each technical literature shall be unit packed submethod IC-1.

5.1.1.4.2 Power cable. Each power cable shall be unit packed in accordance with MIL-C-55442.

Source: https://assist.dla.mil -- Downloaded: 202640-26T03:58Z
Check the source to verify that this is the current version before use

5.1.1.4.3 TD-976 and TD-982. Each item shall be individually packed in accordance with method III as follows: Cushion each unit on all surfaces with cells or pads or both conforming to PPP-F-320, type CF, class weather-resistant, designed to protect all projections and absorb the shock of impact in handling and transit. As an alternate, cushioning material may be fabricated of plastic molding material conforming to MIL-P-19644 or polyurethane foam conforming to MIL-P-26514. Place each individual cushioned item within a close-fitting box conforming to PPP-B-636, type CF, class weather-resistant. Closure of the box shall be as specified in the box specification.

5.1.1.4.4 Consolidation. Consolidate the items, unit packed as specified in 5.1.1.4.2 and 5.1.1.4.3, within a close-fitting box conforming to PPP-B-636,type CF, class weather-resistant. Place the technical literature, unit packed as specified in 5.1.1.4.1, on top of the contents, directly under the lid of the box. Fill all voids with paperboard conforming to PPP-P-291, type III to prevent movement. Closure of the box shall be as specified in the box specification.

5.1.2 Level B.

5.1.2.1 Cleaning, drying, preservative application, unit packing and consolidation. Each item shall be cleaned, dried, preserved, unit packed and consolidated as specified in 5.1.1.

5.1.3 Commercial preservation. Commercial preservation shall be in accordance with FED-STD-356.

Source: https://assist.dla.mil -- Downloaded: 202640-26T03:58Z
Check the source to verify that this is the currentversion before use.

5.2 Packing. Packing shall be Level A, B, or Commercial, as specified (See 6.2). Shipping containers for Level A and B shall be capable of stacking and supporting superimposed loads during shipment and storage without damaging the container(s) or its contents.

5.2.1 Level A. A quantity of consolidation containers unit packed as specified in 5.1, shall be packed within a close-fitting box conforming to PPP-B-601, overseas type; PPP-B-621, Style 4, Class 2; or PPP-B-585, Style 2 or 3, Class 3. When the gross weight exceeds 200 pounds, or the container length and width is 48 x 24 inches or more and the weight exceeds 100 pounds, 3 x 4 inch skids, laid flat, shall be applied in accordance with the requirements of the container specification, or if not specified in the specification, in a manner which will adequately support the item and facilitate the use of material handling equipment. Closure and strapping shall be in accordance with the applibable container specification or appendix thereto except that metal strapping shall conform to QQ-S-781, Type I, Finish A.

5.2.1.1 Unitization. Palletization shall be required when: containers specified in 5.2.1 do not require skids; quantities per destination comprise a pallet load; and container size permits use of one of the pallet patterns of MIL-STD-147. A quantity of containers packed as specified in 5.2.1, except that container strapping may be omitted, shall be placed on a pallet, load Type I, conforming to MIL-STD-147. The pallet shall conform to NN-P-71, Type IV, Group I or II woods. The load shall be "bonded" to the pallet by strapping conforming to QQ-S-781, Type I, Finish A, or shrink film.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:58Z
Check the source to verify that this is the current version before use.

5.2.2 Level B. A quantity of consolidation containers, unit packed as specified in 5.1, shall be packed within a close-fitting fiberboard box conforming to PPP-B-640, class 2, style E, or PPP-B-636, type CF, class weather-resistant, variety DW. The gross weight of boxes conforming to PPP-B-640 shall not exceed 250 pounds. Closure shall be in accordance with the appendix of the applicable box specification. Reinforcing shall be by pressure-sensitive filament tape banding or nonmetallic strapping conforming to PPP-T-97 and PPP-S-760, respectively; selection of the material and application shall be in accordance with the appendix of the applicable box specification. When the gross weight exceeds 200 pounds, or the container length and width is 48 x 24 inches or more and the weight exceeds 100 pounds, containers will be pallet-mounted on pallets conforming to NN-P-71, group I or II woods. The load shall be "bonded" to the pallet by strapping conforming to QQ-S-781, type I, finish A, or shrink film. When a single unit is shipped to a single destination no packing is required; the unit container shall serve as the shipping container. Reinforcing shall be as specified above.

5.2.2.1 Unitization. Palletization shall be required when quantities per destination comprise a pallet load. A quantity of consolidation containers unit packed as specified in 5.1.1, shall be placed on a pallet, load type I, conforming to MIL-STD-147. The pallet shall conform to NN-P-71, type IV, group I or II woods. A fiberboard cap shall be employed over the load having two sides extending down the stacked load at least 12 inches to accommodate marking requirements. The cap shall be fabricated of fiberboard conforming to PPP-F-320, class weather-resistant, W5c or V3c. The load shall be "bonded" to the pallet by strapping conforming to QQ-S-781, type I, finish A, or shrink film.

5.2.3 Commercial packing. Packing shall be in accordance with FED-STD-356.

Source: https://assist.dla.mil -- Downloaded: 202640-26T03:58Z
Check the source to verify that this is the current version before use.

5.3 Marking.

5.3.1 Military marking. In addition to any special marking required by the contract or order, interior packs and exterior shipping containers shall be marked in accordance with MIL-STD-129.

5.3.2 Commercial marking. In addition to any special marking required by the contract or order, interior packs and exterior shipping containers shall be marked in accordance with FED-STD-356.

Source: https://assist.dla.mil -- Downloaded: 202640-26T03:58Z
Check the source to verify that this is the current version before use.

6. NOTES

6.1 Intended use. Multiplexer TD-976 ( )/G provides 48/96 channel traffic capability for transmission through up to 1/2 mile of CX-11230 cable to another TD976 ( )/G or to the AN/GRC-144 Radio. Pulse Form Restorers, TD-982 ( )/G are used at 1/2 mile intervals in the CX-11230 cable system to extend the range up to 5 miles. The equipments operate in the integrated radio and cable systems as part of the AN/TSQ-85 Video Technical Control Center, per EL-CP 0070-0001, and the AN/TCC-73 Telephone Terminal per MIL-S-55307. They operate in conjunction with Radio Set AN/GRC-144, per EL-CP0000-0079, Cable Assembly, Special Purpose, Electrical CX-11230 ( )/G, per MIL-M-55649; and associated video interconnect facilities in the AN/TSQ-85. The ancillary items are used, as follows:

a. Extender panel for TD-976 ( )/G is used as a test fixture to check out operation of printed circuit cards.

b. A special adapter cable is used to interconnect the AN/PTM-7 test set and the TD-982 ( )/G. This allows signalling and talk/listen functions between the TD-982 ( )/G and the TD-976 ( )/G without interrupting traffic.

c. An interconnect cable is provided to connect the Remote Access Unit to the TD-976 ( )/G to simulate a rack mounted installation for both units.

d. An interconnect cable is provided to supply DC power from an external power supply to the Remote Access Unit to simulate the shelter mounted configuration

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:58Z
Check the source to verify that this is the current version before use.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification and any amendment thereto.

b. A list of equipments to be Government loaned for testing purposes.

c. Level A, level B, or Commercial preservation and packing (see Section 5).

d. When rough handling test is required (see 4.23).

e. Training program required during first article inspection.

6.3 Group C inspection. Approval to ship may be withheld, at the discretion of the Government, pending the decision from the contracting officer on the adequacy of corrective action. (See 4.5.3.3).

6.3.1 Initial shipments. Initial shipment of end items will be contingent only upon successful completion of all Group C tests. (See 4.5.3).

6.4 Nomenclature. The parenthesis in the nomenclature will be deleted or replaced by a letter identifying the particular design, for example: TD-976B/G. The contractor should apply for nomenclature in accordance with the applicable clause in the contract.

6.5 Verification inspection. Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this specification and the contract. The amount of verification inspection by the Government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product.

Source: https://assist.dla.mil -- Downloaded: 2026-10-28T03:58Z
Check the source to verify that this is the current version before use.

6.6 Description of Radio Repeater Set AN/TRC-138 ( ). This set is an air or vehicular transportable assemblage used to provide 48/96 channel radio communication facilities. It consists of Shelter S-393/TRC-138, 3 each Radio Set AN/GRC-144, and 3 each Telephone Set TA-312/PT. It operates on a 115V ±10%, 60 Hz single phase power source.

6.7 Description of Telephone Terminal AN/TCC-73 ( ) (V) 2. This terminal has a 96 channel input, time division multiplexing and pulse code modulation. It consists of Shelter S-419 ( )/TCC-73, 8 each Multiplexer TD-660 ( )/G, 8 each Converter, Telephone, Signal CV-1548 ( )/G, 4 each Multiplexer TD-754 ( )/G, 1 each Telephone Set TA-312/PT and 1 each Intercommunication Station LS-1471 ( )/F1. It contains facilities to accept 8 each TSEC/KG-27. It operates on a 115V, 60 Hz, single phase power source.

6.8 Definitions.

6.8.1 Branching. Branching is a connected arrangement of filaments (hypyae) formed by shoots or secondary stems growing from the main stem or filament (hypha).

6.8.2 Exfoliation. Exfoliation is corrosion along the grain boundaries of the metal resulting in the peeling or separating, or both, of successive layers of the metal. The appearance resembles loose book pages or onion skin peeling.

6.8.3 Growth colonization. Growth colonization is a mass of individual plants, generally of one species, living together; or a group of hyphae which is formed from one spore or cell and may be one individual plant. Colonization which completely covers the surface of the nutrient material constitutes abundant growth.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:58Z
Check the source to verify that this is the current version before use.

6.8.4 Microbial growth. Microbial growth is the growth of very minute organisms. Such organisms when present in large numbers may provide a colony visible to the naked eye.

6.8.5 Sporulation. Sporulation is the formation of minute unicellular reproductive or dormant bodies, called spores.

6.8.6 Tubular germination. Tubular germination is partial growth by the production of hyphae, which are tubular shaped fungal filaments. Tubular germination constitutes restricted individual spore growth not proceeding to colonization.

6.9 Environmental pollution. Environmental pollution measures are contained in the packaging material specifications referenced herein. Refer to material specifications or Preparing Activity for recommended disposability methods.

Custodian:

Army - (EL)

Preparing Activity

Army - (EL)

Project No. 5805-A237

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:58Z
Check the source to verify that this is the current version before use.

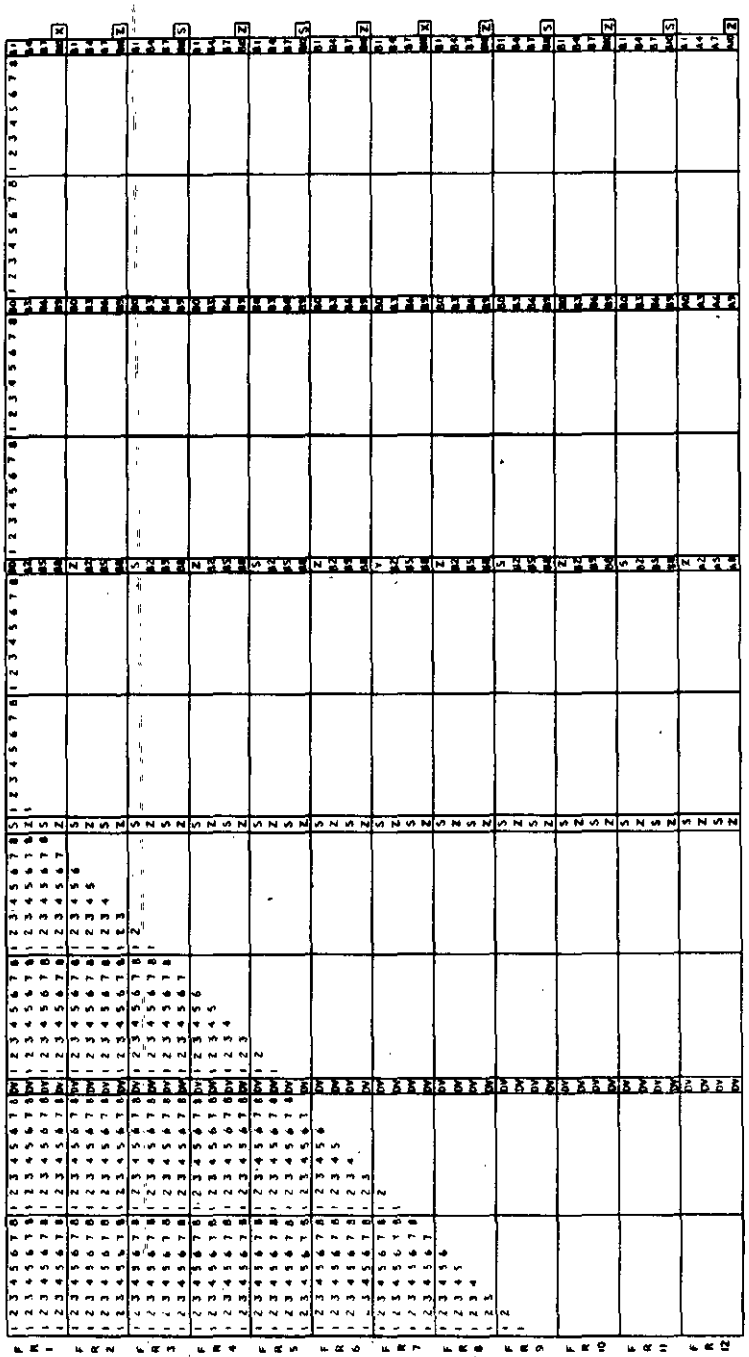
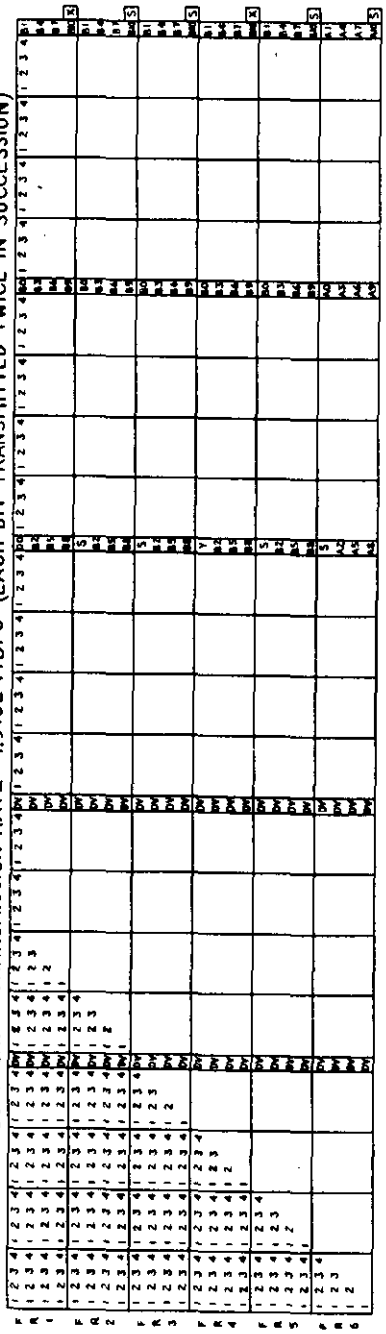


FIGURE 1a 96-CHANNEL MODE MAJOR FRAME FORMAT

Source: https://assist.dla.mil -- Downloaded: 202640-26T03:58Z
Check the source to verify that this is the current version before use.



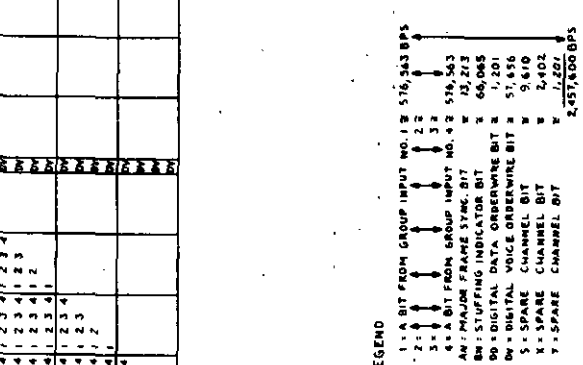
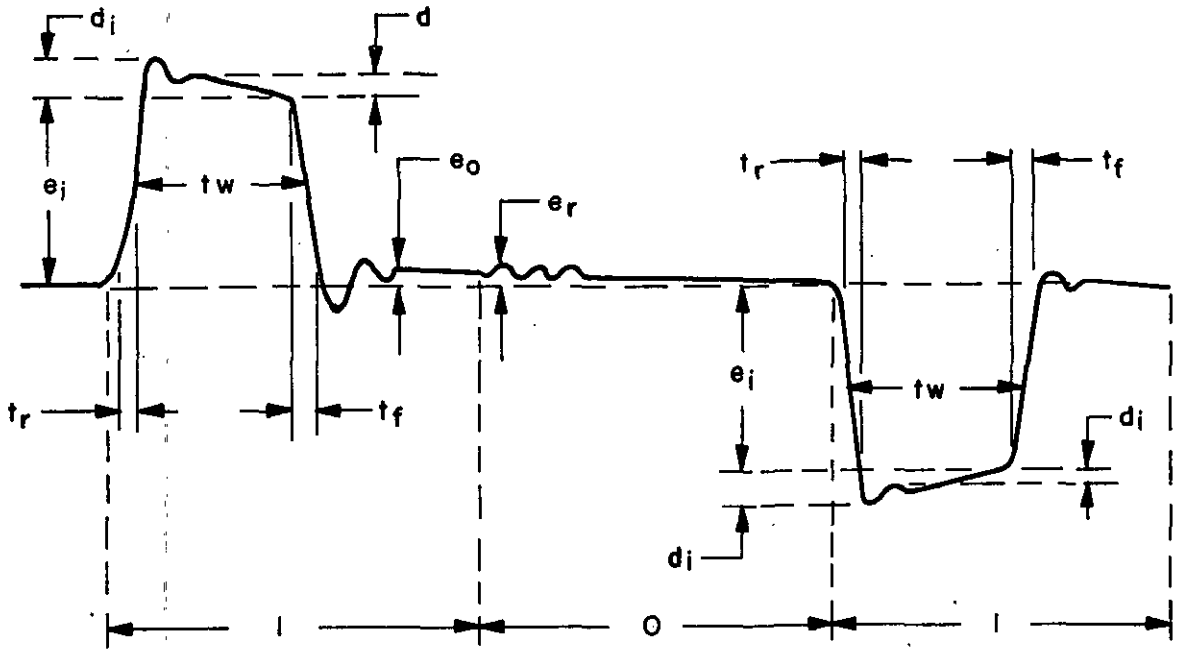


FIGURE 1b 48-CHANNEL MODE MAJOR FRAME FORMAT

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:58Z
Check the source to verify that this is the current version before use.



SYMBOL	DESCRIPTION	VALUE
t_w	PULSE WIDTH	100 NX ± 10 NS
e_i	PULSE AMP	0.9 ± 0.1 VOLTS
d	DROOP	.2 VOLTS MAX
t_r	RISE TIME	25 NSEC MAX
t_f	FALL TIME	25 NSEC MAX
e_o	BACK SWING	.2 VOLTS MAX
e_r	RIPPLE AND NOISE	5% e_i
d_i	PEAK TO LEVEL	.3 VOLTS

FIGURE 2 BIPOLAR SIGNAL

Source: https://assist.dla.mil -- Downloaded: 202640-26T03:58Z
Check the source to verify that this is the currenversion before use.

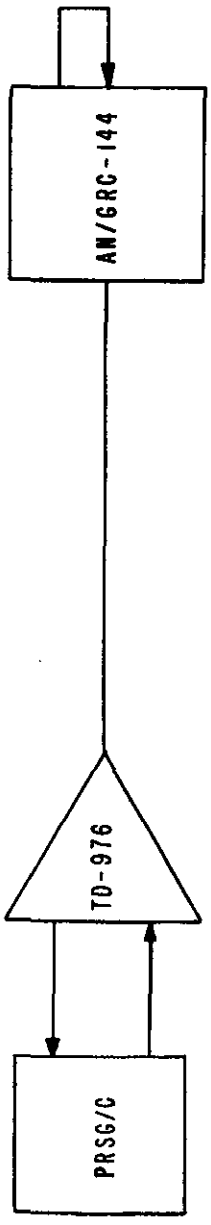


FIGURE 3 OPERATIONAL TEST TD-976

Source: https://assist.dla.mil -- Downloaded: 202640-26T03:58Z
Check the source to verify that this is the current version before use.

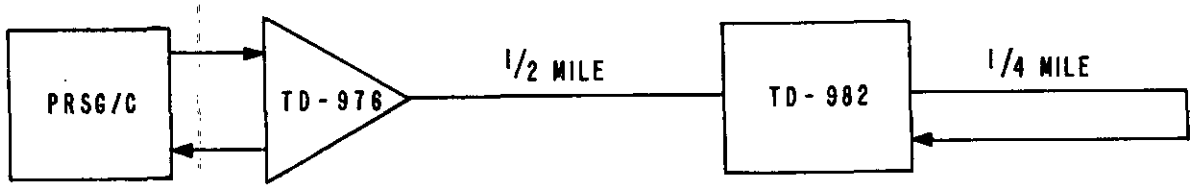


FIGURE 3b OPERATIONAL TEST TD-982

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:58Z
Check the source to verify that this is the current version before use.

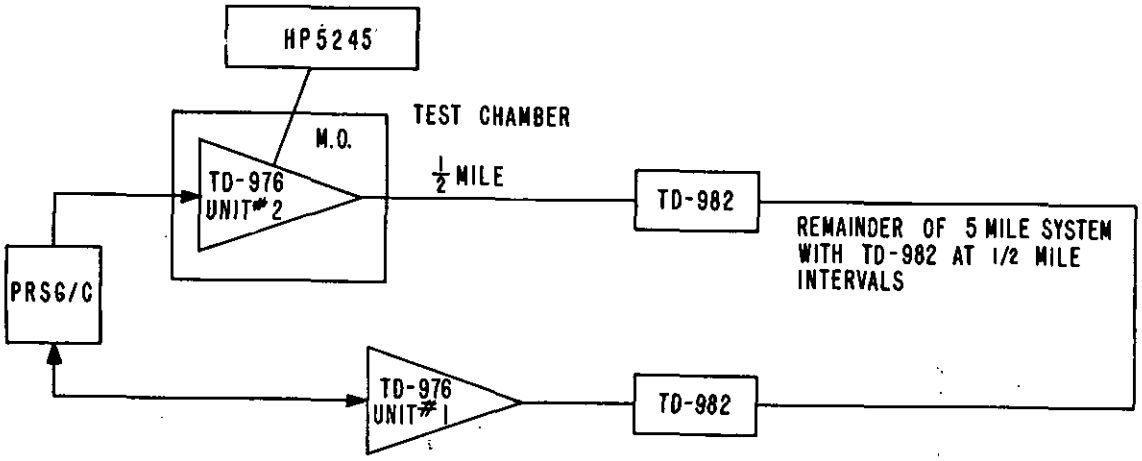


FIGURE 4a SERVICE CONDITION PERFORMANCE VERIFICATION TD-976

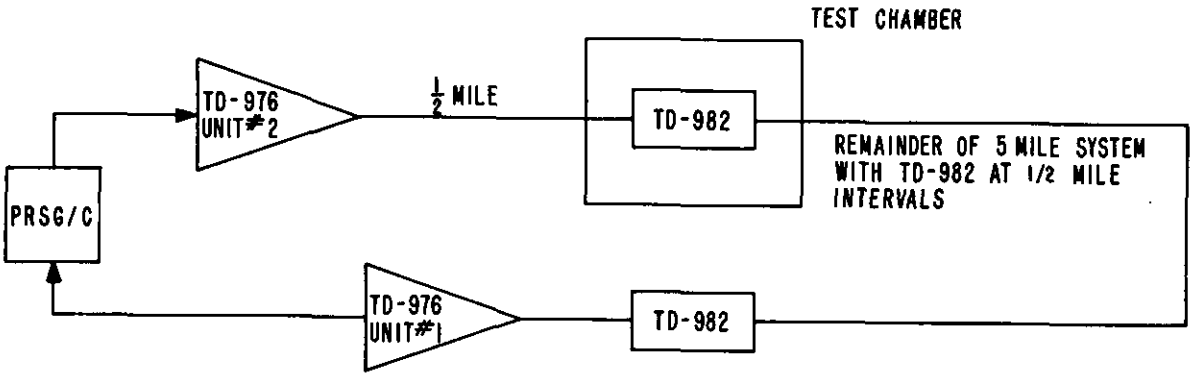


FIGURE 4b SERVICE CONDITION PERFORMANCE VERIFICATION TD-982

Source: https://assist.dla.mil -- Downloaded: 202640-26T03:58Z
Check the source to verify that this is the current version before use.

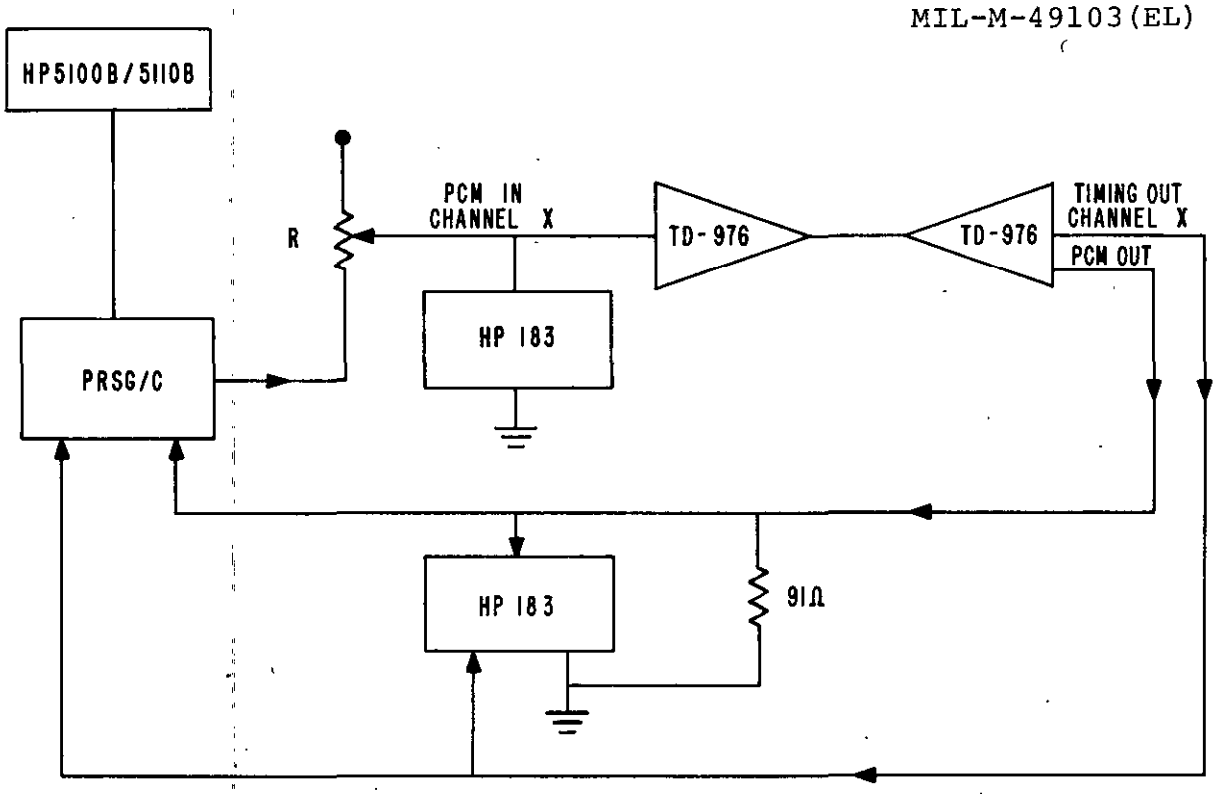


FIGURE 5 PCM INPUT SIGNAL TEST, PCM OUTPUT SIGNAL TEST

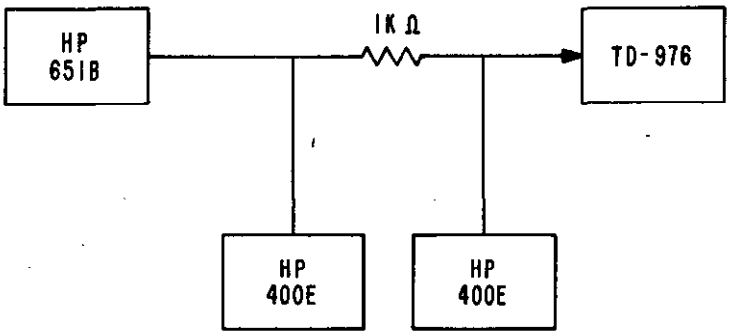


FIGURE 6 PCM INPUT IMPEDANCE

Source: https://assist.dla.mil -- Downloaded: 202640-26T03:58Z
Check the source to verify that this is the current version before use.

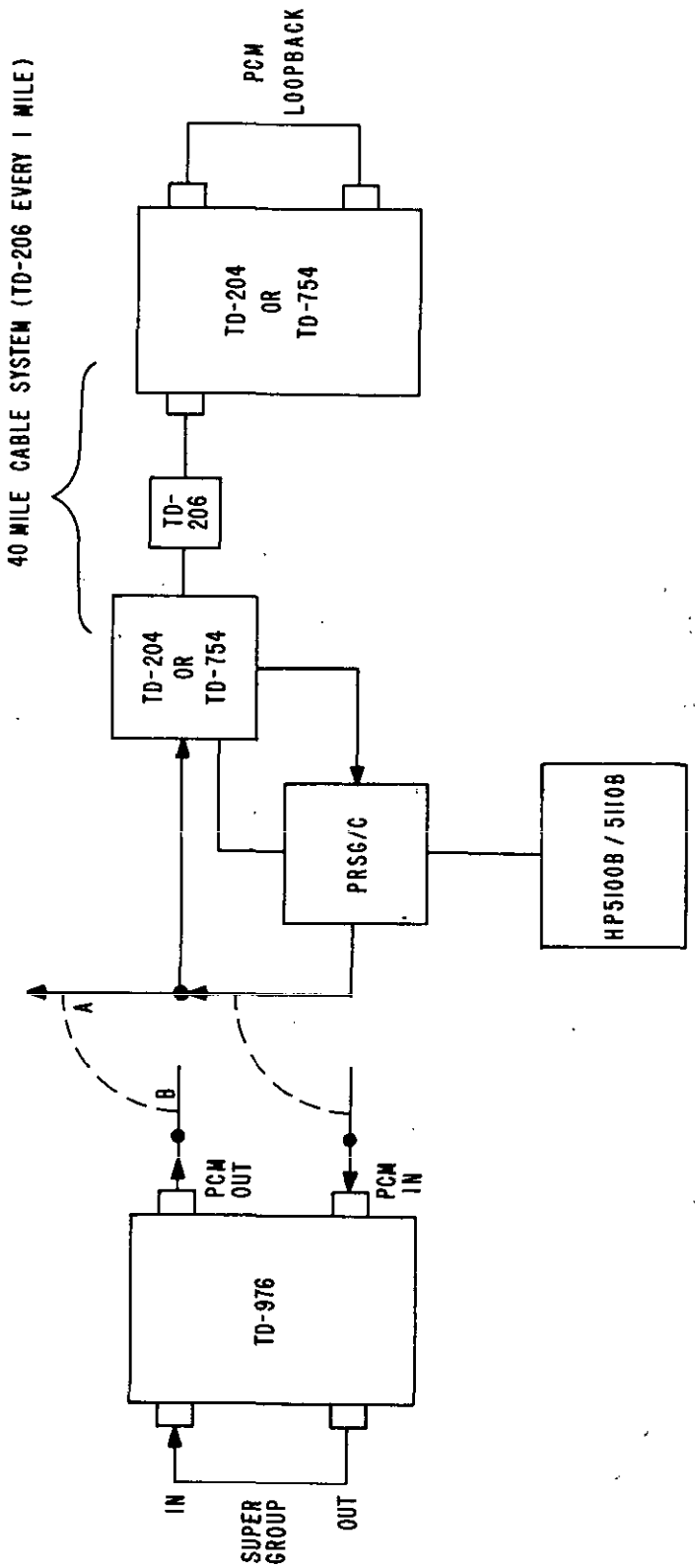


FIGURE 7 PCM OUTPUT JITTER TEST

Source: https://assist.dla.mil -- Downloaded: 202640-26T03:58Z
Check the source to verify that this is the current version before use.

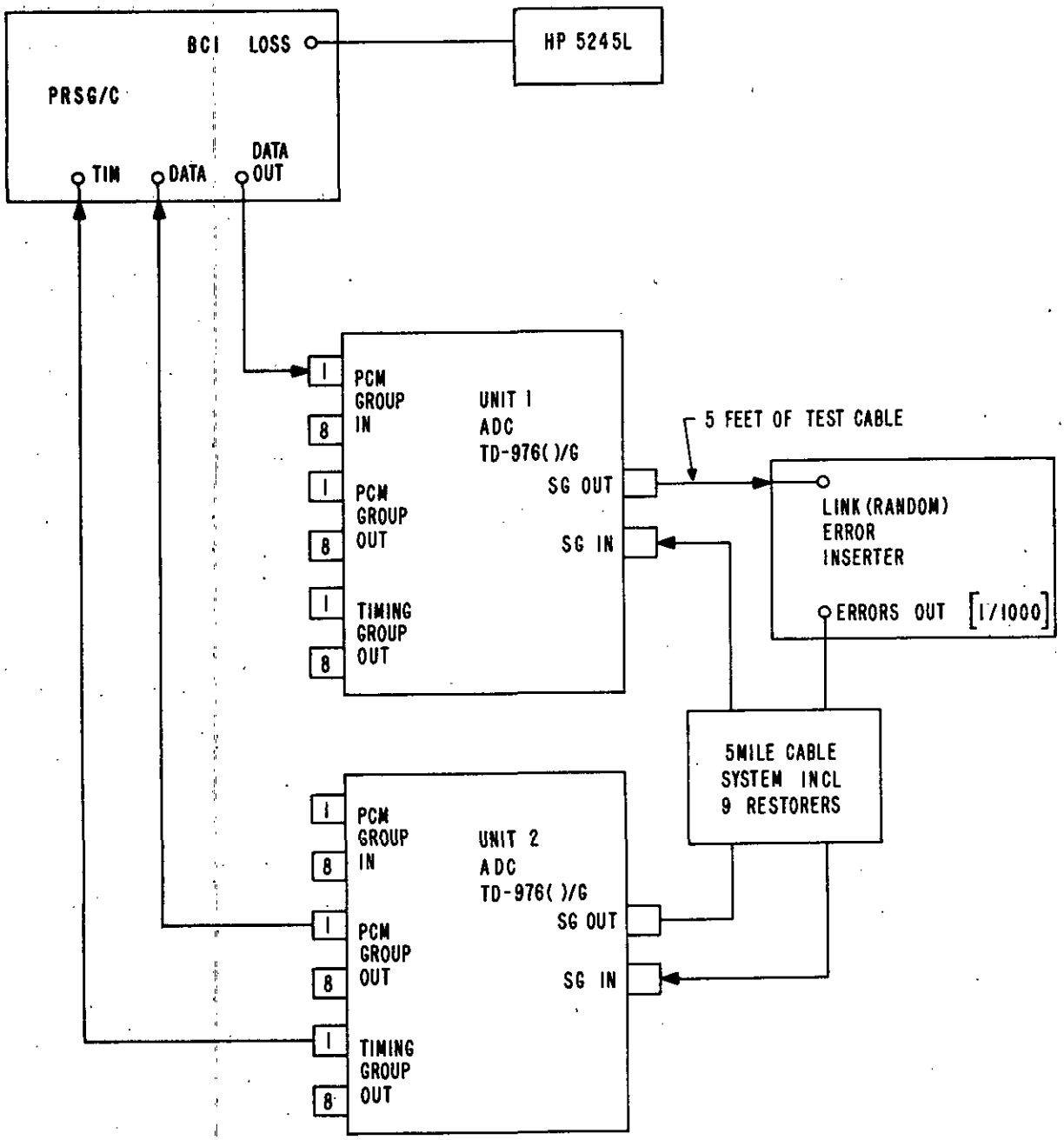


FIGURE 8 GROUP BIT COUNT INTEGRITY TEST, ALSO CABLE POWER SWITCH OPERATION TEST

Source: https://assist.dla.mil -- Downloaded: 202640-26T03:58Z
Check the source to verify that this is the current version before use.

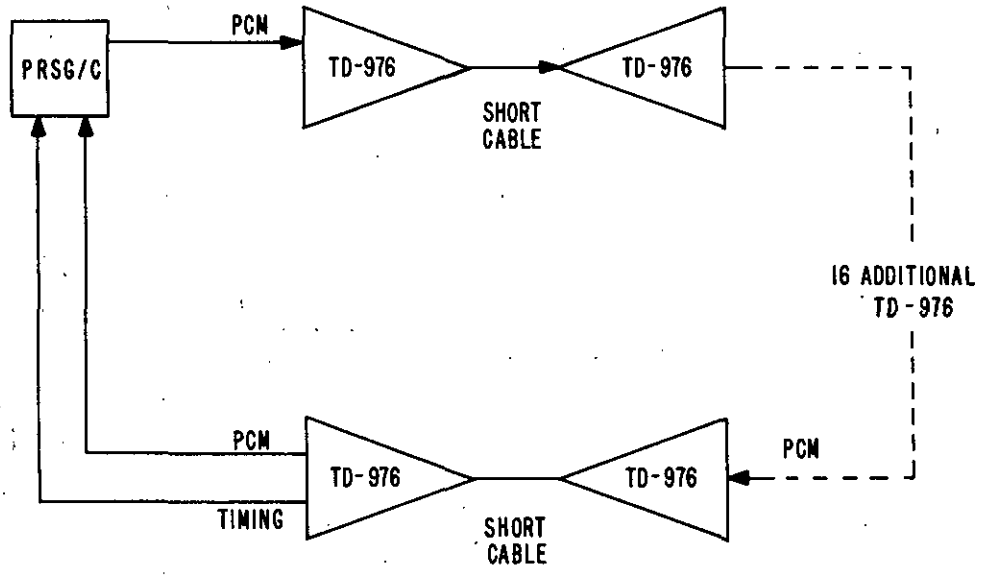


FIGURE 8A - 10 TANDEM HOP TEST

Source: https://assist.dla.mil -- Downloaded: 202640-26T03:58Z
Check the source to verify that this is the current version before use.

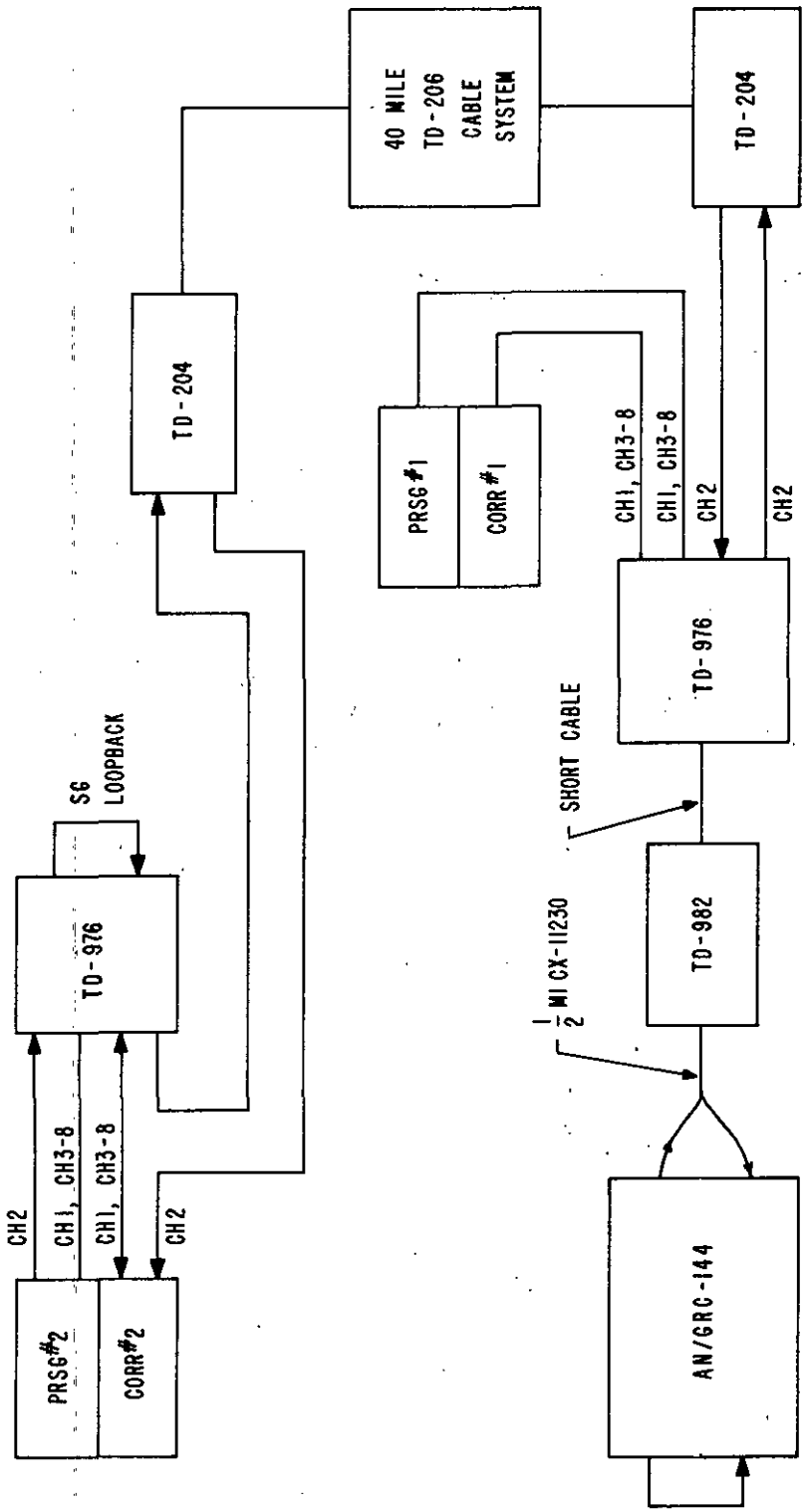


FIGURE 8B SYSTEM OPERATION TEST

Source: https://assist.dla.mil -- Downloaded: 202640-26T03:58Z
Check the source to verify that this is the current version before use.

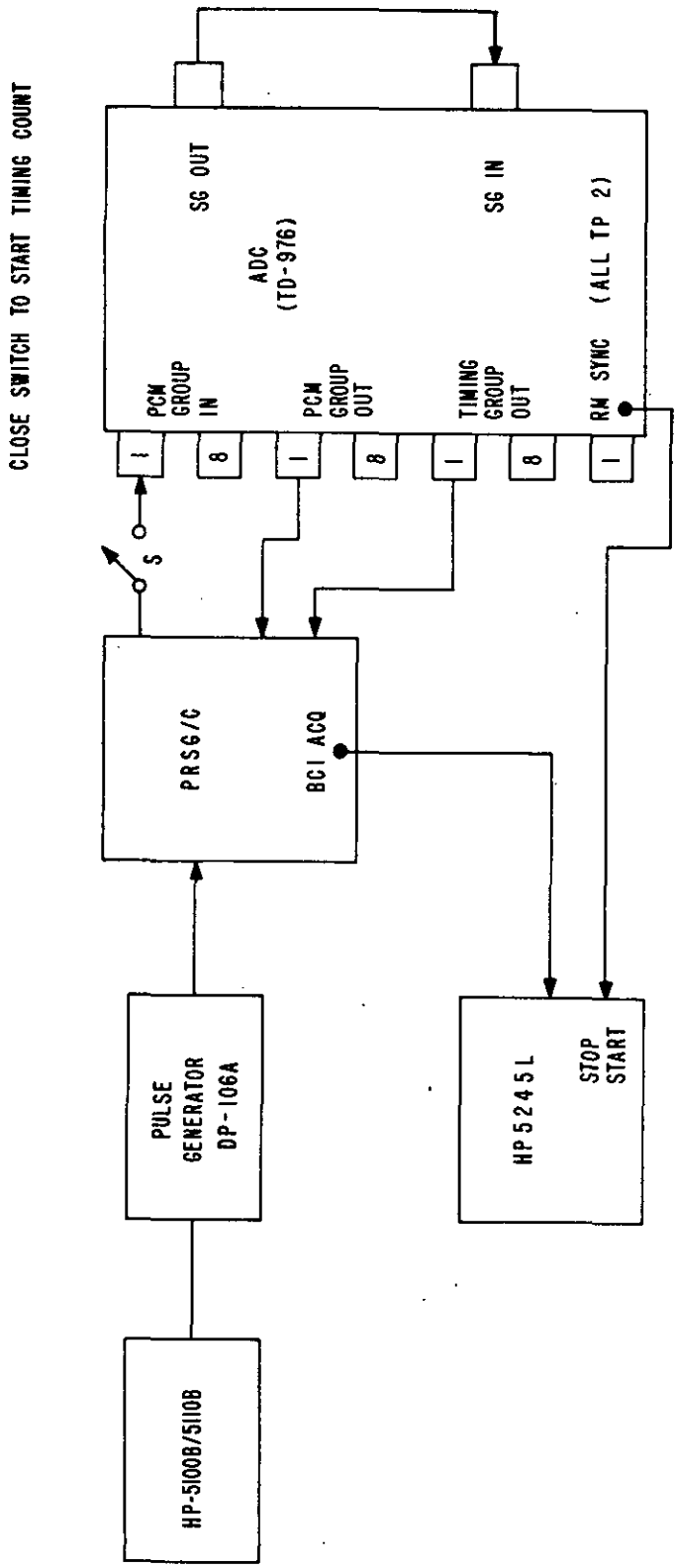


FIGURE 9 TIME-TO-GROUP-BIT-COUNT INTEGRITY TEST SETUP

Source: https://assist.dla.mil -- Downloaded: 202640-26T03:58Z
Check the source to verify that this is the current version before use.

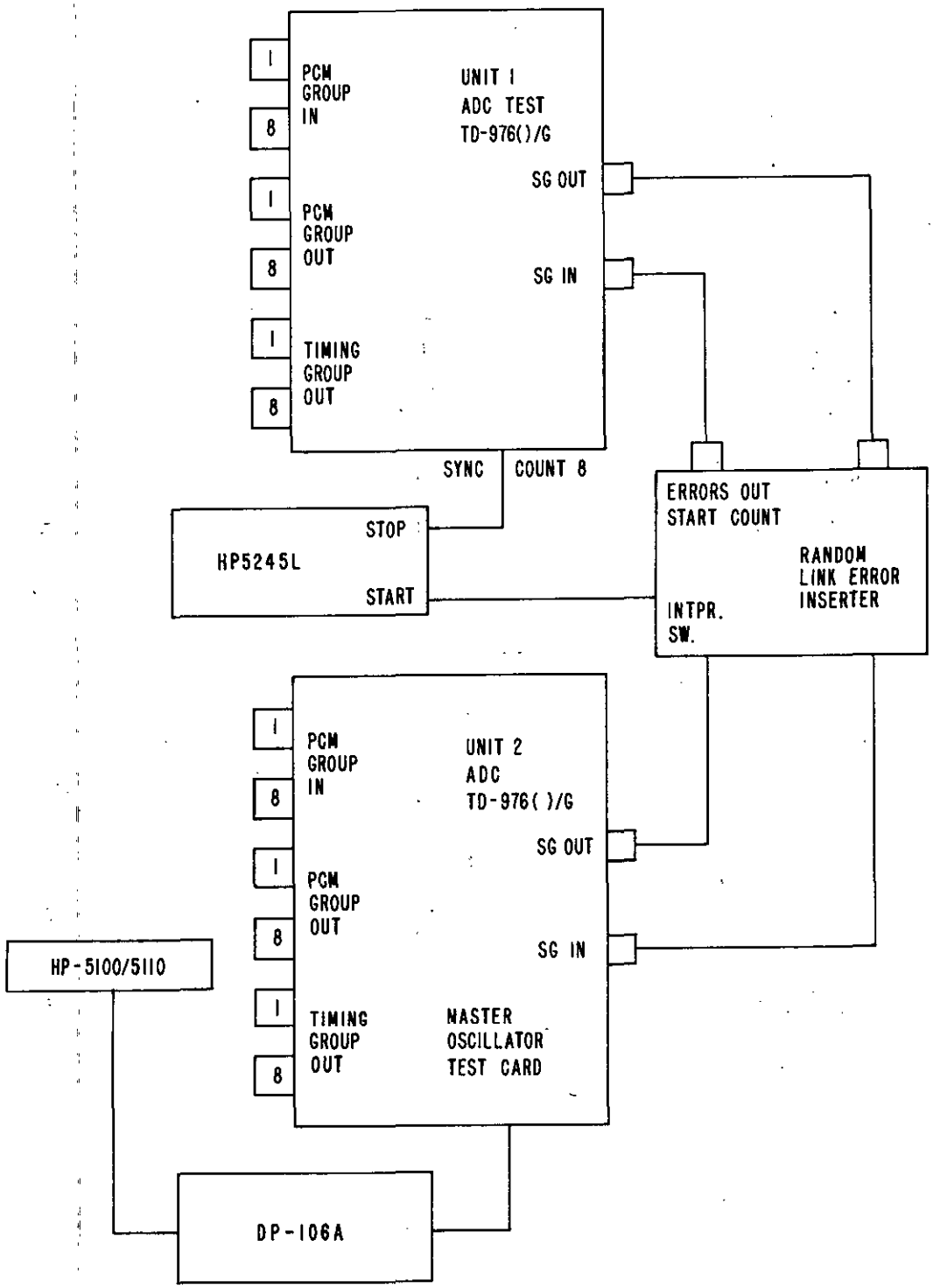


FIGURE 10 SUPER GROUP SYNCHRONIZATION TIME TEST

Source: https://assist.dla.mil -- Downloaded: 202640-26T03:58Z
Check the source to verify that this is the currentversion before use.

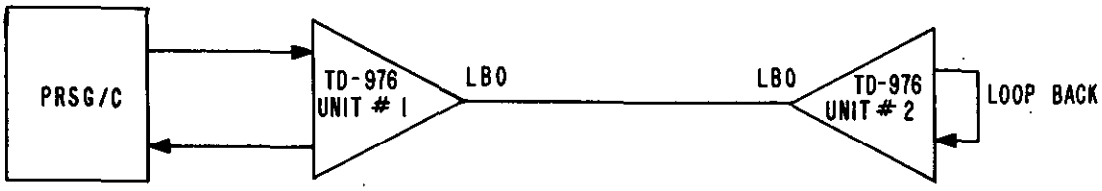


FIGURE 11 SG OUTPUT-INPUT SIGNAL TEST AND SG LOOP BACK SWITCH TEST

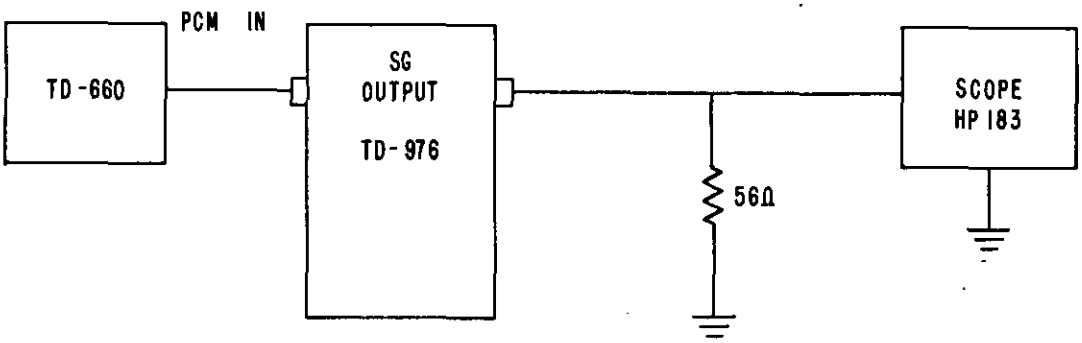


FIGURE 12 SUPER GROUP OUTPUT SIGNAL TEST AND DATA OUTPUT PULSE SHAPE TEST

Source: https://assist.dla.mil -- Downloaded: 202640-26T03:58Z
Check the source to verify that this is the current version before use.

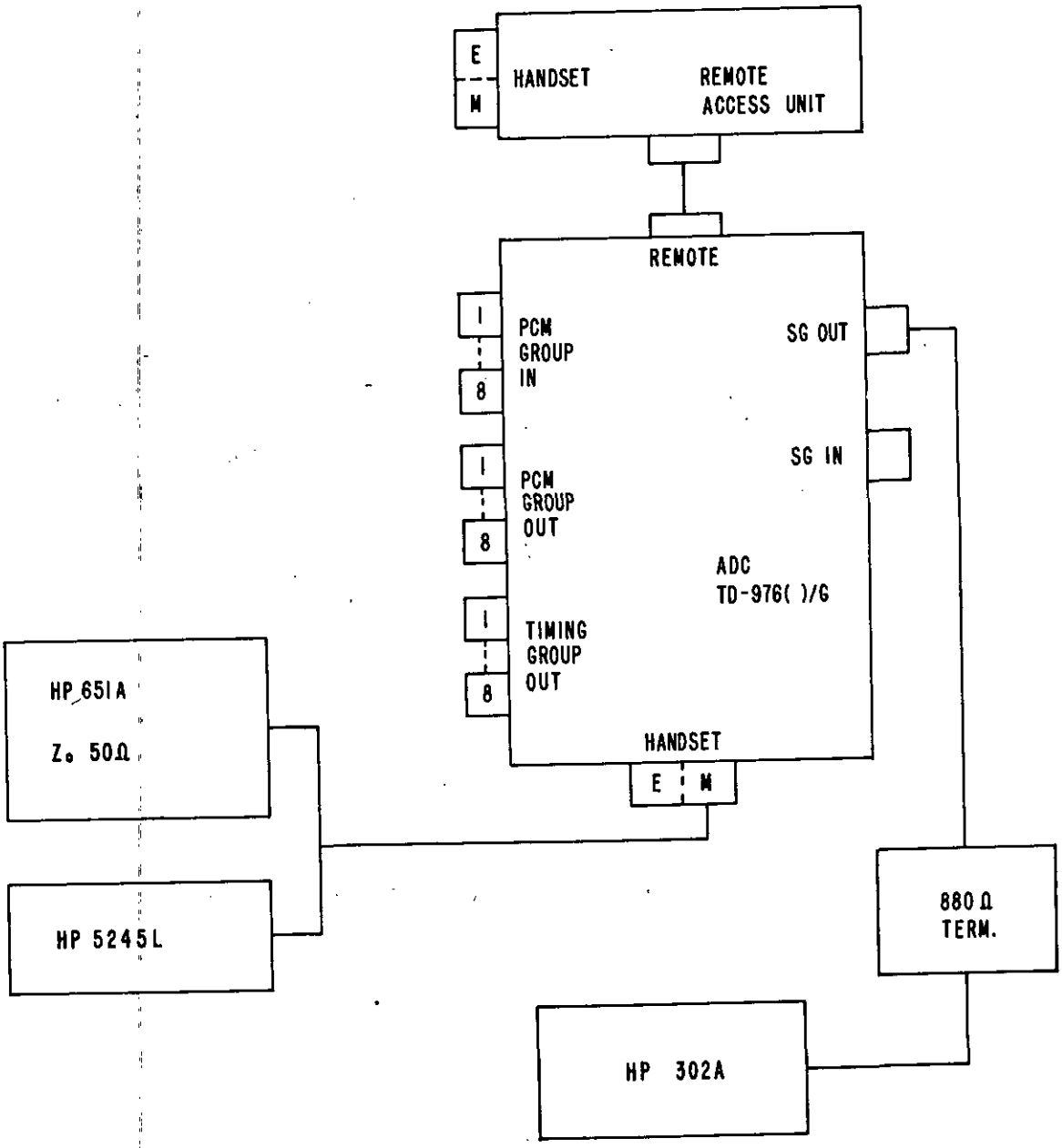
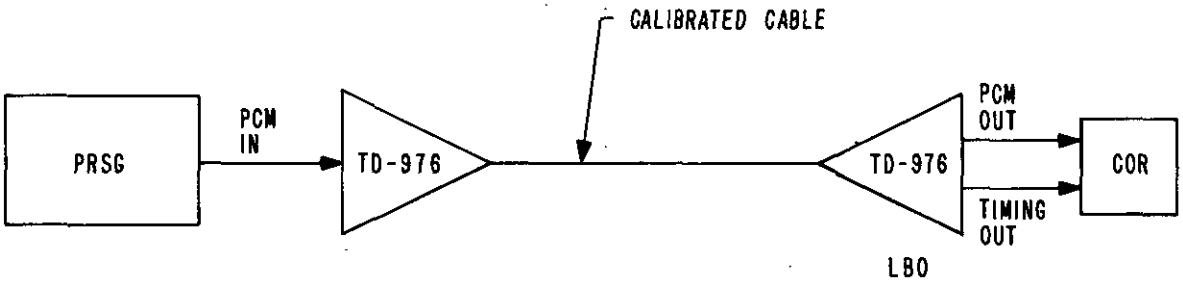


FIGURE 13 CABLE ORDERWIRE SIGNAL LEVEL TEST

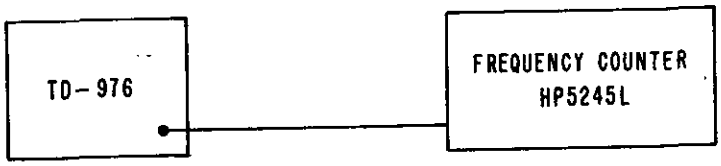
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:58Z
Check the source to verify that this is the current version before use.



LBO SETTINGS
FOR $\frac{1}{2}$ MILE OF CABLE ("0" ATTENUATION)

FIGURE 14 DYNAMIC RANGE TEST

Source: https://assist.dla.mil -- Downloaded: 202640-26T03:58Z
Check the source to verify that this is the current version before use.



NOTE: TO MEASURE FREQUENCY ON PRESENT MODELS, CONNECT TO TEST POINT 3 CARD A4.

FIGURE 15 MASTER OSCILLATOR FREQUENCY ACCURACY

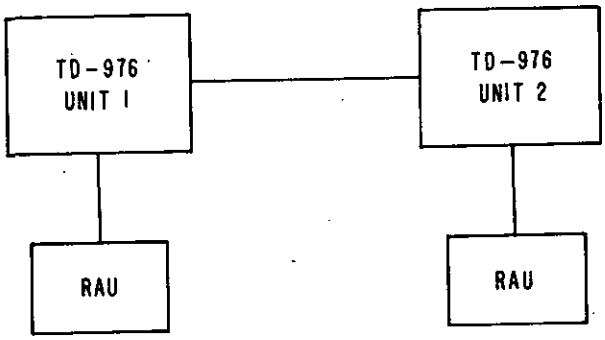


FIGURE 16 PCM GROUP SWITCH, ON/OFF TEST

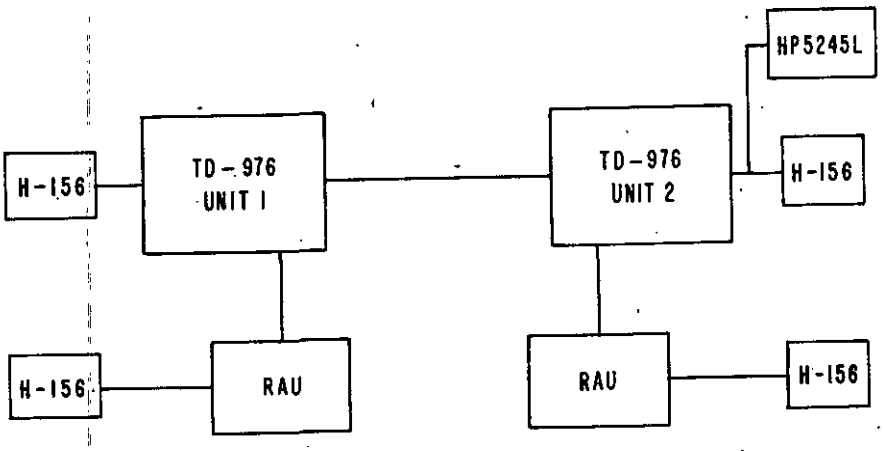


FIGURE 17 ORDERWIRE APPEARANCE TEST

Source: https://assist.dla.mil -- Downloaded: 202640-26T03:58Z
Check the source to verify that this is the current version before use.

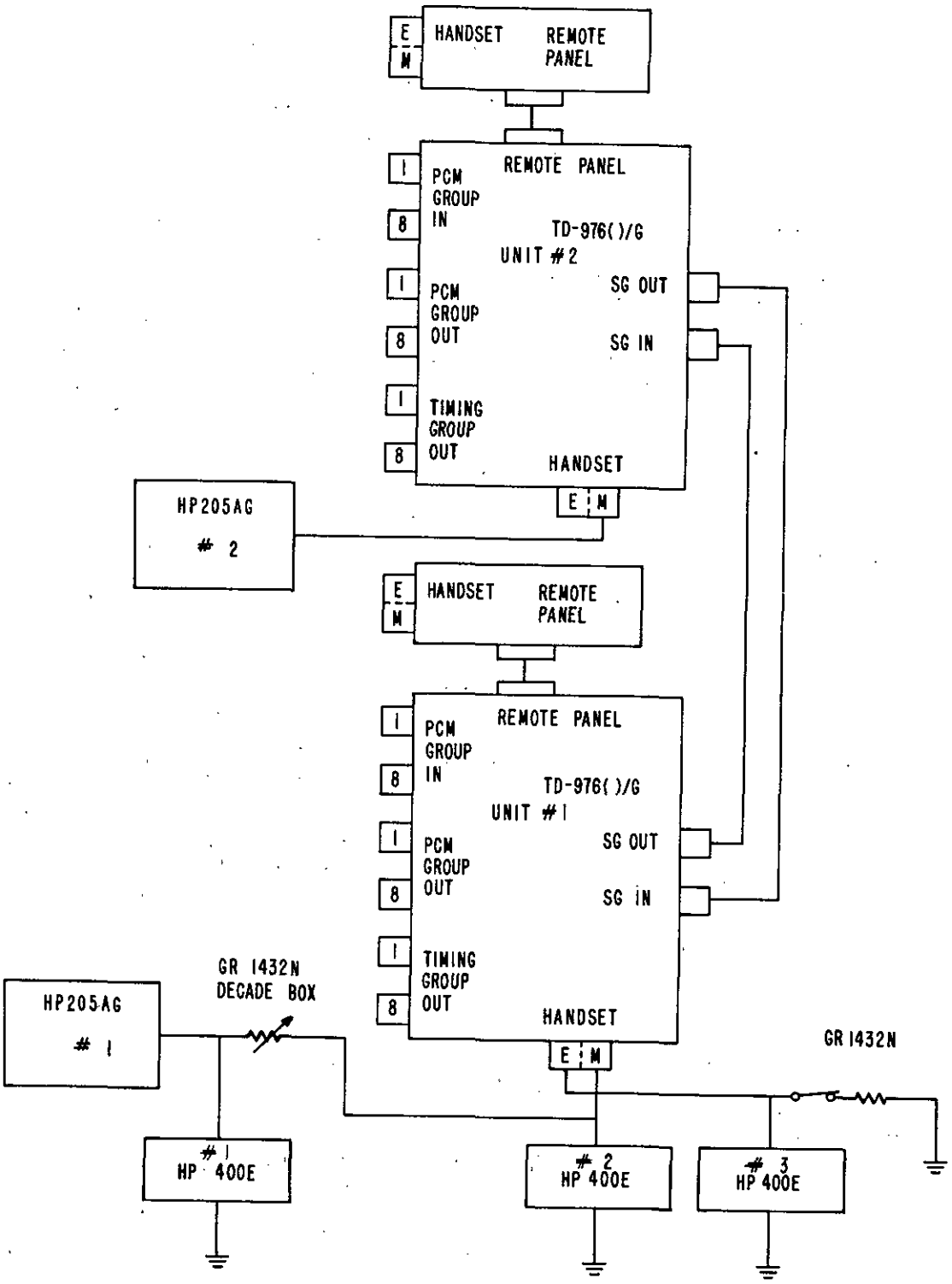


FIGURE 18 AVOW AND DVOW CIRCUIT IMPEDANCE TEST SETUP

Source: https://assist.dla.mil -- Downloaded: 202640-26T03:58Z
Check the source to verify that this is the current version before use.

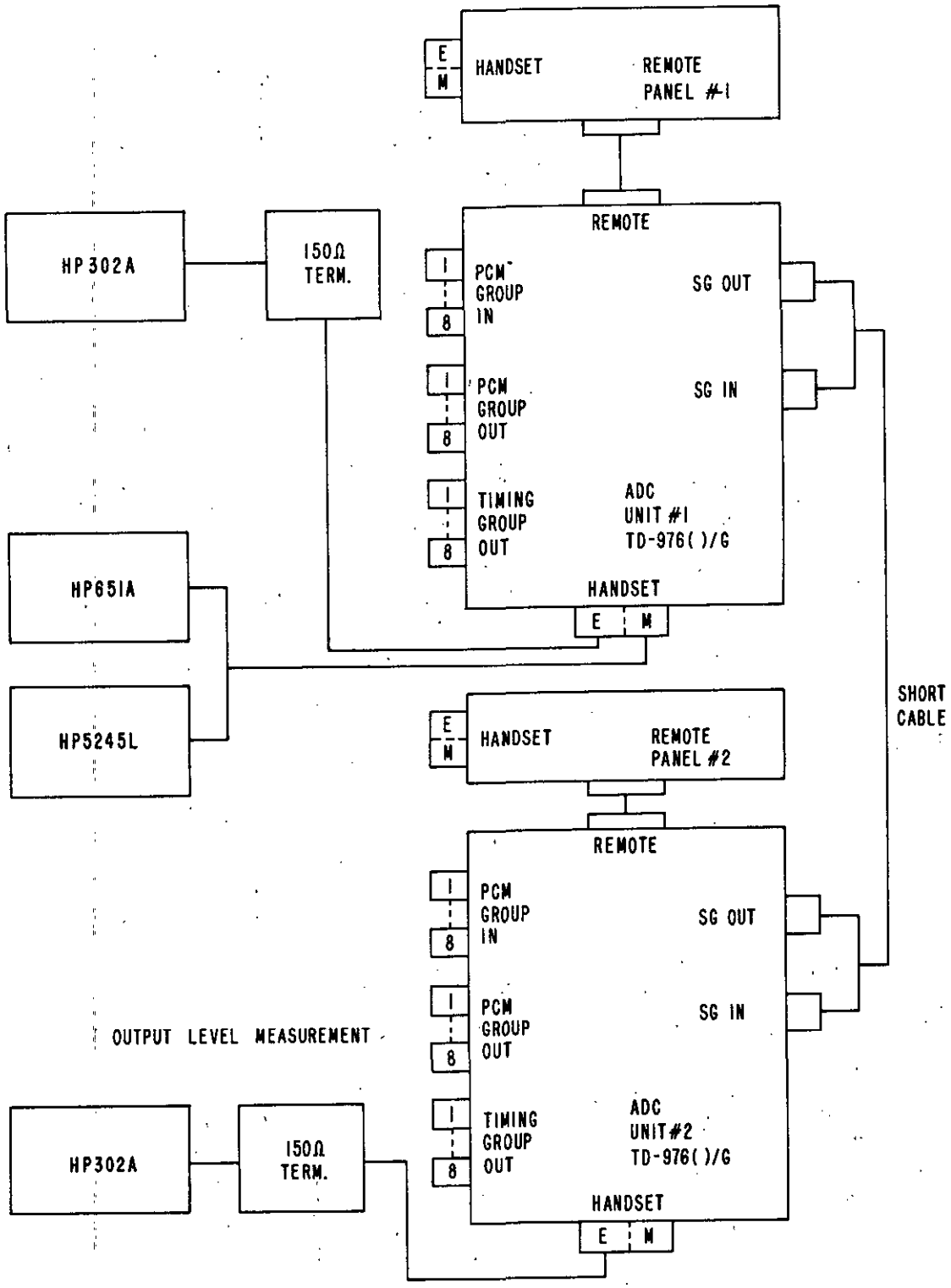


FIGURE 19 AVOW CIRCUIT OUTPUT LEVEL TEST SETUP

Source: https://assist.dla.mil -- Downloaded: 202640-26T03:58Z
Check the source to verify that this is the current version before use.

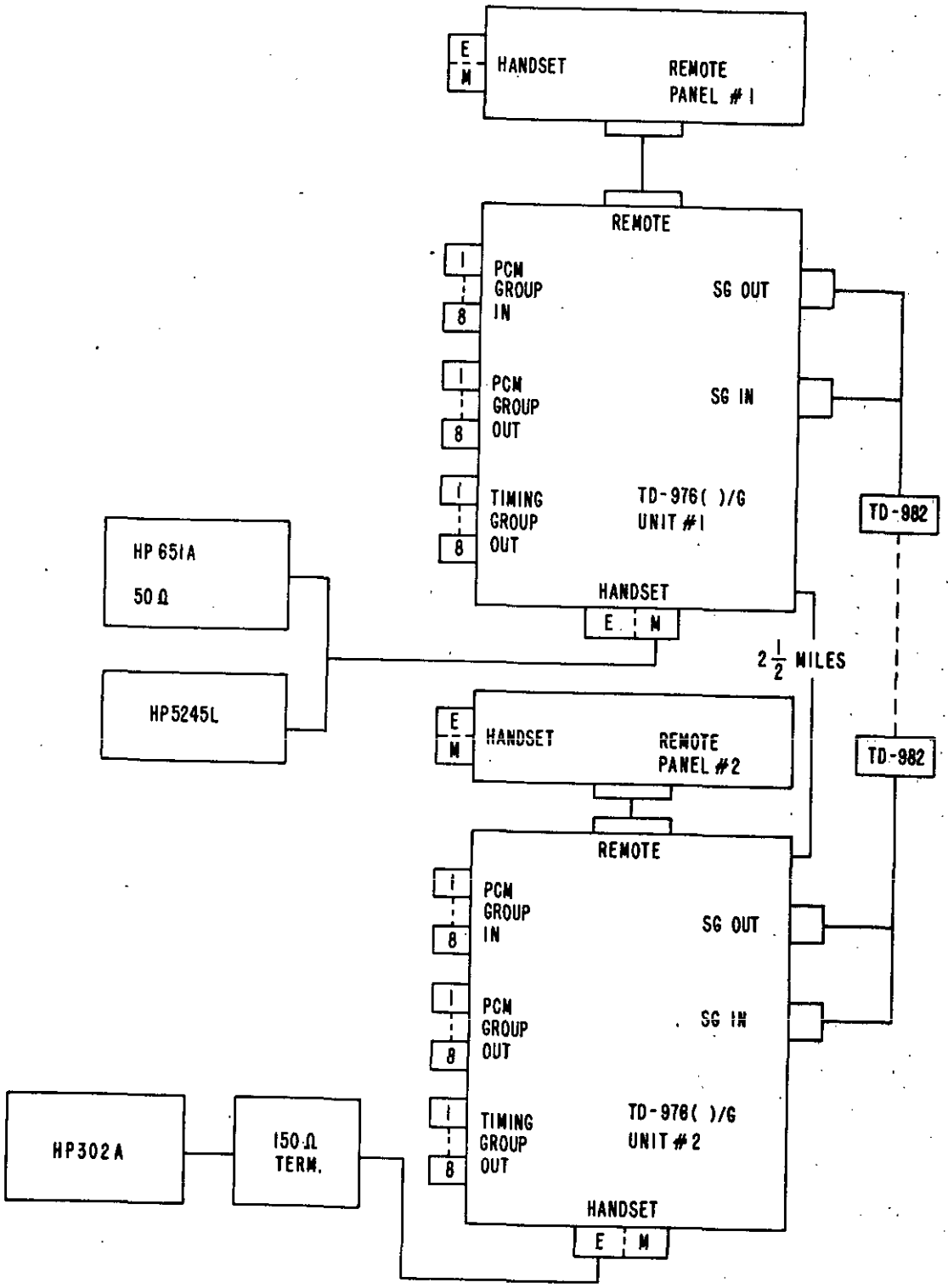


FIGURE 20 AVOW FREQUENCY RESPONSE TEST SETUP

Source: https://assist.dla.mil -- Downloaded: 202640-26T03:58Z
Check the source to verify that this is the current version before use.

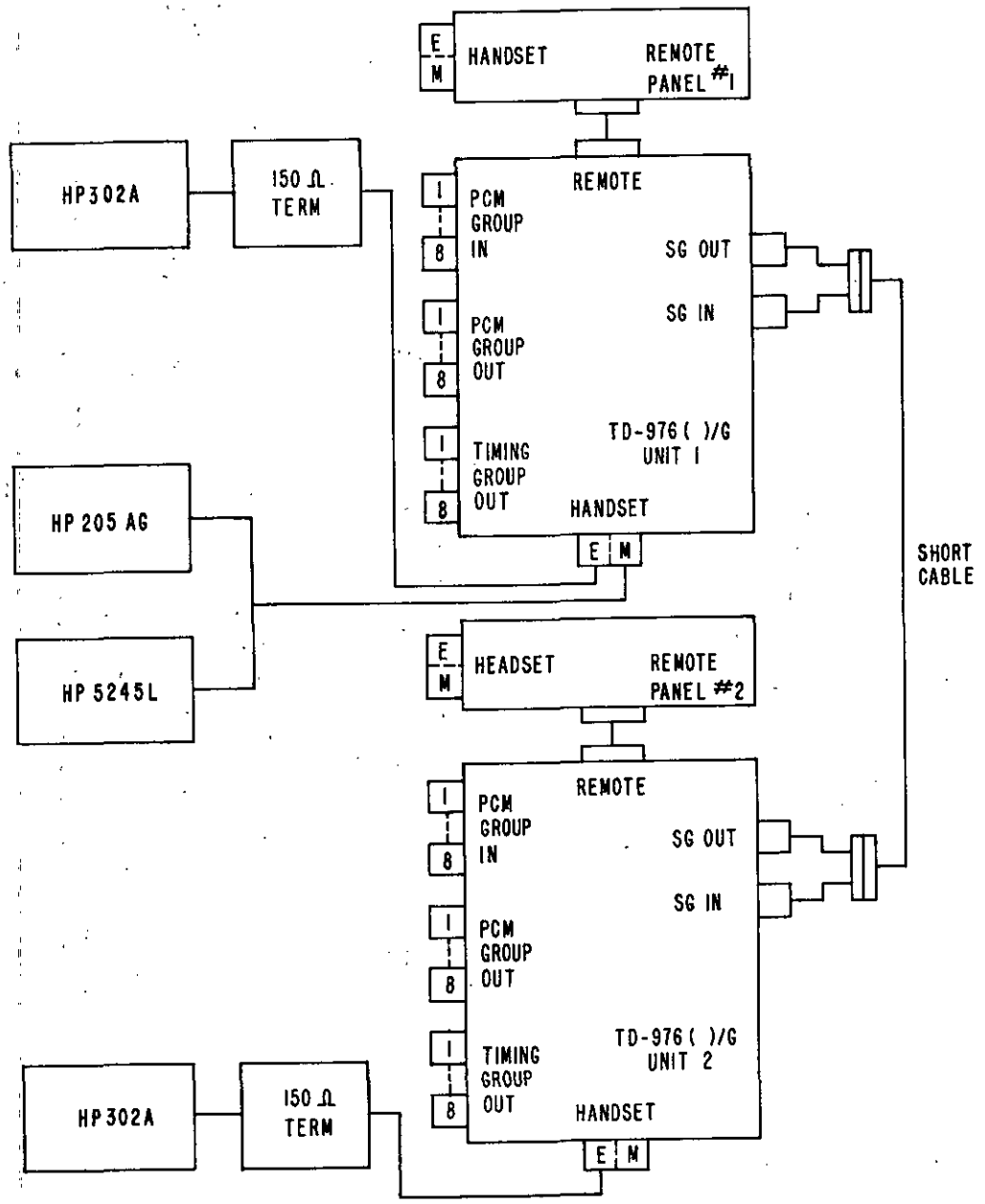


FIGURE 21. AVOW OUTPUT SIGNAL LEVEL TEST SETUP

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:58Z
Check the source to verify that this is the current version before use.

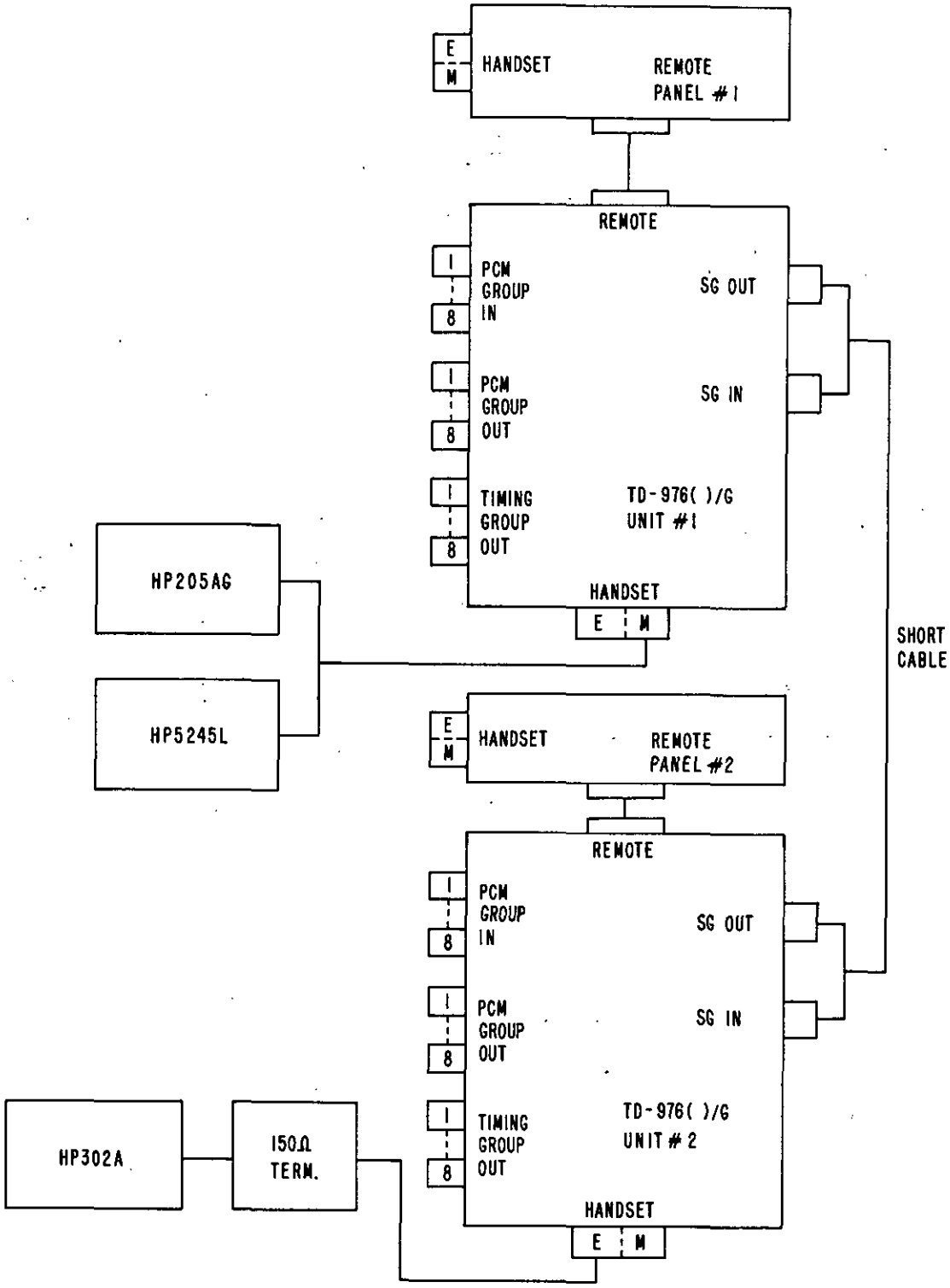


FIGURE 22 DVOW FREQUENCY RESPONSE TEST SETUP

Source: https://assist.dla.mil -- Downloaded: 202640-26T03:58Z
Check the source to verify that this is the current version before use.

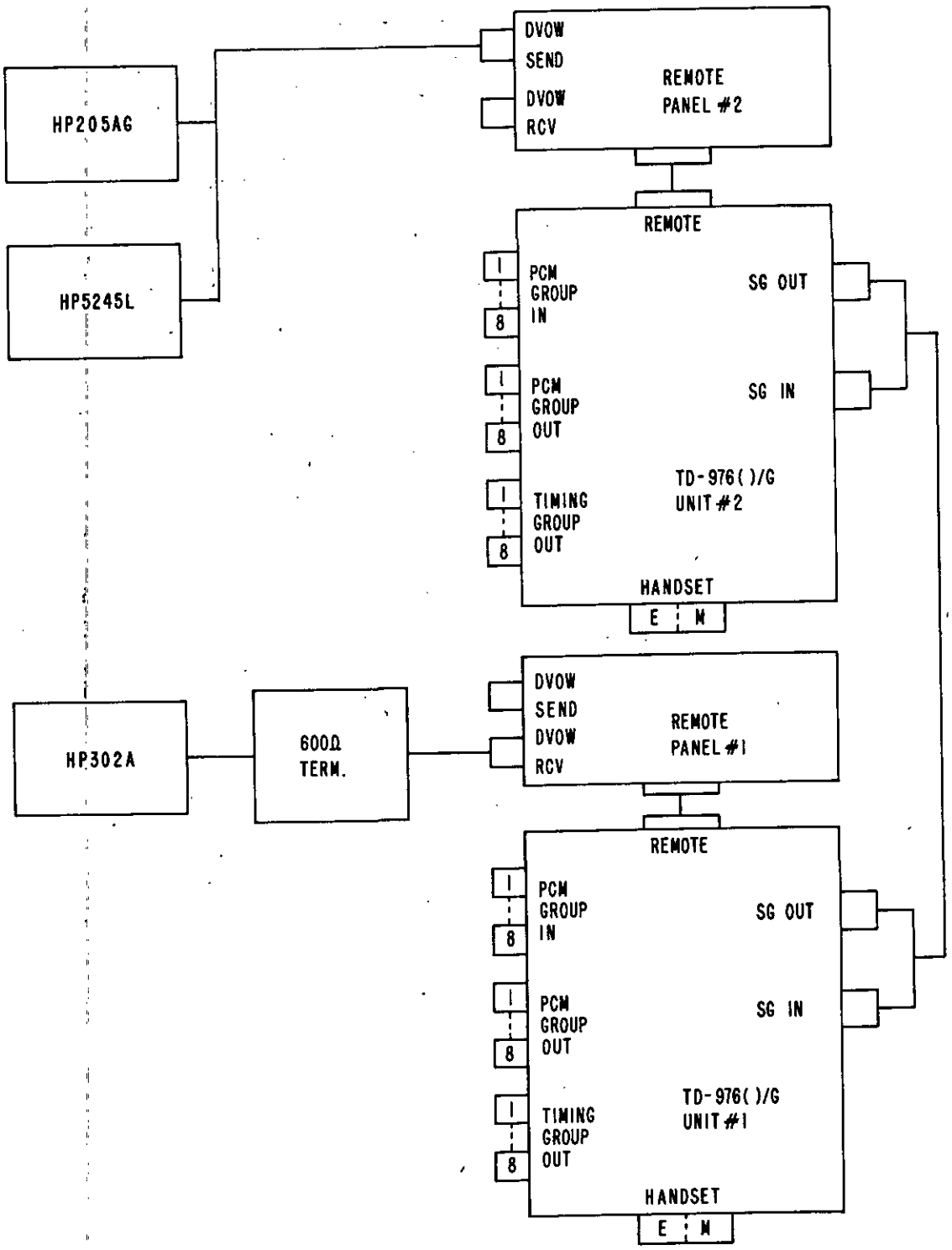


FIGURE 23 DVOW 4-WIRE FREQUENCY RESPONSE TEST SETUP

Source: https://assist.dla.mil -- Downloaded: 202640-26T03:58Z
Check the source to verify that this is the current version before use.

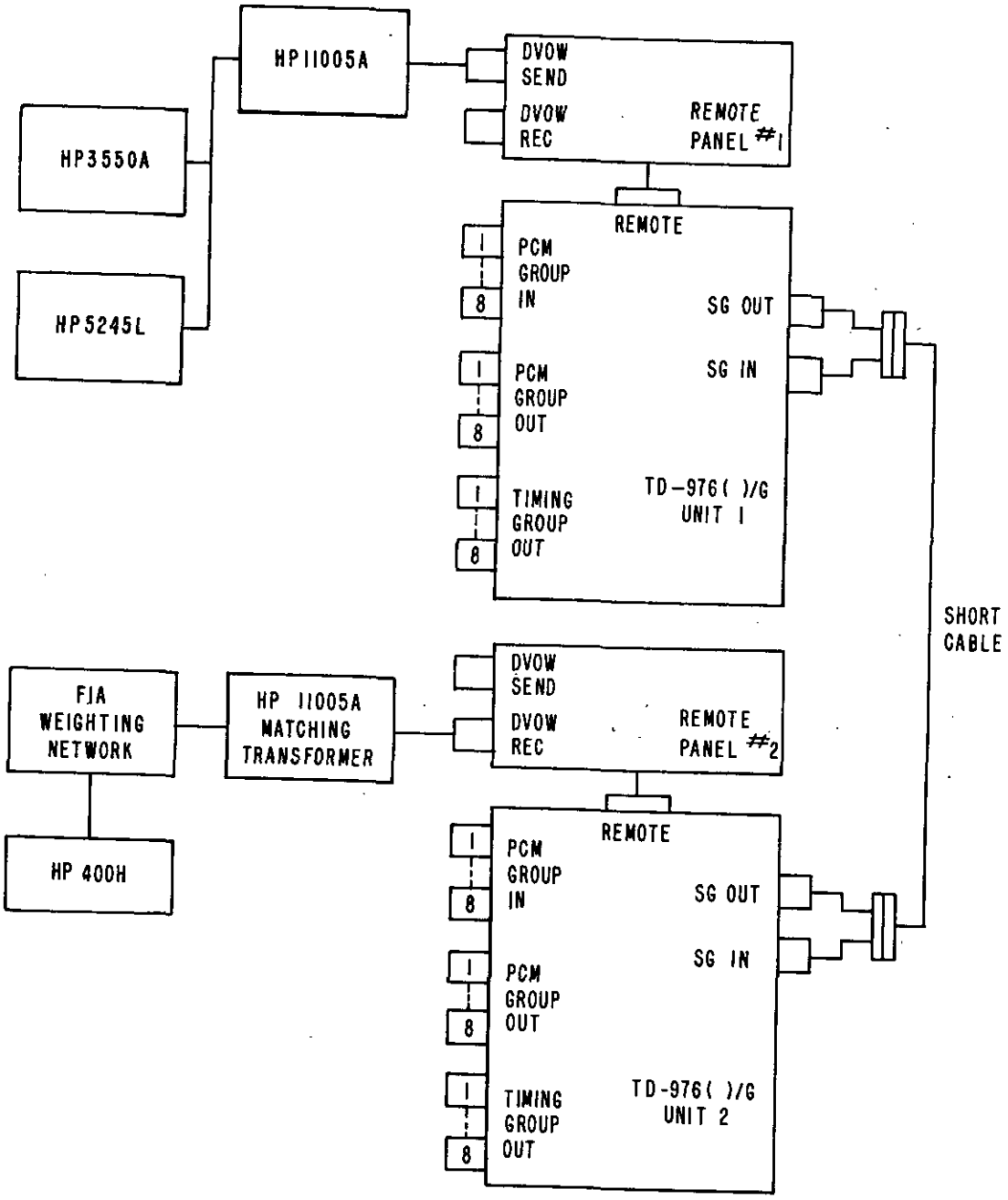


FIGURE 24 DVOW SIGNAL-TO-NOISE TEST SETUP

Source: https://assist.dla.mil -- Downloaded: 202640-26T03:58Z
Check the source to verify that this is the current version before use.

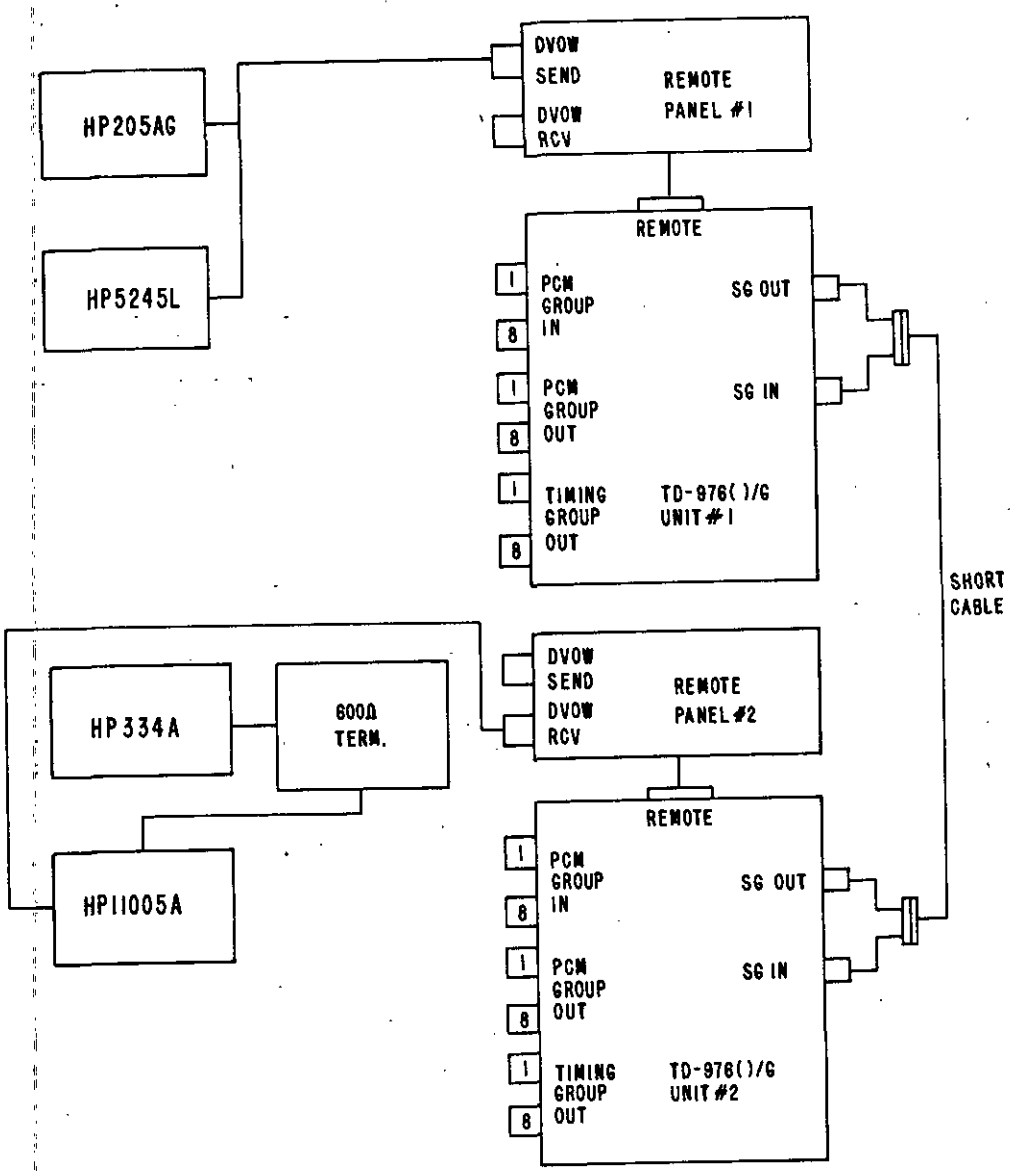


FIGURE 25 DVOW DISTORTION TEST SETUP

Source: https://assist.dla.mil -- Downloaded: 202640-26T03:58Z
Check the source to verify that this is the current version before use.

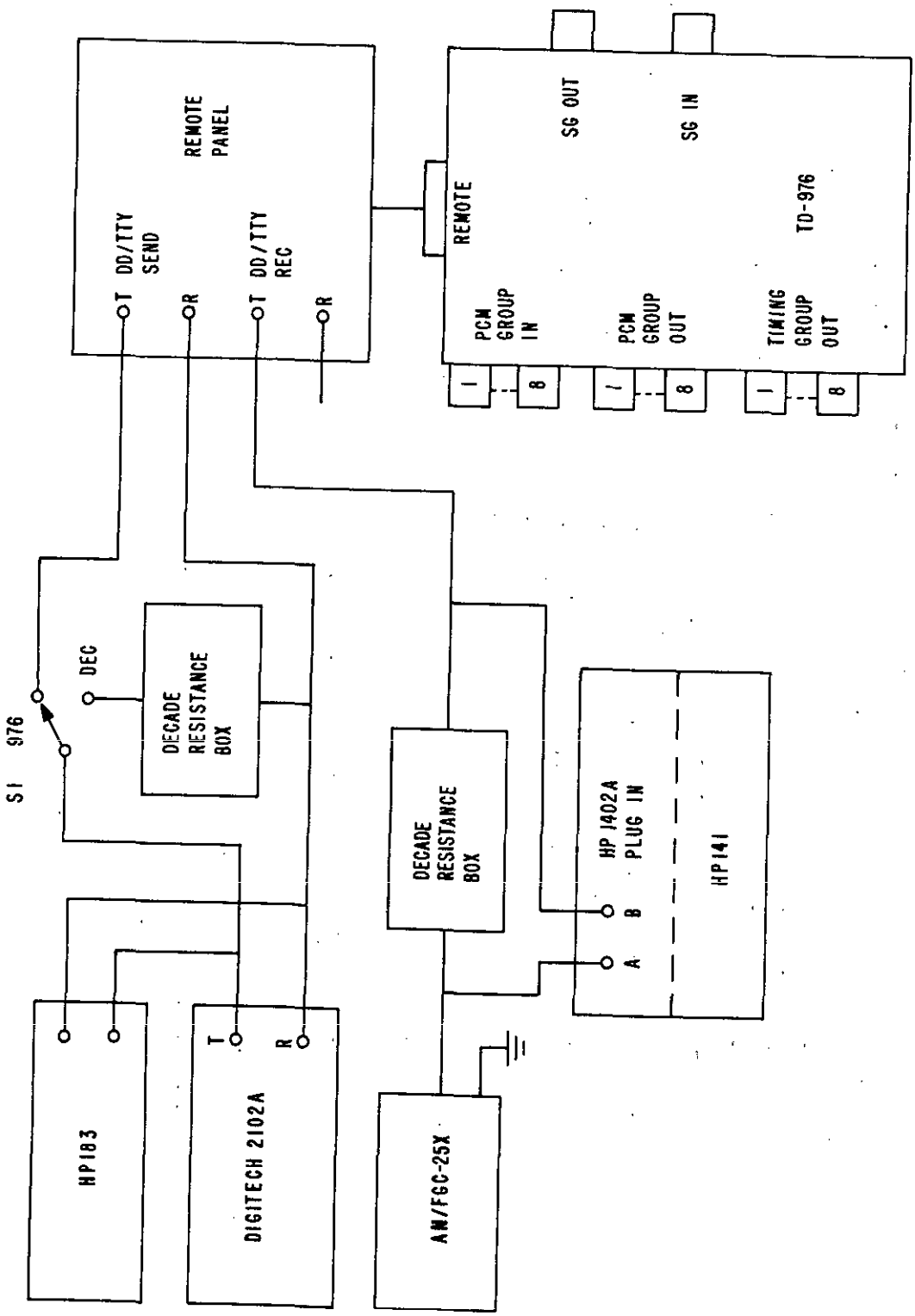


FIGURE 26 DD/TTY ORDER WIRE CIRCUIT IMPEDANCE TEST SETUP A

Source: https://assist.dla.mil -- Downloaded: 202640-26T03:58Z
Check the source to verify that this is the current version before use.

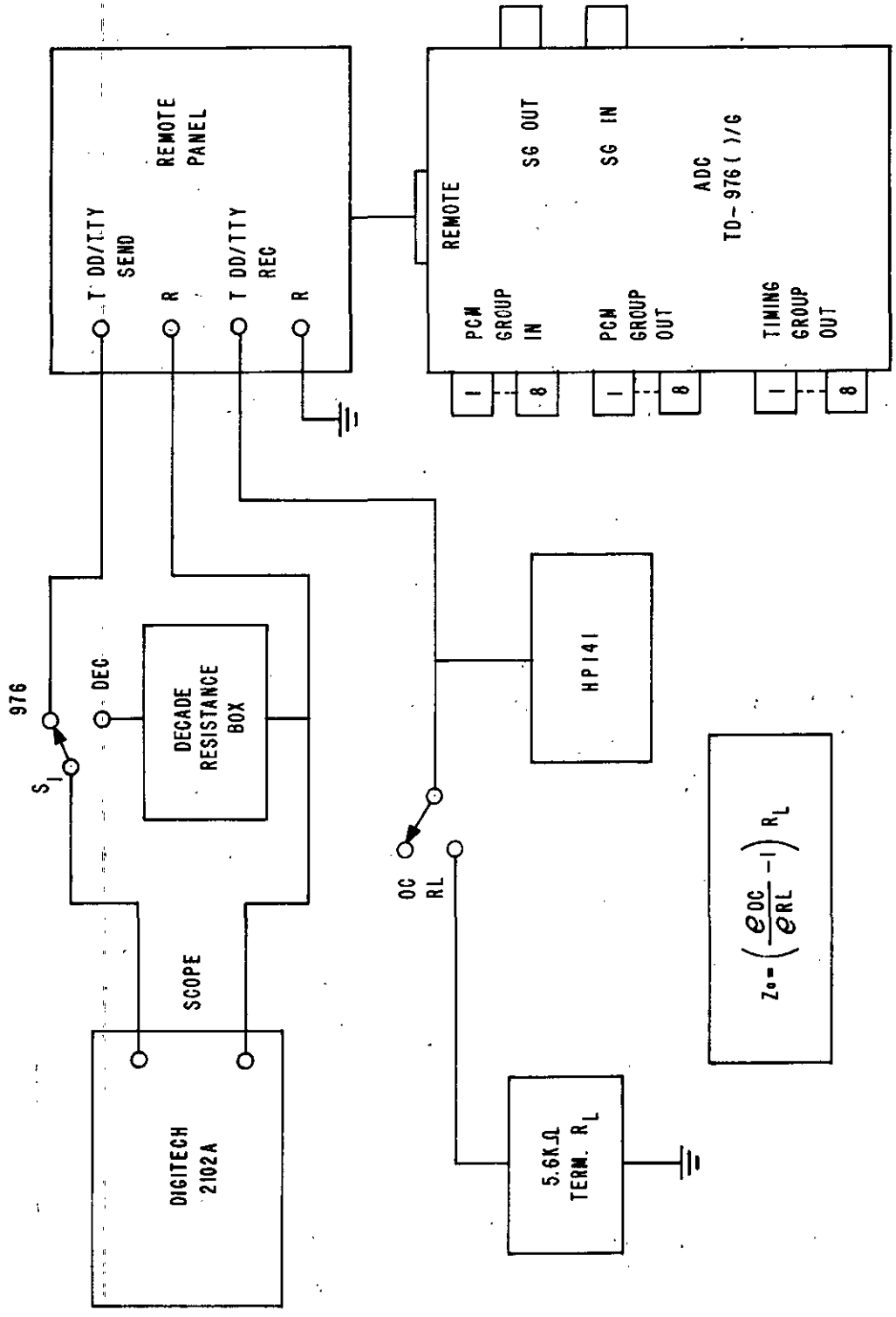


FIGURE 27 DD/TTY ORDER WIRE CIRCUIT IMPEDANCE TEST SETUP B

Source: https://assist.dla.mil -- Downloaded: 202640-26T03:58Z
Check the source to verify that this is the current version before use.

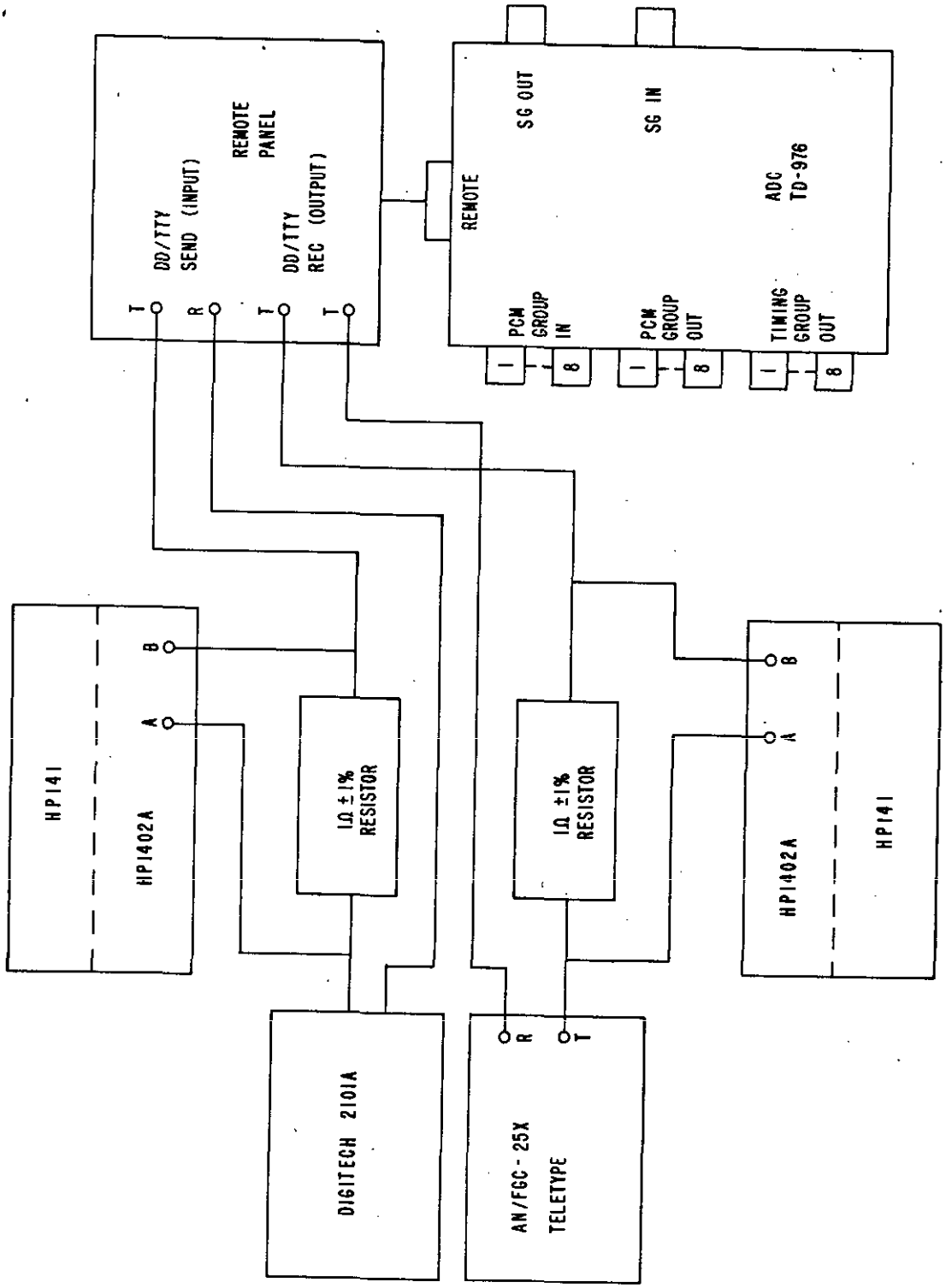


FIGURE 28 DD/TTY ORDER WIRE SIGNAL LEVEL TEST SETUP

Source: https://assist.dla.mil -- Downloaded: 202640-26T03:58Z
Check the source to verify that this is the current version before use.

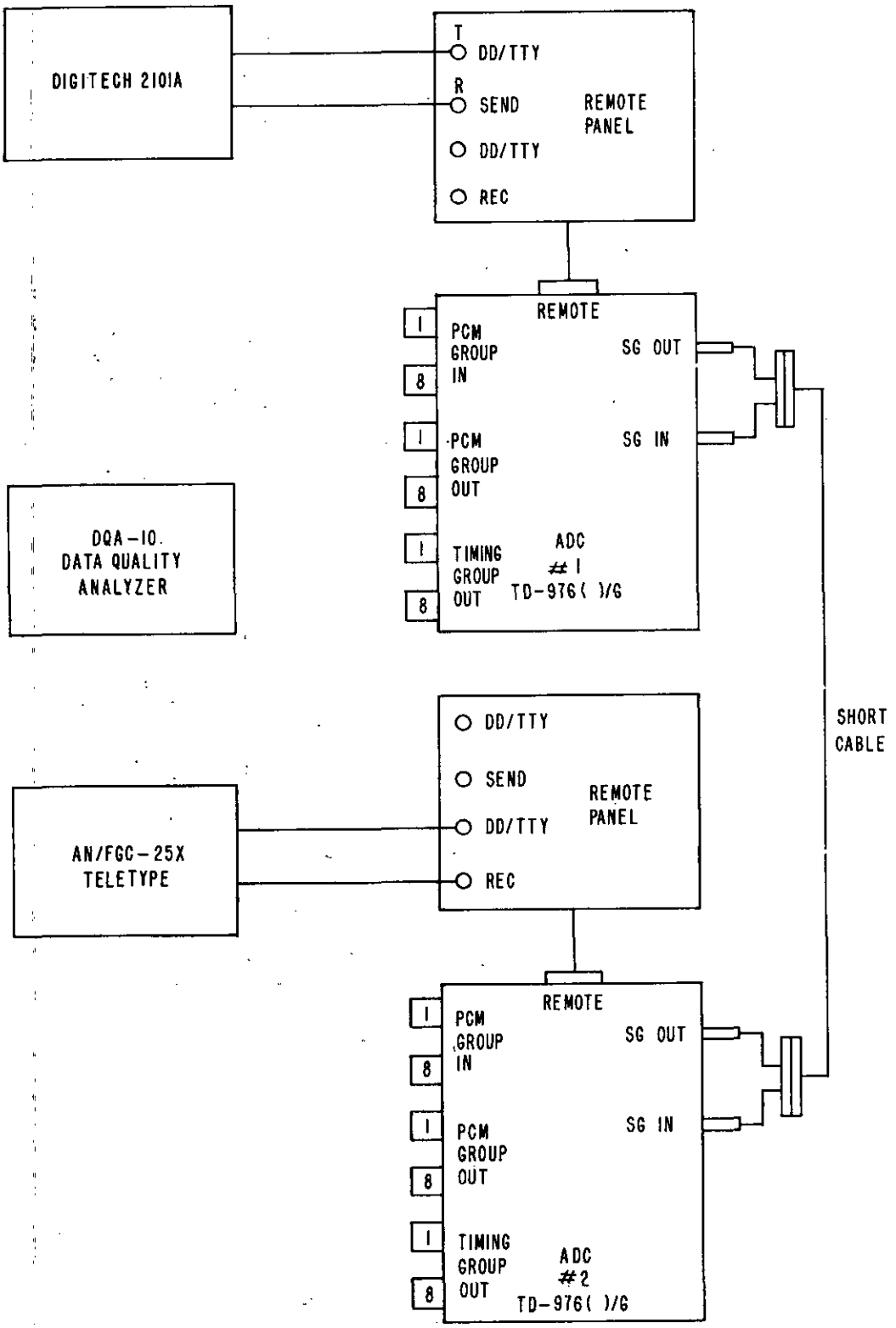


FIGURE 29 DD/TTY CIRCUIT BIAS DISTORTION TEST SETUP

Source: https://assist.dla.mil -- Downloaded: 202640-26T03:58Z
Check the source to verify that this is the current version before use.

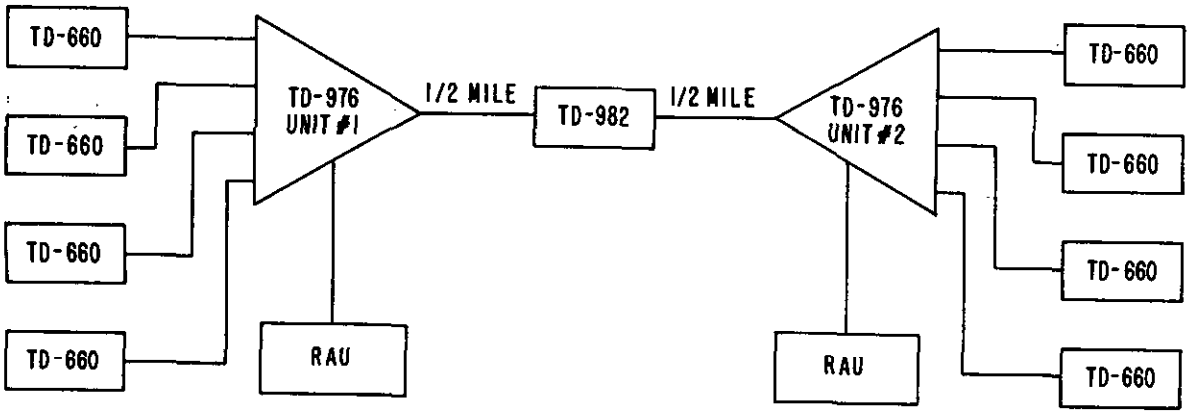


FIGURE 30 ALARM TEST

Source: https://assist.dla.mil -- Downloaded: 202640-26T03:58Z
Check the source to verify that this is the current version before use.

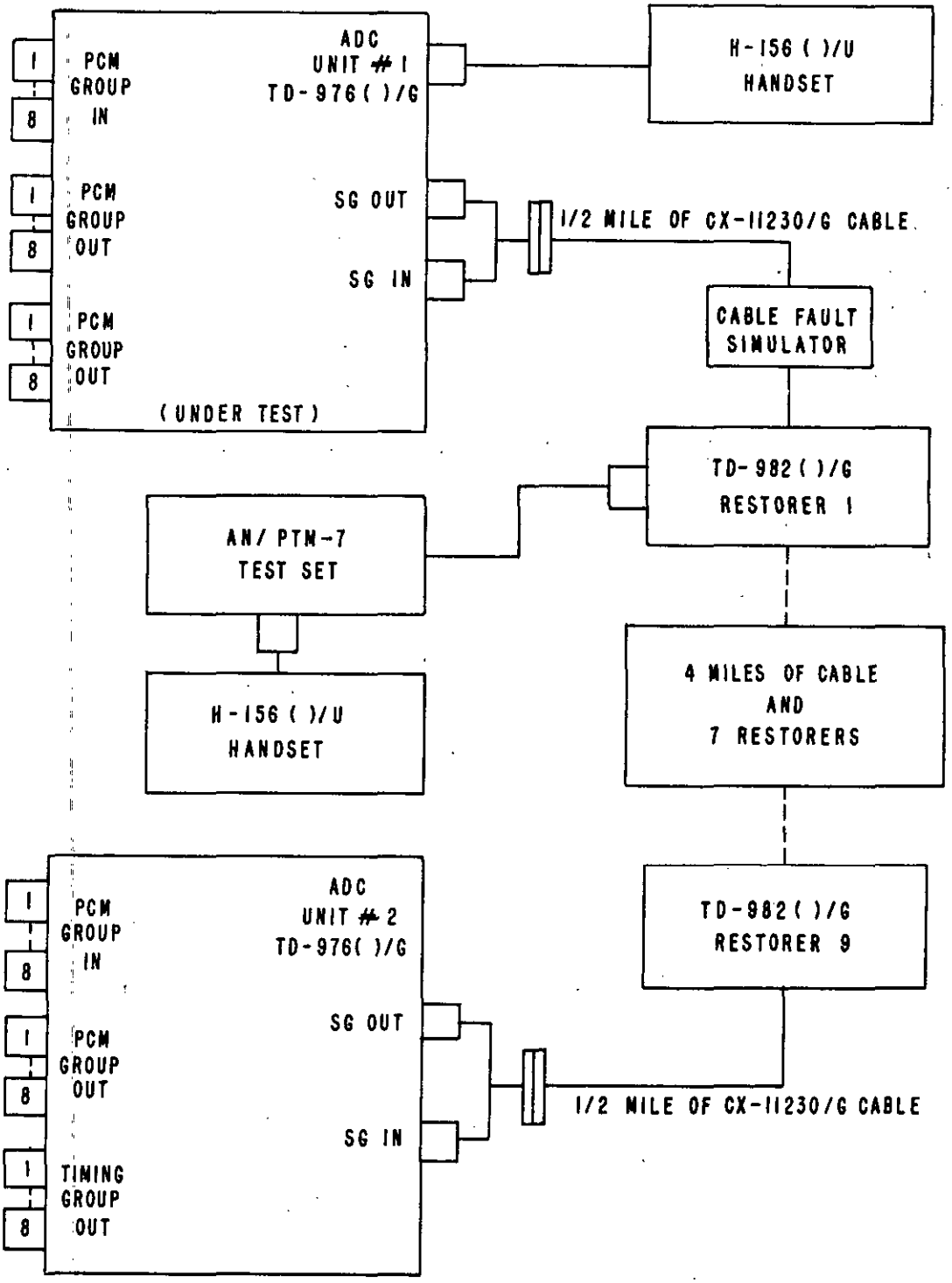


FIGURE 31 CABLE FAULT LOCATION TEST SETUP

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:58Z
Check the source to verify that this is the current version before use.

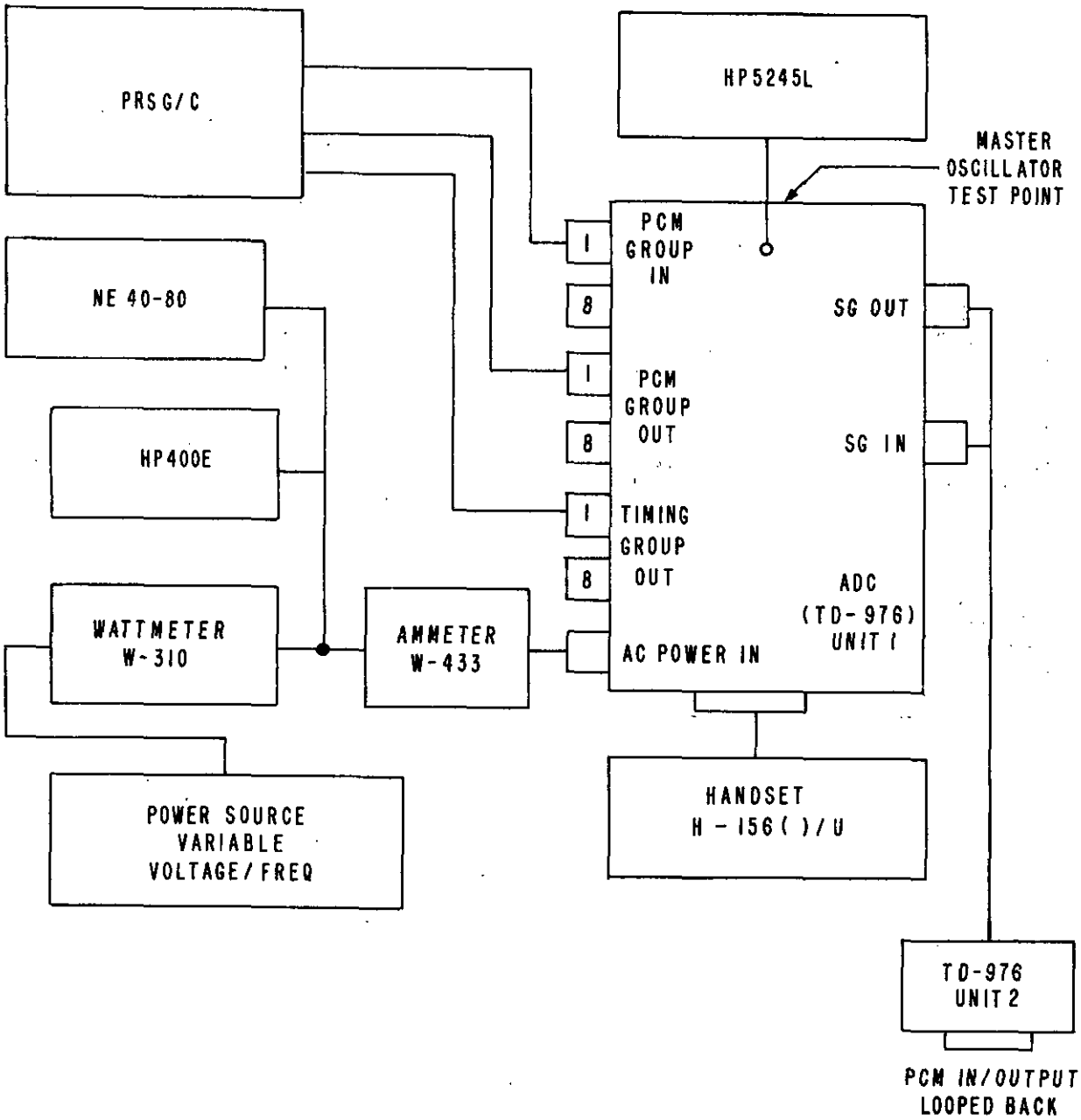


FIGURE 32 INPUT POWER TEST SETUP

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:58Z
Check the source to verify that this is the current version before use.

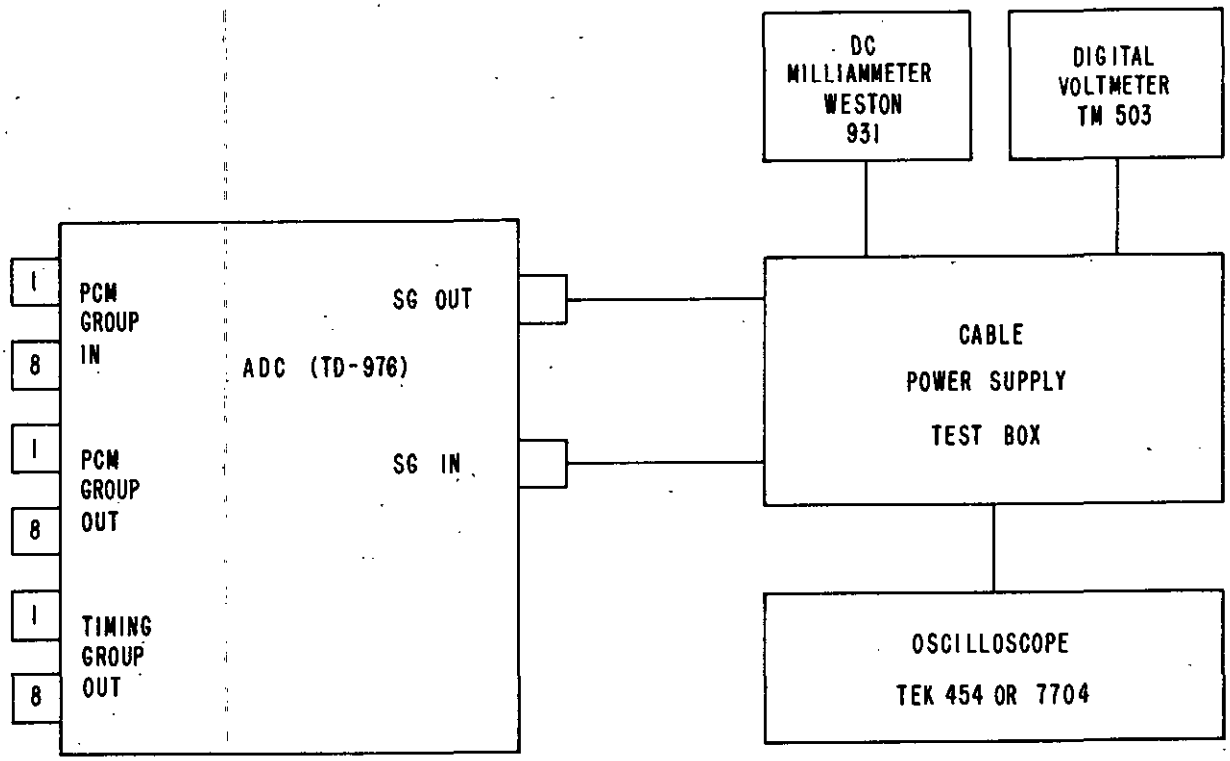


FIGURE 33 CABLE POWER SUPPLY TEST SETUP

Source: https://assist.dla.mil -- Downloaded: 202640-26T03:58Z
Check the source to verify that this is the current version before use.

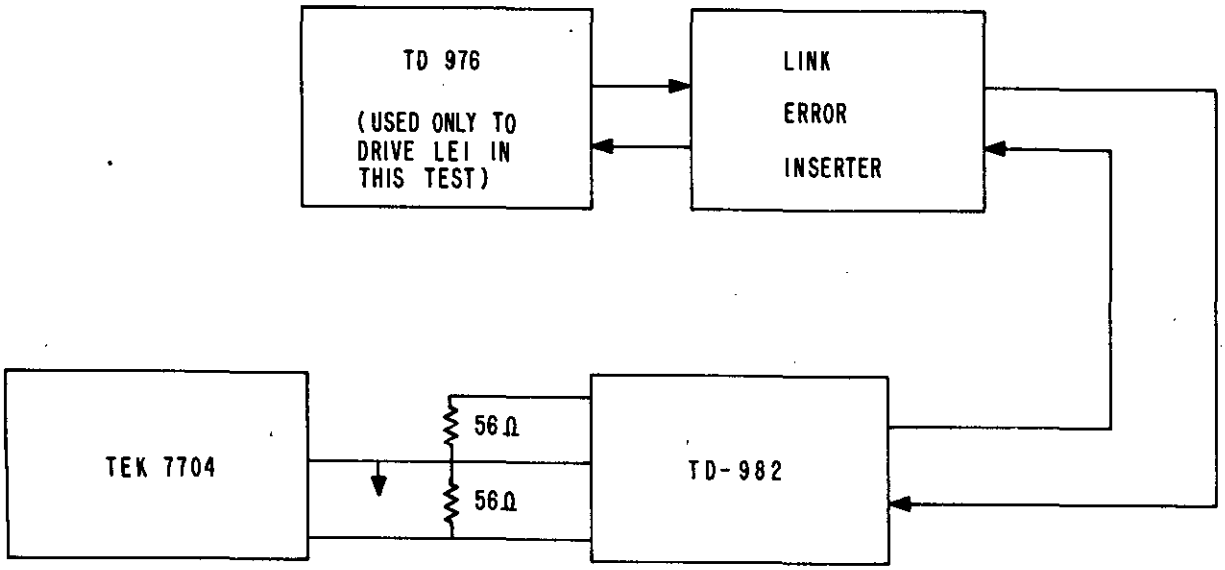


FIGURE 34 RESTORER OUTPUT SIGNAL CHARACTERISTICS TEST BLOCK DIAGRAM

Source: https://assist.dla.mil -- Downloaded: 202640-26T03:58Z
Check the source to verify that this is the current version before use.

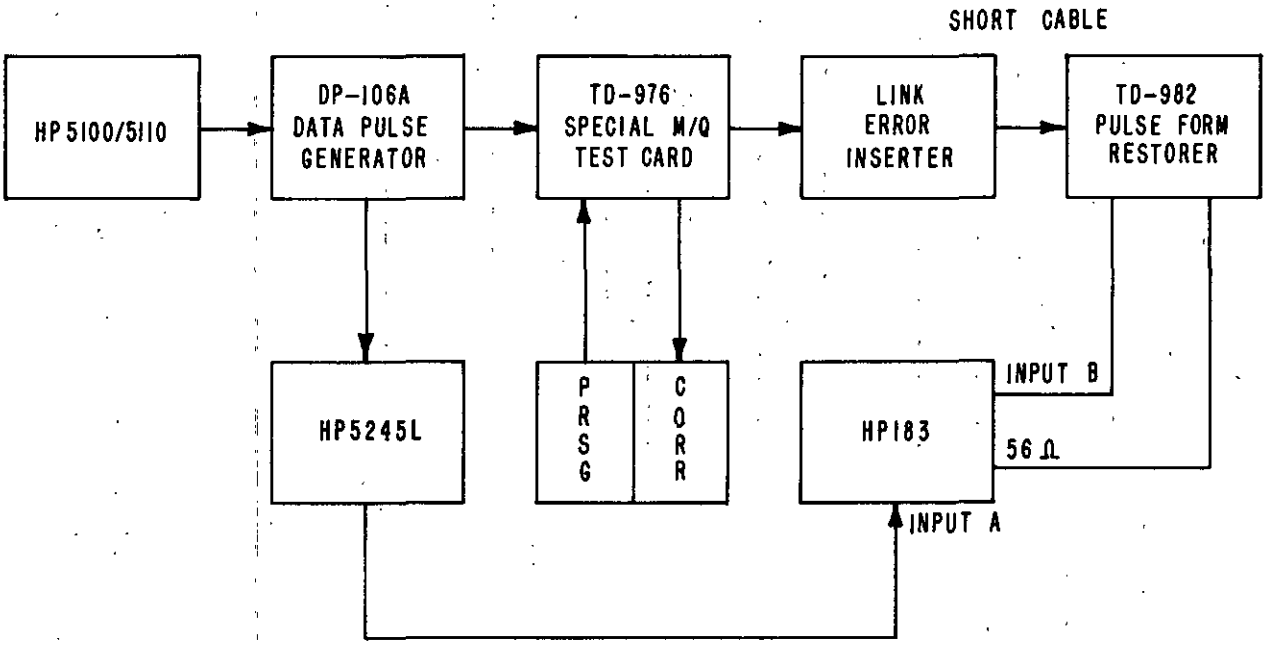
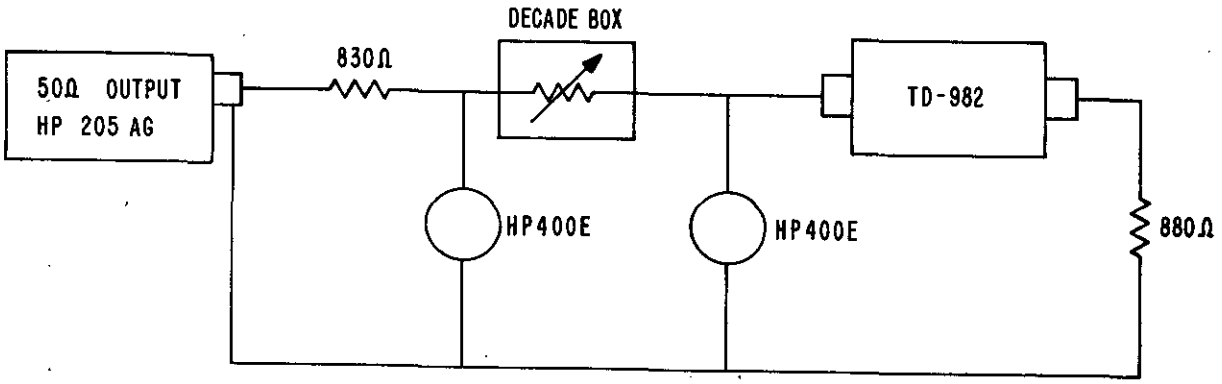
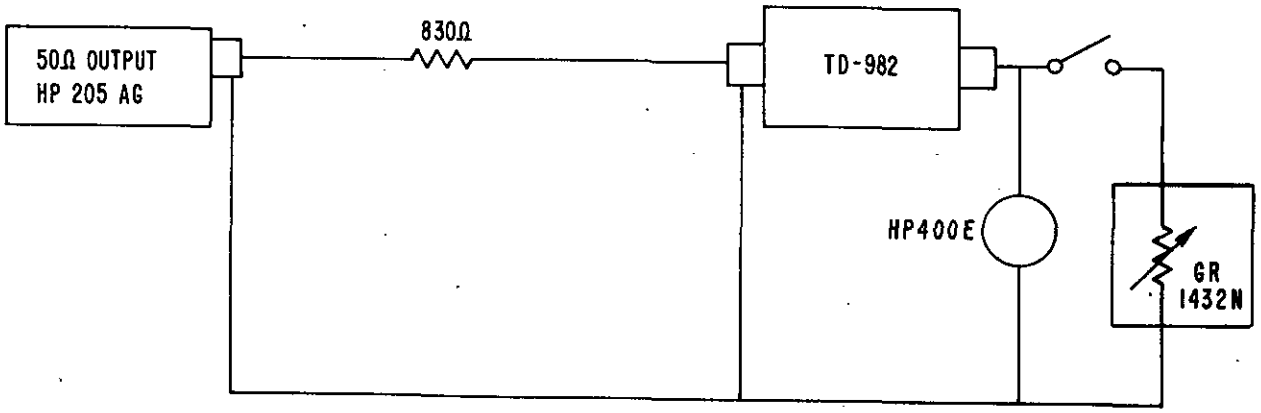


FIGURE 35 RESTORER TIMING RECOVERY TEST BLOCK DIAGRAM

Source: https://assist.dla.mil -- Downloaded: 202640-26T03:58Z
Check the source to verify that this is the currentversion before use.



INPUT IMPEDANCE TEST SETUP



OUTPUT IMPEDANCE TEST SETUP

FIGURE 36 ORDER WIRE CHARACTERISTIC TEST SETUP

Source: https://assist.dla.mil -- Downloaded: 202640-26T03:58Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-M-49103 (EL) MULTIPLEXER, ASYNCHRONOUS DIGITAL COMBINER TD-976()/G, RESTORER, PULSE FORM TD-962()/G AND ANCILLARY ITEMS

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

ESC-FM 1068-68

Source: https://assist.dla.mil -- Downloaded: 202610-26T03:58Z
Check the source to verify that this is the current version before use.

FOLD

Department of the Army
Headquarters
U.S.Army Electronics Command
Fort Monmouth, New Jersey 07703

POSTAGE AND FEES PAID

OFFICIAL BUSINESS

Commanding General
U.S.Army Electronics Command
ATTN: [redacted] DRSEL-RD-TS-S
Fort Monmouth, New Jersey 07703

FOLD

To detach this form, cut along this line

Source: https://assist.dla.mil -- Downloaded: 202640-26T03:58Z
Check the source to verify that this is the current version before use.